

Obs & Gyn Pharmacology

Drug handbook

The drug handbook for the labour ward and gynae ward. For each drug you need its indication, dose and route, what to monitor, what makes it dangerous and what reverses it. Magnesium sulphate toxicity, uterotonic contraindications and drug safety in pregnancy are examined every year.

Must know before the ward round

- MgSO₄ Pritchard:** 4 g of 20% IV over 5–10 min + 10 g of 50% IM (5 g each buttock), then **5 g IM 4-hourly**, alternate buttocks, for **24 h after delivery or the last fit**.
- Before every MgSO₄ dose check R-R-U:** RR ≥ 16 /min, patellar **reflexes present**, urine ≥ 30 mL/h. Loss of reflexes comes before respiratory depression (4–5 vs 6–7.5 mmol/L).
- MgSO₄ antidote:** **calcium gluconate 10% 10 mL (1 g) IV over 10 min**, plus stopping the magnesium and supporting breathing.
- Acute severe hypertension ($\geq 160/110$):** nifedipine IR **10 mg orally** (swallowed, never sublingual), or hydralazine **5 mg IV slowly** every 20–30 min (max 20 mg), or labetalol **20 mg IV** escalating (max 300 mg). Target **140–150/90–100**.
- Never ergometrine** in hypertension, pre-eclampsia or heart disease; **never carboprost** in asthma; **never labetalol** in asthma.
- Oxytocin:** 10 IU IM for AMTSL; PPH = 10 IU IV slowly + 20 IU in 1 L at 60 drops/min. Its toxicities are **tachysystole → fetal distress/rupture** and **water intoxication**.
- Tranexamic acid 1 g IV over 10 min within 3 h of birth**, repeat once if bleeding continues after 30 min. No benefit after 3 h.
- Misoprostol** doses fall as gestation rises: 800 µg for PPH treatment, 400 µg every 3 h at 13–24 weeks, **25 µg** for induction at term. **Avoid for third-trimester induction after a caesarean**.
- Dexamethasone 6 mg IM 12-hourly × 4** (or betamethasone 12 mg IM 24-hourly × 2) at **24–34 weeks** when birth is likely within 7 days. Tocolysis only buys 48 h: **nifedipine**.
- Avoid in pregnancy:** ACE inhibitors/ARBs, warfarin (6–12 weeks), valproate, tetracyclines, fluoroquinolones, methotrexate, misoprostol (unless intended), statins, isotretinoin.
- Sepsis triple:** ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV daily + metronidazole 500 mg IV 8-hourly. **PPROM: erythromycin**, never co-amoxiclav (NEC).
- Lignocaine max: 3 mg/kg plain** (≈ 20 mL of 1% for 60 kg), **7 mg/kg with adrenaline**. Toxicity: perioral tingling → fits → arrhythmia; treat with ABC and **20% lipid emulsion**.

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In a nutshell

On the labour ward a small number of drugs do most of the work, and most of the harm. **Magnesium sulphate** prevents eclamptic fits but can stop a woman breathing. **Oxytocin** saves women from haemorrhage but can rupture a uterus or drown a baby in hypoxia. **Ergometrine** contracts the uterus and can also cause a stroke in a hypertensive woman. **Misoprostol** is the most versatile drug in obstetrics, and its dose changes a hundred-fold (25 µg to 800 µg) depending on the gestation and the indication.

This handbook is organised by drug group. Every key drug has a **drug card** covering class, mechanism, indications, dose and route, side effects, contraindications, monitoring, pregnancy and breastfeeding, and antidote. Learn the cards, then test yourself on the flashcards and questions.

CLINICAL PEARL

For any drug the examiner asks about, answer in this order: **what it is, why we give it, how much and by which route, what we watch for, when we must not give it, and what reverses it.** That structure alone earns most of the marks.

UGANDA CONTEXT

Doses here follow the **Uganda Clinical Guidelines 2023** and WHO recommendations, and match the **Quick reference** page. Where UK practice differs (labetalol, carboprost, anti-D doses) this is flagged. Availability matters: labetalol, carboprost and atosiban are often unavailable in Ugandan public facilities, while MgSO₄, oxytocin, misoprostol, hydralazine, nifedipine and methyldopa are on the essential medicines list.

EXAM FAVOURITE

Pregnancy changes how drugs behave (bigger volume of distribution, faster renal clearance, lower albumin); the table is in **Drug safety** near the end. The classic viva: "Why can magnesium toxicity develop even when the dose was correct?" Because magnesium is **excreted only by the kidneys**. In severe pre-eclampsia with **oliguria** or rising creatinine, the same maintenance dose accumulates. That is why urine output is checked before every dose.

Clinical acumen

Focused history

Before prescribing anything to a pregnant or postpartum woman, ask these questions (each one changes the drug you choose):

- **Gestation, or is she postpartum?** It decides teratogen risk (organogenesis weeks 3–8 after conception), NSAID safety (avoid after ~20 weeks), misoprostol dose and whether ACE inhibitors are allowed (postpartum only).

- **Is she breastfeeding?** Most drugs are fine; check the few that aren't.
- **Allergies**, especially **penicillin**: what exactly happened? True anaphylaxis changes the antibiotic plan; a rash on amoxicillin may not.
- **Asthma** rules out carboprost, labetalol and other beta-blockers, and makes NSAIDs cautious.
- **Hypertension, pre-eclampsia, heart disease or Raynaud's** rule out ergometrine and make oxytocin boluses slow.
- **Kidney function / urine output** affects MgSO₄, gentamicin, NSAIDs and nitrofurantoin.
- **Epilepsy, HIV (which ARVs?), TB (rifampicin?), sickle cell, diabetes, myasthenia gravis** (MgSO₄ is contraindicated).
- **Previous caesarean or uterine surgery**: misoprostol for induction at term is contraindicated, oxytocin needs extra caution.
- **Drugs already given**, and when: how much oxytocin has run in, when the last MgSO₄ dose was, how much lignocaine has already been infiltrated.

Examination

- **Before MgSO₄**: respiratory rate, patellar reflexes, urine output (catheter), SpO₂, level of consciousness.
- **Before a uterotonic**: blood pressure (ergometrine), chest (asthma, pulmonary oedema), and is the **placenta out** and **is there a second twin?** (ergometrine and syntometrine must never be given with an undelivered twin).
- **Before oxytocin for augmentation**: contraction frequency, fetal heart, **cephalopelvic disproportion** (never augment an obstructed labour), parity and scars.
- **Weight**: for gentamicin (5 mg/kg), lignocaine maximum, enoxaparin and cefazolin (3 g if >120 kg).

Red flags

RED FLAG

- **Loss of patellar reflexes or RR <16/min on MgSO₄**: stop the magnesium; calcium gluconate if breathing is depressed.
- **More than 5 contractions in 10 minutes or a fetal bradycardia on oxytocin**: stop the drip and resuscitate the fetus.
- **Headache, confusion, fits or hyponatraemia after hours of oxytocin in dextrose**: water intoxication.
- **Perioral numbness, tinnitus, metallic taste, then fits** after perineal infiltration: local anaesthetic toxicity.
- **Wheeze, stridor, facial swelling or hypotension** after an antibiotic: anaphylaxis, give **adrenaline 0.5 mg IM** (0.5 mL of 1:1000).
- **Crushing headache or BP surge after ergometrine**: hypertensive crisis; treat as severe hypertension.

Thinking like a clinician

- **"Could the drug be the problem?"** A drowsy woman on MgSO₄, a baby who won't breathe after pethidine, a woman who fits after 15 mL of 2% lignocaine, a hypoglycaemic woman on IV quinine: always think of the drug first.
- **"What is the safest effective option here?"** Hypertensive + PPH → oxytocin, misoprostol, TXA; **not ergometrine**. Asthmatic + PPH → oxytocin, ergometrine (if normotensive), misoprostol; **not carboprost**. Asthmatic + severe hypertension → nifedipine or hydralazine; **not labetalol**.
- **Doses change with purpose**. Oxytocin 10 IU IM (AMTSL) is not the same as 2.5 IU in 500 mL titrated for induction. Misoprostol 25 µg (induction) is not 800 µg (PPH). A decimal error with these drugs kills.

- **Timing is part of the dose.** TXA only works within 3 h of birth; anti-D within 72 h; CS antibiotics within 60 min *before* incision; steroids need 24–48 h to help, which is why tocolysis buys 48 h.
- **Common student mistakes:** calling methyldopa an emergency drug (it takes hours); giving nifedipine sublingually; forgetting to check the undelivered twin before ergometrine; prescribing doxycycline or ciprofloxacin to a pregnant woman; giving 5 mg folic acid with SP (it blocks SP's antimalarial effect).

Clinical workup

For drugs, the "workup" is the set of checks you do **before** prescribing and the monitoring you do **after**. Use this every time you write a drug chart on the labour ward.

Step 0: First 5 minutes

In an emergency (eclamptic fit, PPH, anaphylaxis), the drug goes in first and the paperwork follows:

1. **Call for help** and assign someone to draw up drugs and record the time of each dose.
2. **ABC**, left lateral tilt if pregnant, oxygen, IV access with two large-bore cannulae.
3. Give the **first-line drug from memory**: MgSO₄ loading dose (fit), oxytocin 10 IU and TXA 1 g (PPH), adrenaline 0.5 mg IM (anaphylaxis), calcium gluconate 1 g IV (MgSO₄ toxicity with respiratory depression).
4. **Say the dose out loud** and have a second person confirm the drug, strength and route (closed-loop communication). Most labour ward drug errors are ampoule mix-ups (e.g. 50% vs 20% MgSO₄, ergometrine vs oxytocin in the fridge).

Step 1: Focused history

- ☐ **Allergies** (and the reaction): changes the antibiotic choice.
- ☐ **Gestation or days postpartum**: teratogen window, NSAIDs, ACE inhibitors, misoprostol dose.
- ☐ **Breastfeeding**: affects a few drugs (methotrexate, cabergoline, some antiepileptics).
- ☐ **Asthma, hypertension, heart disease, myasthenia, epilepsy, renal disease**: each removes specific drugs.
- ☐ **Current medicines**: ARVs (efavirenz/rifampicin interact with hormonal contraception), antiepileptics, anticoagulants.
- ☐ **Doses already given**: especially oxytocin, MgSO₄, lignocaine and opioids.

Step 2: Focused examination

- **Vital signs**: BP (ergometrine, antihypertensives), pulse (hydralazine tachycardia), RR and SpO₂ (MgSO₄, opioids), temperature (misoprostol fever vs sepsis).
- **Neurological**: patellar reflexes (MgSO₄), level of consciousness.
- **Chest**: wheeze (asthma: avoid carboprost and beta-blockers), crackles (pulmonary oedema: restrict fluids and oxytocin volume).
- **Obstetric**: uterine tone and contractions, fetal heart, placenta delivered, second twin excluded.
- **Weight** for weight-based doses.

Step 3: Bedside tests

- **Urine output** by catheter (hourly) on MgSO₄ and in PPH.
- **Urine dipstick** for protein and nitrites (pre-eclampsia, UTI before nitrofurantoin).
- **Random blood glucose** on insulin, IV quinine, or after steroids in diabetics.
- **CTG or intermittent fetal heart auscultation** on oxytocin, misoprostol and after antihypertensive boluses (a sudden BP drop can cause fetal distress).

- **Bedside clotting test** before regional anaesthesia or LMWH decisions when a coagulopathy is possible.

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
Serum magnesium	On MgSO ₄ with oliguria, absent reflexes or confusion (regional/national referral; rarely at HC IV)	Therapeutic 2–3.5 mmol/L ; reflexes lost at 4–5 ; respiratory depression at 6–7.5	Withhold or reduce doses; clinical monitoring is the standard where levels are unavailable
Creatinine / urea	Before and during MgSO ₄ , gentamicin, NSAIDs	Rising creatinine (>90 µmol/L in pregnancy is abnormal)	Halve or withhold MgSO ₄ maintenance; avoid NSAIDs; once-daily gentamicin with care or switch
Serum sodium	Long oxytocin infusions, especially in dextrose	<130 mmol/L = water intoxication	Stop oxytocin, restrict fluids, use saline-based infusions
Blood group and Rh, Coombs	Every booking; any sensitising event	Rh-negative, non-sensitised	Anti-D within 72 h
Hb / full blood count	Before iron, after PPH, on hydroxyurea/cotrimoxazole	Low Hb; low platelets (HELLP)	Iron dose; avoid NSAIDs and regional anaesthesia if platelets <75 × 10 ⁹ /L
LFTs	Methotrexate, pre-eclampsia, TB drugs, nevirapine	ALT/AST rise	Methotrexate contraindicated; review drug choice
Serum β-hCG	Before and day 4 and 7 after methotrexate	Needs a ≥15% fall day 4 → 7	Repeat dose or surgery if the fall is <15%
Blood glucose profile	GDM on insulin or metformin	Fasting >5.3 mmol/L, 1 h >7.8 or 2 h >6.4	Titrate insulin
G6PD (where available)	Before primaquine, nitrofurantoin, sulphonamides in known deficiency	Deficient	Avoid oxidant drugs (haemolysis)

Step 5: Putting it together

Model summary before prescribing: "Mrs K, 24, G2P1 at 36 weeks, 62 kg, with severe pre-eclampsia (BP 168/112, proteinuria 3+). No asthma, no allergies, catheterised with urine 40 mL/h, RR 18, reflexes present. **Plan:** MgSO₄ Pritchard loading dose, nifedipine 10 mg orally now and repeat in 30 min if BP ≥160/110, R-R-U before every dose, hourly BP and fetal heart; calcium gluconate drawn up at the bedside; no ergometrine at delivery."

Worked example: is it safe to give the next MgSO₄ dose? Four hours after loading: RR 14/min, knee jerks absent, urine 60 mL over the last 4 h, drowsy.

- RR <16 and reflexes absent → **withhold the dose**.
- Urine 15 mL/h (<30) explains the accumulation (renal excretion).
- Drowsy with RR 14 → **oxygen**, ask for a senior, **calcium gluconate 1 g IV over 10 min** is indicated if respiratory depression progresses (give it now if RR falls further or SpO₂ drops).
- Send creatinine and magnesium level if available; restart at a lower dose only when reflexes return and urine output improves.

Worked example: maximum lignocaine 60 kg woman, 1% lignocaine plain (10 mg/mL). Max 3 mg/kg = **180 mg = 18 mL**. With 2% lignocaine that is only **9 mL**. If more is needed (e.g. a big tear), switch to 1% with adrenaline (max 7 mg/kg) or dilute, or move to theatre.

Magnesium sulphate: the anticonvulsant

Magnesium sulphate (MgSO_4) is the single most important drug on the Ugandan labour ward after oxytocin. It **prevents** eclampsia in pre-eclampsia with severe features, **treats** eclamptic fits, and **protects the preterm baby's brain** before 32 weeks. It is cheap, on the essential medicines list, and can be given IM at a health centre III. What kills women is not the drug: it is **not giving it, giving it without checks, or having no calcium gluconate** when toxicity appears.

EXAM FAVOURITE

MgSO_4 is **not an antihypertensive**. A woman on MgSO_4 with a BP of 170/115 still needs **nifedipine or hydralazine now**. Examiners love this trap.

Magnesium sulphate (MgSO_4)

Class	Anticonvulsant (electrolyte); calcium antagonist; NMDA-receptor antagonist
Mechanism	(1) NMDA (glutamate) receptor antagonist : blocks the excitatory channel, raising the seizure threshold and limiting excitotoxic neuronal injury. (2) Physiological calcium antagonist : competes with Ca^{2+} at voltage-gated channels, relaxing vascular smooth muscle → cerebral vasodilatation , reversing the vasospasm and ischaemia behind eclampsia. (3) Reduces blood-brain-barrier permeability → less vasogenic cerebral oedema (PRES). (4) At the neuromuscular junction it reduces acetylcholine release: this is why reflexes disappear and, at higher levels, breathing stops
Indications (Obs/Gyn)	Eclampsia (treatment and prevention of recurrence); pre-eclampsia with severe features (prophylaxis, and intrapartum/24 h postpartum per unit protocol); HELLP ; fetal neuroprotection when birth is expected <32 weeks . Not recommended as a tocolytic (ineffective)
Dose & route (Uganda)	Pritchard (standard) : 4 g of 20% IV over 5–10 min + 10 g of 50% IM (5 g deep IM each buttock, each with 1 mL of 2% lignocaine) = 14 g loading ; then 5 g of 50% IM every 4 h , alternate buttocks. Zuspan (IV) : 4 g IV over 5–20 min , then 1 g/h infusion . Recurrent fit : 2 g of 20% IV over 5 min . Neuroprotection : 4 g IV over 20–30 min , then 1 g/h until birth or max 24 h. Duration (eclampsia/PE): 24 h after delivery or the last fit, whichever is later
Onset / key PK	IV: onset immediate . IM: onset about 1 h , lasts 3–4 h (hence 4-hourly doses). Distributed in extracellular fluid, crosses the placenta freely (fetal level $\approx$ maternal). Excreted almost entirely by the kidneys (unchanged); half-life about 4 h with normal renal function, much longer in renal impairment. Oliguria = accumulation = toxicity
Side effects	Warmth and flushing during IV loading (expected), nausea/vomiting, headache, sweating, muscle weakness, drowsiness, diplopia , slurred speech, pain/abscess at IM site , mild hypotension. Toxicity: areflexia → respiratory depression → cardiac arrest
Contraindications / cautions	Myasthenia gravis (can precipitate a myasthenic crisis: absolute). Heart block / significant conduction disease. Renal impairment/oliguria : reduce or withhold. Caution with neuromuscular blockers (potentiates both depolarising and non-depolarising), aminoglycosides (gentamicin: additive neuromuscular block), nifedipine (theoretical hypotension/neuromuscular block; rare in practice)
Monitoring	Before every IM dose and hourly on infusion: R-R-U: RR $\geq 16/\text{min}$, patellar reflexes present, urine $\geq 30 \text{ mL/h}$ ($\geq 100 \text{ mL}$ in 4 h) . Also SpO_2 (keep $\geq 95\%$), BP, pulse, level of consciousness; FHR. Serum Mg^{2+} only if available (oliguria, rising creatinine, suspected toxicity)
Pregnancy/breastfeeding	Safe and life-saving in pregnancy; compatible with breastfeeding. Neonate may be hypotonic or have respiratory depression /low Apgar: tell the resuscitator. Prolonged use (>5–7 days, as tocolysis) linked to fetal bone demineralisation: not relevant to a 24-hour course
Antidote/ reversal	Calcium gluconate 10%, 10 mL (1 g) IV slowly over 10 min . Keep it at the bedside before you start

Ward pearls

- **Label syringes in grams, not mL.** "4 g" of 50% is **8 mL**; "4 g" of 20% is **20 mL**. Mix-ups between grams and millilitres are the classic fatal error.
- The **50% solution is for IM use only**; never push 50% as an IV bolus. IV loading is always given as **20%**, slowly (**not faster than 5 min**: rapid IV injection can cause hypotension, arrhythmia and arrest).
- If she has had a **spinal**, the patellar reflex may be absent from the block: **test the biceps reflex** instead.
- A woman who had her **loading dose at an HC III** 3 hours ago needs her **maintenance** dose, **not a second loading dose**. Always read the referral note for the time and dose given.

Preparations and dilution maths

Preparation	Concentration	What you need to know
MgSO₄ 50% (10 mL ampoule, 5 g)	50 g per 100 mL = 0.5 g/mL	10 mL = 5 g ; 2 mL = 1 g. Use neat for IM (with lignocaine)
MgSO₄ 20% (made on the ward)	20 g per 100 mL = 0.2 g/mL	8 mL of 50% (4 g) + 12 mL water for injection (or NS) = 20 mL of 20% containing 4 g. Use for IV loading and recurrent fits
Recurrent-fit dose	2 g of 20%	4 mL of 50% + 6 mL water = 10 mL of 20%

DRUG DOSING

Worked example 1: making the IV loading dose. You need **4 g** as **20%**.

1. 50% = 0.5 g/mL, so 4 g ÷ 0.5 g/mL = **8 mL of 50%**.
2. 20% = 0.2 g/mL, so 4 g must sit in 4 ÷ 0.2 = **20 mL** total volume.
3. Add 20 – 8 = **12 mL of water for injection**. Give the **20 mL over 5–10 min** (about 2–4 mL per minute).

Worked example 2: Zuspan infusion at 1 g/h. Put **5 g (10 mL of 50%)** in **250 mL NS** → 5 g in about 260 mL ≈ **19 mg/mL**. For 1 g/h you need 1,000 ÷ 19 ≈ **52 mL/h** (most units round to **50 mL/h** on a pump). If urine output falls below 30 mL/h, **halve to 0.5 g/h (≈25 mL/h)**.

Worked example 3: IM maintenance. 5 g of 50% = **10 mL**; draw up with **1 mL of 2% lignocaine** (11 mL in total) and give deep IM into the upper outer quadrant of the buttock. Big volume: use a long (21 G, 38 mm) needle and alternate sides.

RED FLAG

Never run MgSO₄ by gravity drip "by eye" without a pump or burette: a free-flowing bag can deliver many grams in minutes. If you have no pump, use **Pritchard (IM)**.

Regimens at a glance

Regimen	Loading	Maintenance	Duration	Where
Pritchard (IM): Uganda standard	4 g 20% IV over 5–10 min + 10 g 50% IM (5 g each buttock) = 14 g	5 g 50% IM 4-hourly , alternate buttocks	24 h after delivery or last fit (whichever is later)	Any level: needs no pump
Zuspan (IV)	4 g IV over 5–20 min	1 g/h IV infusion	Same	Hospitals with pumps and 1:1 monitoring
Recurrent fit	2 g of 20% IV over 5 min (some protocols 4 g if >70 kg)	Continue current regimen	If fits continue, think of another cause (CT head)	All

Regimen	Loading	Maintenance	Duration	Where
Fetal neuroprotection (<32 wk)	4 g IV over 20–30 min	1 g/h	Until birth or max 24 h	Hospitals
Pre-referral (HC II/III)	Full loading dose (10 g IM ± 4 g IV if trained)	Send the next 5 g IM with her; due 4 h after loading , even on the road	Continue at the receiving hospital	HC II/III

UGANDA CONTEXT

Uganda (UCG 2023 and the MoH Essential Maternal and Newborn Clinical Care Guidelines) uses **Pritchard** because IM dosing needs no infusion pump and fits the staffing reality: one midwife, many mothers. In practice the commonest failures are **missed 4-hourly doses** and **undocumented R-R-U checks**. As the clerk, own the MgSO₄ chart: write the time the next dose is due in big letters.

Monitoring: what to check and what to do

Check	How and when	Normal	If abnormal: action
Respiratory rate	Count a full minute; before each IM dose , hourly on infusion (every 15–30 min in the first hours after loading in sick women)	≥16/min	RR <16: withhold the dose, call for help, oxygen, check reflexes and SpO ₂ . RR <12 or apnoea: stop MgSO₄, bag-mask ventilate, calcium gluconate 1 g IV over 10 min
Patellar reflex	Tap both knees before each dose (biceps if spinal)	Present (may be brisk in PE)	Absent: withhold the dose; this is the first sign of toxicity (≈4–5 mmol/L). Recheck hourly; resume only when reflexes return
Urine output	Foley catheter + hourly measurement	≥30 mL/h (≥100 mL in 4 h)	<30 mL/h: withhold the next IM dose (or halve the infusion to 0.5 g/h); check creatinine (± serum Mg); strict fluids (80 mL/h); no big fluid boluses (pulmonary oedema). Resume when output improves
SpO₂	Pulse oximeter, continuous if available	≥95%	Falling SpO ₂ : oxygen, check RR/airway, think toxicity or pulmonary oedema
Level of consciousness, speech	With each set of observations	Alert	Drowsiness, slurred speech: early toxicity → check reflexes and RR; also consider stroke/post-ictal state

Toxicity: levels and signs

Toxicity follows a predictable ladder: **reflexes go first, then breathing, then the heart**. Because reflexes disappear well before breathing is affected, a **present reflex is a reliable safety check**.

Serum Mg ²⁺ (mmol/L)	Approx. mg/dL	Clinical picture
0.7–1.0	1.7–2.4	Normal (non-pregnant)
2–3.5	4.8–8.4	Therapeutic for seizure prophylaxis. Warmth, flushing, nausea common
4–5	~10–12	Loss of patellar reflexes ; drowsiness, slurred speech, diplopia , muscle weakness
~5–7.5	~12–18	ECG changes (prolonged PR, widened QRS), hypotension, somnolence
6–7.5	~15–17	Respiratory depression → respiratory paralysis

Serum Mg ²⁺ (mmol/L)	Approx. mg/dL	Clinical picture
>7.5	>18	Cardiac conduction defects (heart block), profound hypotension
>12	>30	Cardiac arrest (asystole)

NOTE

Sources quote slightly different bands (RCOG and several UK units give a therapeutic range of about **2–4 mmol/L**; Williams quotes 4.8–8.4 mg/dL, i.e. **2–3.5 mmol/L**). This handbook uses **2–3.5 mmol/L** to match the quick reference. The **order** of signs is what matters clinically, and it is the same in every source. Conversion: **mmol/L × 2.43 = mg/dL**.

Early symptoms to ask about (she will tell you before the reflexes go): feeling hot and **flushed**, **nausea**, **slurred speech**, **drowsiness**, **double vision**, muscle weakness ("my legs feel heavy").

Step-by-step management of magnesium toxicity

1. **Stop the infusion / withhold the IM dose.** Do not give the next dose.
2. **Call for help** (senior midwife, doctor, anaesthetist).
3. **A-B-C:** open the airway, left lateral tilt if antenatal; **oxygen by face mask**; SpO₂ monitor.
4. If breathing is depressed (RR <12, falling SpO₂) or has stopped: **bag-valve-mask ventilation**, and prepare for intubation.
5. **Calcium gluconate 10%, 10 mL (1 g) IV slowly over 10 min** (it transiently antagonises Mg at the neuromuscular junction and heart). May be repeated if needed. Giving it too fast causes bradycardia and arrhythmia.
6. **Cardiac arrest:** start CPR (with left uterine displacement) and give calcium gluconate as above.
7. **Find the cause:** check **urine output** (oliguria is the usual culprit), **creatinine**, and **serum Mg²⁺** if available. Review the chart for **dosing errors** (double loading dose, 50% given IV, mL/g confusion).
8. **Enhance excretion** (senior decision): with adequate volume status, furosemide can increase renal Mg loss; **dialysis** for toxicity with renal failure at referral centres.
9. **When to restart:** only when **reflexes have returned**, **RR ≥16** and **urine ≥30 mL/h**, usually at a **reduced dose** (e.g. infusion 0.5 g/h, or the next IM dose delayed/halved per unit protocol) with hourly checks.

DRUG DOSING

Calcium gluconate 10%: 10 mL = **1 g** (≈2.2 mmol Ca²⁺). **Dose: 1 g IV over 10 min.** Draw it up and tape it to the MgSO₄ chart or bed before the first dose. Calcium effect is **short-lived**: ventilation, not calcium, is what keeps her alive if she stops breathing.

UGANDA CONTEXT

Calcium gluconate is the item most often **missing** when MgSO₄ is on the shelf. At HC III/IV, check stock at the start of every shift and include it in the eclampsia box. If there is none, the life-saver for respiratory depression is still **bag-and-mask ventilation**.

Why MgSO₄ beats diazepam and phenytoin

Trial	Comparison	Result
Collaborative Eclampsia Trial (Lancet 1995)	MgSO ₄ vs diazepam in eclampsia	52% fewer recurrent fits with MgSO ₄ ; lower maternal death trend, better neonatal outcomes

Trial	Comparison	Result
Same trial	MgSO ₄ vs phenytoin	67% fewer recurrent fits with MgSO ₄
Magpie Trial (Lancet 2002; ~10,000 women in 33 countries)	MgSO ₄ vs placebo in pre-eclampsia	Halved the risk of eclampsia (~58% reduction); NNT ~ 63 in severe pre-eclampsia; no clinically significant harm to the baby

Diazepam **sedates** and depresses breathing in both mother and neonate, masks the level of consciousness and is a poorer anticonvulsant for eclampsia. Use it **only if MgSO₄ is unavailable** (10 mg IV slowly over 2 min; rectal 20 mg if no IV access).

CLINICAL PEARL

Magnesium for the fetal brain: given before birth <32 weeks it reduces **cerebral palsy by about one third** (NNT about 63). Give when birth is expected within **24 h**, ideally **within 4 h** of birth, but **never delay an urgent delivery** to give it.

Antihypertensives in pregnancy

The aim is to **prevent maternal stroke** (the main cause of death in severe pre-eclampsia), not to "cure" pre-eclampsia (only delivery does that). **Treat urgently if BP ≥160/110** (confirm within 15 min; treat within **30–60 min**). **Acute target: 140–150/90–100**. Maintenance target: **~135/85**. **Do not drop the diastolic below 80**: the placenta has no autoregulation, so a rapid fall causes **fetal distress**.

Situation	First choice in Uganda	Alternatives
Acute severe hypertension	Nifedipine IR 10 mg oral or hydralazine 5 mg IV	Labetalol IV (rarely available)
Maintenance antenatal	Methyldopa ± nifedipine MR (retard)	Labetalol oral
Postpartum	Nifedipine MR, amlodipine, enalapril	Labetalol; switch methyldopa within 2 days

Hydralazine

Class	Direct arteriolar vasodilator
Mechanism	Relaxes arteriolar smooth muscle (interferes with intracellular calcium release), lowering peripheral resistance; reflex rise in heart rate and cardiac output
Indications (Obs/Gyn)	Acute severe hypertension (≥160/110) in pregnancy and postpartum
Dose & route (Uganda)	5 mg IV slowly over 2–5 min (10 mg if very high). Check BP every 5 min for 20 min; repeat 5–10 mg every 20–30 min . Max 20 mg IV per episode (WHO alternative: 12.5 mg IM every 2 h). Dilution: 20 mg ampoule (powder) reconstituted and diluted to 20 mL with water for injection or NS = 1 mg/mL , so 5 mg = 5 mL
Onset / key PK	IV onset 10–20 min , peak ~20 min, lasts 2–4 h. Hepatic acetylation (slow acetylators get higher levels)
Side effects	Reflex tachycardia , palpitations, headache (mimics imminent eclampsia), flushing, nausea, maternal hypotension → fetal distress (more often than with nifedipine or labetalol); lupus-like syndrome with long-term use
Contraindications / cautions	Maternal tachycardia >120, SLE; caution with hypovolaemia (e.g. APH) and with other antihypertensives given together (stacking)

Monitoring	BP and pulse every 5 min for 20–30 min after each dose, then every 15–30 min; FHR (CTG if available)
Pregnancy /breastfeeding	Long safety record in pregnancy; compatible with breastfeeding
Antidote/r reversal	None specific; for overshoot hypotension: left lateral, IV crystalloid 250–500 mL cautiously, stop further doses

Ward pearls

- Always **dilute and give slowly**: bolus injection of the neat ampoule causes precipitous hypotension.
- New **headache after hydralazine** may be the drug, not worsening disease; but still check reflexes, clonus and GCS.
- Don't give the next dose on a timer: **give only if BP is still $\geq 160/110$** .

Labetalol

Class	Combined α_1 and non-selective β blocker
Mechanism	α_1 blockade → vasodilatation; β blockade → prevents reflex tachycardia. Preserves uteroplacental flow
Indications (Obs/Gyn)	Acute severe hypertension (IV); maintenance (oral). First line in the UK (NICE)
Dose & route (Uganda)	IV: 20 mg over 2 min , then 40, 80, 80 mg every 10 min if needed; max 300 mg (or infusion 20–160 mg/h). Oral: 100–400 mg 8–12-hourly (max 2.4 g/day)
Onset / key PK	IV onset 5 min, peak 10–15 min; oral onset 1–2 h. Hepatic metabolism
Side effects	Bradycardia, postural hypotension, fatigue, bronchospasm, scalp tingling; neonatal bradycardia and hypoglycaemia (tell the paediatrician)
Contraindications / cautions	Asthma (bronchospasm), heart failure , heart block/bradycardia, cocaine use
Monitoring	BP and pulse every 5–10 min during IV titration; neonatal glucose after birth
Pregnancy/breast feeding	Safe in pregnancy and breastfeeding
Antidote/reversal	Atropine for bradycardia; (glucagon for severe β -blocker overdose)

Ward pearls

- **Ask about asthma before every labetalol dose**: it is the viva question.
- **Rarely available** in Ugandan public facilities; know it for exams and private hospitals.

Nifedipine

Class	Dihydropyridine calcium-channel blocker
Mechanism	Blocks L-type calcium channels in arteriolar smooth muscle (vasodilatation) and myometrium (relaxation: hence also a tocolytic)
Indications (Obs/Gyn)	Acute severe hypertension (IR); maintenance (MR/retard); postpartum hypertension; tocolysis (see below)
Dose & route (Uganda)	Acute: nifedipine IR 10 mg orally (swallowed) ; recheck after 20–30 min , repeat 10–20 mg if still $\geq 160/110$; about 30–40 mg in the first hour , then MR. Maintenance: nifedipine MR (retard) 20–40 mg 12-hourly (max 120 mg/day)

Onset / key PK	IR onset 10–20 min orally; MR smooth over 12 h. Hepatic (CYP3A4)
Side effects	Headache, flushing , feeling hot, palpitations/tachycardia, dizziness, ankle oedema , hypotension
Contraindications / cautions	Hypotension, aortic stenosis, cardiac failure; caution when combined with MgSO ₄ (see below) and when it is already being used for BP and someone proposes it for tocolysis
Monitoring	BP and pulse every 15 min for the first hour after an acute dose; FHR
Pregnancy/breastfeeding	Safe in pregnancy and breastfeeding
Antidote/reversal	Calcium gluconate IV and fluids for severe overdose/hypotension

Ward pearls

- **Never sublingual, never bite or pierce the capsule.** Sublingual/chewed nifedipine gives an **unpredictable, precipitous fall in BP** → placental hypoperfusion, **fetal distress**, maternal MI or stroke. The woman **swallows** the capsule whole.
- The "**sensation**" after a dose (hot, flushed, throbbing headache) is expected: warn her so she does not panic, and do not confuse it with worsening pre-eclampsia (still check reflexes).
- **Nifedipine + MgSO₄**: an old fear of profound hypotension and neuromuscular blockade comes from case reports; a large cohort study (Magee 2005) found no meaningful increase. The combination is **standard in Uganda**: give both, and **monitor**.
- **Most available** acute agent in Uganda (on the shelf in ~93% of HC III+ facilities) and needs **no IV access**: ideal at HC III before referral.

Methyldopa

Class	Centrally acting α₂-agonist
Mechanism	Converted to α-methylnoradrenaline, which stimulates central α ₂ receptors → reduced sympathetic outflow
Indications (Obs/Gyn)	Maintenance treatment of chronic or gestational hypertension/pre-eclampsia antenatally
Dose & route (Uganda)	250 mg orally 8-hourly , increase to 500–750 mg 8-hourly (max 3 g/day)
Onset / key PK	Slow: onset 4–6 h , full effect in 24–48 h. Useless for acute control
Side effects	Sedation , tiredness, depression , dry mouth, postural hypotension, nasal stuffiness; positive direct Coombs test (up to ~20%), rarely haemolytic anaemia ; drug-induced hepatitis
Contraindications / cautions	History of depression , active liver disease, phaeochromocytoma; caution in the postpartum period
Monitoring	BP; mood; LFTs if symptoms; a positive Coombs can complicate blood cross-matching (tell the lab)
Pregnancy/breastfeeding	Longest safety record in pregnancy (child follow-up data); passes into breast milk in small amounts but stop/switch within 2 days postpartum because of postnatal depression
Antidote/reversal	None

Ward pearls

- **Never give methyldopa for BP $\geq 160/110$.** Give nifedipine IR or hydralazine first, then start methyldopa for maintenance.
- On the postnatal ward, switch to **nifedipine MR, amlodipine or enalapril**. Examiners ask "why?": **depression**.

Enalapril (postpartum only)

Class	ACE inhibitor
Mechanism	Blocks conversion of angiotensin I to II $\rightarrow$ vasodilatation, less aldosterone
Indications (Obs/Gyn)	Postpartum hypertension (especially with proteinuria, diabetes or cardiomyopathy)
Dose & route (Uganda)	5–20 mg orally once or twice daily
Onset / key PK	Onset ~ 1 h, peak 4–6 h; renally excreted (active metabolite enalaprilat)
Side effects	Dry cough, first-dose hypotension, hyperkalaemia , renal impairment , angio-oedema
Contraindications / cautions	Contraindicated antenatally ; AKI, hyperkalaemia, bilateral renal artery stenosis
Monitoring	Creatinine and potassium within about 1 week; BP
Pregnancy/breastfeeding	Pregnancy: avoid (ACE-inhibitor fetopathy). Breastfeeding: safe (preferred ACE inhibitor, with captopril)
Antidote/reversal	None; stop and give fluids for hypotension

RED FLAG

Drugs to AVOID in pregnancy

- **ACE inhibitors and ARBs** (enalapril, captopril, lisinopril, losartan): **2nd/3rd-trimester fetopathy**: fetal renal failure, **oligohydramnios**, **skull hypoplasia**, limb contractures, pulmonary hypoplasia, FGR, stillbirth. If a woman conceives on one, **switch as soon as pregnancy is known** (early exposure is not an indication for termination).
- **Atenolol**: associated with **fetal growth restriction** (especially when started early).
- **Diuretics** (furosemide, thiazides): reduce the already contracted plasma volume in pre-eclampsia; use only for **pulmonary oedema**.
- **Ergometrine** in any hypertensive woman: severe hypertension and stroke. Use oxytocin.

Low-dose aspirin (prevention)

Class	Antiplatelet (irreversible COX-1 inhibitor)
Mechanism	At low dose, selectively suppresses platelet thromboxane A₂ while sparing prostacyclin $\rightarrow$ improves placentation/reduces placental thrombosis
Indications (Obs/Gyn)	Prevention of pre-eclampsia in women at high risk: one high-risk factor (previous pre-eclampsia, chronic hypertension, chronic kidney disease, diabetes, SLE/antiphospholipid syndrome, multiple pregnancy) or two or more moderate factors (first pregnancy, age ≥ 40 , BMI ≥ 35 , pregnancy interval >10 years, family history of pre-eclampsia)
Dose & route (Uganda)	75–150 mg orally at night (Uganda usually has 75 mg tablets)
Onset / key PK	Start from 12 weeks , ideally before 16 weeks (benefit is small if started after 20 weeks); continue to 36 weeks

Side effects	Dyspepsia; minor bleeding
Contraindications / cautions	Aspirin allergy/asthma triggered by NSAIDs, active peptic ulcer, bleeding disorders
Monitoring	Adherence (the most common failure)
Pregnancy/breastfeeding	Low dose is safe in pregnancy
Antidote/reversal	None needed at this dose

Ward pearls

- **Stop at 36 weeks** (to reduce bleeding risk around delivery and neuraxial anaesthesia).
- Every woman who has had pre-eclampsia should leave the postnatal ward knowing: "**aspirin from 12 weeks next time.**"

Calcium supplementation

Class	Mineral supplement
Mechanism	Low calcium intake raises parathyroid hormone and intracellular calcium in vascular smooth muscle → vasoconstriction; supplementation reduces this
Indications (Obs/Gyn)	Prevention of pre-eclampsia in populations with low dietary calcium (most of Uganda); WHO ANC 2016 recommendation
Dose & route (Uganda)	1.5–2 g elemental calcium daily , orally, in divided doses, from 20 weeks until delivery
Onset / key PK	Absorption reduced by iron and tea
Side effects	Constipation, bloating
Contraindications / cautions	Hypercalcaemia, renal stones
Monitoring	Adherence; tablet burden
Pregnancy/breastfeeding	Safe
Antidote/reversal	Not applicable

Ward pearl: take calcium and iron **at least 2 hours apart** (calcium blocks iron absorption); a practical schedule is iron in the morning and calcium at lunch and supper.

Tocolytics and antenatal corticosteroids

In threatened preterm labour we **cannot reliably stop labour**, and tocolysis on its own does **not** improve newborn outcomes. Its only job is to **buy 48 hours** to complete **antenatal corticosteroids (ACS)** and/or **transfer the mother in utero** to a unit that can care for the baby.

RED FLAG

Contraindications to tocolysis: chorioamnionitis/maternal infection; **antepartum haemorrhage** (abruption); **severe pre-eclampsia/eclampsia**; **fetal distress**; **intrauterine fetal death** or lethal anomaly; **GA ≥ 34 weeks**; advanced labour (e.g. cervix ≥ 4 –6 cm per protocol); maternal cardiac disease or hypotension; severe FGR. **PPROM** is a relative contraindication (brief use for steroids/transfer only if no infection).

Nifedipine (as a tocolytic)

Class	Calcium-channel blocker
Mechanism	Blocks calcium entry into myometrial cells → uterine relaxation
Indications (Obs/Gyn)	Threatened preterm labour 24–34 weeks to allow ACS or transfer (WHO 2022 : the only tocolytic recommended, conditionally)
Dose & route (Uganda)	20 mg orally (loading) , then 10–20 mg every 6–8 h for up to 48 h
Onset / key PK	Onset 20–30 min; no benefit from maintenance tocolysis beyond 48 h
Side effects	Headache, flushing, hypotension , tachycardia, dizziness
Contraindications / cautions	As above; maternal BP $<90/50$; cardiac disease; caution if she is already on nifedipine for hypertension (senior input)
Monitoring	BP and pulse before each dose (withhold if BP $<90/50$) ; contractions; FHR
Pregnancy/breastfeeding	Safe
Antidote/reversal	Fluids; calcium gluconate for severe overdose

Ward pearl: at HC II/III, give **dexamethasone 6 mg IM + nifedipine 20 mg** and refer **the mother with the baby inside her**: the uterus is the best transport incubator.

Indomethacin

Class	NSAID (COX inhibitor)
Mechanism	Inhibits prostaglandin synthesis → reduces myometrial contractility
Indications (Obs/Gyn)	Specialist use in preterm labour <32 weeks (also polyhydramnios)
Dose & route (Uganda)	50–100 mg loading (oral or rectal), then 25 mg 6-hourly, max 48 h
Onset / key PK	Crosses the placenta
Side effects	Maternal: dyspepsia, GI bleeding, platelet dysfunction. Fetal: premature constriction/closure of the ductus arteriosus (pulmonary hypertension), oligohydramnios (reduced fetal urine), neonatal NEC and IVH with prolonged use
Contraindications / cautions	≥ 32 weeks , >48 h use , oligohydramnios, peptic ulcer, asthma, renal impairment, bleeding disorders
Monitoring	Liquor volume (and ductal Doppler) if used >48 h

Pregnancy/breastfeeding	Avoid after 32 weeks
Antidote/reversal	Stop the drug

Ward pearl: the numbers are "**32 and 48**": never after **32 weeks**, never beyond **48 hours**.

Salbutamol / terbutaline (β_2 -agonists)

Class	β_2 -adrenergic agonists (betamimetics)
Mechanism	Raise cAMP in myometrium → relaxation
Indications (Obs/Gyn)	No longer recommended for tocolysis (WHO 2022). Terbutaline 250 mcg SC is still used in some units for acute tocolysis during intrauterine resuscitation (e.g. uterine hyperstimulation with fetal distress) while preparing for CS
Dose & route (Uganda)	Not recommended for preterm labour. (Acute uterine relaxation: terbutaline 250 mcg SC once, per unit protocol)
Onset / key PK	Rapid; systemic β_1/β_2 effects
Side effects	Maternal tachycardia , palpitations, tremor, hypokalaemia , hyperglycaemia , myocardial ischaemia, pulmonary oedema (especially with IV fluids, steroids, twins), maternal deaths reported; fetal tachycardia, neonatal hypoglycaemia
Contraindications / cautions	Cardiac disease , diabetes, hyperthyroidism, APH, multiple pregnancy
Monitoring	Pulse (stop if >120), fluid balance, glucose, potassium, chest
Pregnancy/breastfeeding	Inhaled salbutamol for asthma is completely safe: the problem is tocolytic IV doses
Antidote/reversal	Stop; β -blocker for severe tachyarrhythmia (specialist)

Ward pearl: why not salbutamol? **More side effects, no better outcomes:** they buy time no better than nifedipine and can cause **pulmonary oedema**.

Atosiban

Class	Oxytocin-receptor antagonist
Mechanism	Competitively blocks myometrial oxytocin (and vasopressin V1a) receptors
Indications (Obs/Gyn)	Preterm labour 24–33+6 weeks (Europe)
Dose & route (Uganda)	6.75 mg IV bolus over 1 min , then 18 mg/h for 3 h , then 6 mg/h for up to 45 h (max 48 h)
Onset / key PK	Rapid; IV only
Side effects	Few: nausea, headache, injection-site reaction
Contraindications / cautions	As for all tocolytics
Monitoring	Contractions, FHR
Pregnancy/breastfeeding	Safe profile
Antidote/reversal	None

UGANDA CONTEXT

Atosiban is expensive and not available in Ugandan public facilities. WHO 2022 does not recommend it outside research. **Nifedipine is the Ugandan tocolytic.** MgSO₄ is **not** a tocolytic: give it for neuroprotection, not to stop contractions.

Dexamethasone and betamethasone (antenatal corticosteroids)

Class	Fluorinated glucocorticoids (cross the placenta; not inactivated by placental 11 β -HSD2)
Mechanism	Induce type II pneumocytes to make surfactant , mature lung structure, and mature other organs (brain germinal matrix, gut, cardiovascular)
Indications (Obs/Gyn)	Preterm birth likely within 7 days at 24–34 weeks (WHO 2022; Uganda usually 28–34 , lower at NICU centres), with accurate GA , no maternal infection , and adequate childbirth and newborn care available. Also before planned preterm delivery (e.g. severe pre-eclampsia, APH)
Dose & route (Uganda)	Dexamethasone 6 mg IM 12-hourly $\times$ 4 doses (24 mg; first choice in Uganda) or betamethasone 12 mg IM 24-hourly $\times$ 2 doses (24 mg)
Onset / key PK	Benefit begins within hours; maximal if birth occurs 24 h to 7 days after the first dose. Start even if birth may occur before the course is complete
Side effects	Maternal hyperglycaemia (up to 5 days), WBC rise for ~72 h (confuses sepsis assessment), transient reduced fetal movements/FHR variability (1–3 days); neonatal hypoglycaemia ; possible harm if the baby is actually term or the mother is infected
Contraindications / cautions	Systemic maternal infection (including chorioamnionitis, TB), uncertain dates suggesting ≥ 34 weeks; diabetes (monitor and adjust insulin)
Monitoring	Capillary glucose 4-hourly in diabetic women for 3–5 days (increase insulin); temperature (infection); document dose times
Pregnancy/breastfeeding	Safe in pregnancy at this dose; compatible with breastfeeding
Antidote/reversal	Not applicable

Benefits (Cochrane): reduced **neonatal death** (~30%), **respiratory distress syndrome** (~35%), **intraventricular haemorrhage** (~45%) and **necrotising enterocolitis**.

Ward pearls

- **One repeat ("rescue") course only:** if birth has not occurred ≥ 7 days after the first course and she is again at high risk of birth within 7 days, still **<34 weeks** (WHO 2022). Never give multiple repeated courses (reduced fetal growth).
- **Not routine at ≥ 34 weeks** in Uganda (the ACT trial showed **harm** where dating and newborn care were poor; ACTION-I showed benefit when the WHO conditions are met).
- A raised **WBC after steroids** is expected for 72 hours: decide chorioamnionitis on **clinical signs** (fever, tachycardia, tender uterus, offensive liquor).

DRUG DOSING

Worked example 4: drawing up dexamethasone. Ugandan dexamethasone ampoules are commonly **4 mg/mL** (check: some are 5 mg/mL). For **6 mg**: $6 \div 4 = 1.5$ mL IM. Four doses 12 hours apart: e.g. 08:00, 20:00, 08:00, 20:00. Write the dose times on the referral note so the next unit does not restart the course.

EXAM FAVOURITE

Viva triad for threatened preterm labour at 30 weeks: "Steroids, magnesium, and move her":

dexamethasone 6 mg IM 12-hourly × 4; MgSO₄ 4 g IV then 1 g/h if birth expected <32 weeks within 24 h; nifedipine 20 mg to buy 48 h for **in-utero transfer**; plus erythromycin if PPROM and GBS prophylaxis in labour.

Uterotonics: oxytocin, ergometrine, prostaglandins

Uterotonics make the uterus contract. You use them for three different jobs, and the **dose depends on the job**:

1. **Making labour happen** (induction, augmentation): tiny, carefully titrated doses, because too much kills the baby (hyperstimulation) or ruptures the uterus.
2. **Stopping or preventing bleeding after birth** (AMTSL, PPH): big doses, because you want a hard, sustained contraction and the baby is already out.
3. **Emptying the uterus in early pregnancy** (miscarriage, IUFD, abortion care): prostaglandins, because the early uterus has **few oxytocin receptors**.

CLINICAL PEARL

The uterus becomes more sensitive as pregnancy advances. Oxytocin receptors in the myometrium rise from about 20 weeks and are **about 100–200-fold higher by early labour** than in the non-pregnant uterus; prostaglandin sensitivity rises too. So: oxytocin is weak in the first trimester, and a misoprostol dose that is fine at 16 weeks (400 µg) can **rupture the uterus at 36 weeks**. Doses go **down** as gestation goes **up**.

Oxytocin

Class	Posterior pituitary nonapeptide hormone (synthetic; Syntocinon/Pitocin). Differs from ADH (vasopressin) by only 2 amino acids
Mechanism	Binds the Gq-coupled oxytocin receptor on myometrium → phospholipase C → IP ₃ → intracellular Ca²⁺ release → rhythmic contractions. Also makes the decidua release prostaglandins . Receptor density rises steeply near term (weak drug before ~20 weeks). Also causes milk let-down (myoepithelial cells)
Indications (Obs/Gyn)	AMTSL (first choice); treatment of PPH; induction of labour (after ARM or a ripe cervix); augmentation of slow labour; second twin with poor contractions; after evacuation/expulsion of later pregnancies; at caesarean section after delivery of the baby
Dose & route (Uganda)	AMTSL: 10 IU IM (or IV slowly) within 1 min of birth . PPH: 10 IU IV slowly (or IM), then 20–40 IU in 1 L NS/RL (e.g. 20 IU in 1 L at 60 drops/min , maintenance 40 drops/min). Augmentation/induction (WHO MCPC): 2.5 IU in 500 mL NS/RL at 10 drops/min, ↑ by 10 drops/min every 30 min to max 60 drops/min ; if still inadequate, 5 IU in 500 mL at 30 drops/min, ↑ to 60 drops/min (see worked maths below)
Onset / PK	IV: within 1 min ; IM: 2–3 min , lasts 30–60 min. Half-life 3–5 min → a bolus wears off quickly (hence the infusion) and stopping the drip reverses hyperstimulation within minutes . Steady state takes ~30–40 min → increase no faster than every 30 min
Side effects	Tachysystole / hyperstimulation → fetal distress; uterine rupture (grand multipara, scar, obstruction); water intoxication / hyponatraemia (antidiuretic effect at high dose with large volumes); hypotension, tachycardia, flushing with rapid IV bolus (can cause myocardial ischaemia); nausea; atony PPH after prolonged augmentation (receptor desensitisation)

Contraindications / cautions	CI: obstructed labour / CPD, malpresentation (transverse lie), fetal distress , hypertonic uterus, placenta praevia, previous classical CS or myomectomy entering the cavity. Caution: grand multipara (≥ 5), previous lower-segment CS , overdistended uterus (twins, polyhydramnios), severe heart disease (avoid boluses)
Monitoring	In labour: contractions every 30 min (number in 10 min, duration), FHR every 15–30 min (every 5 min in second stage), maternal pulse, BP; record IU/L and drops/min on the partograph/LCG every 30 min; fluid balance. In PPH: tone, loss, vitals every 15 min
Pregnancy / breastfeeding	Physiological hormone; safe in breastfeeding (it <i>is</i> the let-down hormone)
Storage / antidote	Refrigerate 2–8 °C , protect from light; loses potency in heat (cold-chain failures → substandard ampoules). No antidote: stop the infusion (short half-life); tocolytic (terbutaline/salbutamol) for persisting hyperstimulation

Ward pearls

- **Oxytocin is only as safe as the person counting the contractions.** One-to-one care, a partograph/LCG, and a documented rate every 30 min. Never leave an oxytocin drip running unattended at night.
- **Never augment a woman who might be obstructed.** Poor progress in a multipara is obstruction until proven otherwise (moulding 3+, caput, Bandl's ring, haematuria). Oxytocin + obstruction = rupture.
- **Oxytocin before ergometrine or misoprostol** in PPH: first line, cheapest, fewest side effects. But after **many hours of augmentation** the receptors are desensitised, so add a drug from a **different class** early.
- **Uterine inversion: stop oxytocin until the uterus is replaced**, then give it.

Oxytocin toxicity: what to recognise and what to do

Toxicity	Why it happens	Recognise	Management
Tachysystole / hyperstimulation	Dose too high, increased too fast, very sensitive uterus (multipara, after misoprostol)	>5 contractions in 10 min , contractions lasting >60 s , poor relaxation between, then FHR <110 , late or prolonged decelerations , meconium	1. Stop the oxytocin immediately (disconnect, don't just slow). 2. Left lateral position (relieves aortocaval compression). 3. IV fluid (crystalloid, not dextrose) and oxygen if hypoxic/per local protocol. 4. Call a senior; listen to FHR continuously. 5. If contractions/fetal distress persist: tocolysis: terbutaline 250 µg (0.25 mg) SC (or IV slowly over 5 min), or salbutamol 10 mg in 1 L IV fluid at 10 drops/min (WHO MCPC). 6. If FHR does not recover → deliver (CS or assisted delivery if fully dilated). 7. Restart only after senior review, at half the previous rate
Water intoxication / hyponatraemia	Oxytocin acts on V2 (ADH) receptors at high doses (≥ 20 –40 mIU/min) for many hours, especially in large volumes of 5% dextrose (free water)	Headache, nausea, vomiting, confusion, drowsiness, convulsions (mimics eclampsia!), low urine output, oedema; Na⁺ <130 mmol/L ; neonatal hyponatraemia too	Stop oxytocin, restrict fluids , check U&E , stop dextrose; treat seizures; senior/ICU for severe hyponatraemia. Prevention: use NS or RL, not 5% dextrose, and limit oxytocin-containing fluid to about 3 L

Toxicity	Why it happens	Recognise	Management
Hypotension	Rapid IV bolus causes vasodilatation, reflex tachycardia (and ECG ischaemia)	Sudden fall in BP, flushing, chest tightness, especially at caesarean section under spinal or in a hypovolaemic or cardiac patient	Never push IV oxytocin fast: give IV slowly over 1–2 min (or IM). In cardiac disease, use a slow infusion rather than a bolus
Uterine rupture	Oxytocin against an obstruction or scar; grand multipara	Sudden pain between contractions, contractions stop, scar tenderness, FHR abnormality/loss, fetal parts easily felt, shock, vaginal bleeding	Stop oxytocin, resuscitate, laparotomy (see obstructed labour)

RED FLAG

A labouring woman on oxytocin who fits is not automatically eclamptic. Check BP and urine protein, but also think **water intoxication** (low Na⁺, many litres of dextrose) and treat accordingly.

Worked oxytocin drip-rate calculation (20 drops/mL giving set)

Three formulas you must know:

$$\begin{aligned}
 \text{mL/min} &= \text{drops/min} \div 20 && (\text{standard adult set} = 20 \text{ drops/mL}) \\
 \text{drops/min} &= \text{volume (mL)} \times 20 \div \text{time (min)} \\
 \text{mIU/min} &= \text{concentration (mIU/mL)} \times \text{mL/min} \\
 \text{concentration (mIU/mL)} &= \text{IU in bag} \times 1,000 \div \text{bag volume (mL)}
 \end{aligned}$$

Example 1: augmentation starting dose. 2.5 IU in 500 mL = 2,500 mIU ÷ 500 mL = **5 mIU/mL**. At **10 drops/min** → 10 ÷ 20 = 0.5 mL/min → 5 × 0.5 = **2.5 mIU/min**. Each step up of 10 drops/min adds 2.5 mIU/min, so the maximum **60 drops/min = 3 mL/min = 15 mIU/min**.

Example 2: second concentration. 5 IU in 500 mL = **10 mIU/mL**. Why restart at **30 drops/min**? 30 ÷ 20 = 1.5 mL/min × 10 = **15 mIU/min**: exactly the dose she was already on at 60 drops/min of the weaker bag. At **60 drops/min = 3 mL/min × 10 = 30 mIU/min**.

Example 3: PPH infusion. 20 IU in 1 L = **20 mIU/mL**. At **60 drops/min = 3 mL/min × 20 = 60 mIU/min** (a big dose, deliberately), and 3 mL/min = 180 mL/h, so the litre lasts **about 5.5 hours**.

Example 4: a nurse asks "the doctor wrote 500 mL over 4 hours; what rate?" 500 × 20 ÷ 240 min = **42 drops/min**.

If an infusion pump is available: mL/h = mIU/min × 60 ÷ mIU/mL. For 2.5 mIU/min using 10 mIU/mL: 2.5 × 60 ÷ 10 = **15 mL/h**.

EXAM FAVOURITE

Viva favourite: "What dose in mIU/min is 2.5 IU in 500 mL at 10 drops/min?" **2.5 mIU/min**. Then: "and at 60 drops/min?" **15 mIU/min**. Check the drop factor on the giving set: a **paediatric burette is 60 drops/mL**, which changes every answer by a factor of 3.

UGANDA CONTEXT

Many Ugandan wards start with **5 IU in 500 mL** rather than 2.5 IU. Know your unit's protocol, **write concentration and drops/min on the partograph**, and remember that the **WHO MCPC 10 IU in 500 mL step is for primigravidae only**. Multiparas and women with a previous CS who fail at the 5 IU concentration go to **caesarean section**, not to a stronger drip.

Ergometrine and Syntometrine

Class	Ergot alkaloid (ergonovine); Syntometrine = oxytocin 5 IU + ergometrine 0.5 mg per 1 mL ampoule
Mechanism	Partial agonist at α-adrenergic, serotonin (5-HT₂) and dopamine receptors → sustained, tetanic contraction of the whole uterus including the lower segment (unlike oxytocin's rhythmic contractions); also vasoconstriction (the cause of its dangers)
Indications (Obs/Gyn)	Second-line treatment of PPH from atony; AMTSL where oxytocin unavailable (only if normotensive); heavy bleeding from incomplete abortion while awaiting evacuation
Dose & route (Uganda)	0.2 mg IM (or slow IV over at least 1 min); repeat 0.2 mg IM after 15 min , then 0.2 mg every 4 h if needed. Max 5 doses (1 mg) in 24 h . Syntometrine: 1 mL IM (not IV)
Onset / PK	IM 2–7 min , IV ~1 min ; action lasts ~3 h (much longer than oxytocin). Syntometrine combines oxytocin's fast onset with ergometrine's long action
Side effects	Nausea and vomiting (common), hypertension (can precipitate stroke, eclampsia , pulmonary oedema), headache, chest pain/coronary spasm, peripheral vasoconstriction; ergotism with interacting drugs
Contraindications / cautions	Hypertension, pre-eclampsia/eclampsia, cardiac disease (including rheumatic valve disease), retained placenta, undelivered second twin (never before delivery of the last baby), peripheral vascular disease/Raynaud's, severe liver or renal disease, sepsis. Interaction: strong CYP3A4 inhibitors, especially ritonavir/cobicistat-boosted protease inhibitors , clarithromycin/erythromycin, azole antifungals → ergotism (severe vasospasm)
Monitoring	BP before every dose and after; uterine tone; vomiting (aspiration risk if drowsy)
Pregnancy / breastfeeding	Never before birth of the (last) baby . Single doses fine in breastfeeding; repeated doses may reduce prolactin and milk supply
Storage / antidote	Refrigerate 2–8 °C, protect from light (degrades quickly in heat). No antidote; treat hypertension (e.g. hydralazine/nifedipine) and vasospasm

Ward pearls

- **Check the BP before you reach for ergometrine.** "Ergometrine: not for the hypertensive heart."
- **Retained placenta → never ergometrine:** the tetanic contraction closes the cervix and **traps the placenta**. Give oxytocin, empty the bladder, CCT, then manual removal.
- **HIV on a boosted protease inhibitor** (second-line ART such as ATV/r, LPV/r, DRV/r): avoid ergometrine; use oxytocin/misoprostol. On TLD (dolutegravir) there is no such interaction.
- Syntometrine gives slightly less PPH than oxytocin alone but **much more vomiting and hypertension**, which is why oxytocin alone is preferred for AMTSL.

Carboprost (15-methyl PGF2α)

Class	Prostaglandin F2α analogue (Hemabate)
Mechanism	Stimulates FP prostaglandin receptors → strong myometrial contraction; also smooth muscle of bronchi and gut (hence the side effects)
Indications (Obs/Gyn)	PPH from atony refractory to oxytocin (± ergometrine/misoprostol)
Dose & route (Uganda)	250 µg (0.25 mg) deep IM , repeat every 15 min ; max 8 doses (2 mg) . Intramyometrial injection only by a specialist
Onset / PK	Onset within ~15 min, peak effect 15–60 min
Side effects	Diarrhoea, vomiting , fever, flushing, bronchospasm , hypertension, desaturation (ventilation–perfusion mismatch)
Contraindications / cautions	Asthma (absolute: bronchospasm), active cardiac, pulmonary, renal or hepatic disease; caution in hypertension, glaucoma
Monitoring	SpO ₂ , RR and wheeze; BP; uterine tone
Pregnancy / breastfeeding	Used only after birth; breastfeeding acceptable
Storage / antidote	Refrigerate 2–8 °C . Bronchospasm: salbutamol nebuliser, oxygen

Ward pearls

- **"Carboprost: not for the wheezy chest."**
- **Expensive and rarely stocked in Ugandan public facilities:** in practice, second-line after oxytocin is **misoprostol 800 µg SL** (± ergometrine if normotensive), then tamponade, NASG and surgery. Don't waste time looking for carboprost.

Misoprostol

Class	Synthetic prostaglandin E1 (PGE1) analogue (originally for NSAID ulcers). Tablets of 200 µg ; 25 µg vaginal tablets in some places
Mechanism	EP-receptor agonist → myometrial contraction and cervical ripening (collagen breakdown, softening)
Indications (Obs/Gyn)	PPH prevention (where no oxytocin) and treatment; induction of labour (live fetus, unscarred uterus); IUD; missed and incomplete miscarriage; abortion care where legal; cervical priming before MVA/surgery
Dose & route (Uganda)	Depends on indication and gestation: see the table below. Never write "miso 1 tab": write µg, route, interval
Onset / PK	Sublingual: fastest and highest peak (~30 min), good for PPH. Oral: peak ~30 min, shorter action (hence 2-hourly for IOL). Vaginal: slower peak (1–2 h) but most sustained (hence 6-hourly). Buccal: between
Side effects	Shivering and fever (up to 40 °C, commonest with 800 µg SL; not sepsis), diarrhoea , nausea, vomiting, cramps; tachysystole and uterine rupture (dose-related, worse with a scar); teratogenic if pregnancy continues (Möbius syndrome, limb defects)
Contraindications / cautions	Previous CS / uterine scar: avoid for induction in the third trimester; lower doses and caution in the second trimester. Allergy to prostaglandins. Suspected ectopic (misoprostol does not treat it). Caution in severe asthma is not needed (PGE1 is not a bronchoconstrictor)
Monitoring	Contractions and FHR (IOL); temperature (fever expected, but look for sepsis if >24 h); bleeding and expulsion; wait ≥4 h after the last dose before starting oxytocin

Pregnancy / breastfeeding	Abortifacient: never in a wanted ongoing pregnancy. Compatible with breastfeeding
Storage / antidote	Heat-stable at room temperature (in blister pack, keep dry): ideal for Uganda. Hyperstimulation → remove vaginal tablets if possible, left lateral, terbutaline 250 µg SC

Misoprostol regimens by indication (agree with the Quick Reference)

Indication	Gestation	Regimen
PPH prevention (no oxytocin, e.g. home birth)	After birth	600 µg orally once
PPH treatment	After birth	800 µg sublingually once
Induction of labour, live fetus	Term (≥37 weeks), no uterine scar	25 µg vaginally every 6 h or 25 µg orally every 2 h (oral solution: 1 × 200 µg tablet in 200 mL water, give 25 mL ; discard after 24 h)
Incomplete miscarriage/abortion	<13 weeks (uterus ≤12–13 weeks)	600 µg orally once or 400 µg sublingually once
Missed miscarriage	<14 weeks	Mifepristone 200 mg orally, then misoprostol 800 µg vaginally/SL/buccally 1–2 days later ; misoprostol alone: 800 µg vaginally (or 600 µg SL), repeat after 3 h if needed (max 2 doses)
Cervical priming before MVA / surgical evacuation	<13 weeks	400 µg sublingually 1 h or vaginally 3 h before
Incomplete abortion / termination (legal indication)	13–24 weeks	400 µg vaginally, SL or buccally every 3 h until expulsion
IUFD	14–28 weeks	400 µg vaginally/SL every 4–6 h (WHO 2022, ideally after mifepristone 200 mg) or 200 µg every 4–6 h (FIGO); 100 µg every 4 h at 27–28 weeks
IUFD	>28 weeks	25 µg vaginally every 6 h or 25 µg orally every 2 h ; if no response after 2 doses, 50 µg every 6 h; never more than 50 µg per dose

DRUG DOSING

The dose ladder to memorise: first trimester **400–800 µg** → mid-trimester **200–400 µg** → 27–28 weeks **100 µg** → third trimester/term **25 µg** (max 50 µg). **After birth it jumps back up** to 600 µg (prevention) and 800 µg (treatment), because now you *want* a maximal contraction. **Never 200 µg to a woman in the third trimester with a baby still inside.**

Ward pearls

- **Post-misoprostol fever** is expected within hours. A fever **persisting beyond 24 h**, with foul discharge or tenderness, is **sepsis** (septic abortion, endometritis) until proven otherwise.
- Misoprostol is the **Ugandan workhorse**: cheap, heat-stable, on the shelf at HC III. That is exactly why dosing errors are common: check the gestation every time.
- In PPH, misoprostol is **slower** than IV oxytocin; give it **with** the oxytocin infusion, not instead of it (unless no oxytocin). If she already had misoprostol 600 µg for prophylaxis, be wary of high-dose repeats (hyperpyrexia) and prefer another class.

RED FLAG

Previous caesarean at term: no misoprostol for induction (WHO; rupture risk several-fold higher). Use a **Foley catheter** then **oxytocin** with senior supervision, in a hospital with theatre, or plan a repeat CS.

Carbetocin (heat-stable)

Class	Long-acting synthetic oxytocin analogue
Mechanism	Oxytocin-receptor agonist; half-life ~40 min (about 10× longer than oxytocin) → sustained contraction from one dose
Indications (Obs/Gyn)	Prevention of PPH only (AMTSL, including at CS)
Dose & route (Uganda)	100 µg IM or slow IV, single dose , after delivery of the baby
Onset / PK	Onset ~2 min; effect lasts ~1–2 h
Side effects	As oxytocin: nausea, flushing, headache, hypotension with fast IV, tachycardia
Contraindications / cautions	Not for treatment of PPH, not for induction or augmentation, never before birth of the baby. Caution in cardiac disease, epilepsy, hyponatraemia risk
Monitoring	Uterine tone every 15 min for 2 h; BP
Pregnancy / breastfeeding	Postpartum use only; compatible with breastfeeding
Storage / antidote	Heat-stable formulation: no cold chain needed (the whole point). No antidote

Ward pearls

- **CHAMPION trial** (WHO, 2018; ~30,000 women in 10 countries including Uganda): heat-stable carbetocin was **non-inferior** to oxytocin for preventing PPH ≥500 mL.
- WHO recommends it where **oxytocin quality cannot be guaranteed** (cold chain). Available through some programmes in Uganda; oxytocin 10 IU IM remains the standard.

Dinoprostone (PGE2)

Class	Natural prostaglandin E2
Mechanism	Cervical ripening (collagen remodelling) and myometrial contraction
Indications (Obs/Gyn)	Cervical ripening / induction of labour with an unfavourable cervix
Dose & route	Vaginal gel 1–2 mg , repeat after 6 h (max per product), or controlled-release pessary 10 mg left for up to 24 h (removable)
Onset / PK	Onset within 1 h; pessary releases slowly and can be pulled out if hyperstimulation occurs (advantage over misoprostol)
Side effects	Tachysystole, fetal distress, nausea, vomiting, diarrhoea, fever
Contraindications / cautions	Previous CS (UK allows with caution; generally avoided in Uganda), grand multiparity, placenta praevia, malpresentation, fetal distress, active asthma (caution)
Monitoring	CTG/FHR and contractions; wait ~6 h after gel (30 min after pessary removal) before oxytocin
Pregnancy / breastfeeding	Used only to induce labour

Storage / antidote	Refrigerate. Remove pessary; terbutaline 250 µg SC for hyperstimulation
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UGANDA CONTEXT

Dinoprostone is expensive, needs a fridge, and is rarely available in Ugandan public hospitals (mainly private units). In Uganda, cervical ripening means **misoprostol** or a **Foley catheter**. Know dinoprostone for exams comparing UK (NICE) and WHO practice.

Comparison of uterotonics

Drug	Class	Prevention (AMTSL)	PPH treatment	IOL / ripening	Early pregnancy	Key contraindication	Storage
Oxytocin	Peptide hormone	10 IU IM	10 IU IV slow + 20–40 IU/L infusion	Yes (after ripening/ARM), titrated	Weak (<20 weeks)	Obstruction, fetal distress, classical scar	2–8 °C
Carbetocin	Long-acting oxytocin analogue	100 µg IM/IV	No	No	No	Before birth of baby	Heat-stable
Ergometrine	Ergot alkaloid	0.2 mg IM (if normotensive)	0.2 mg IM, max 1 mg/24 h	Never	Bleeding incomplete abortion	HTN, PE, heart disease, retained placenta, 2nd twin	2–8 °C
Syntometrine	Oxytocin 5 IU + ergometrine 0.5 mg	1 mL IM	Occasionally	Never	No	As ergometrine	2–8 °C
Misoprostol	PGE1	600 µg PO	800 µg SL	25 µg (unscarred only)	Yes: drug of choice	Scarred uterus at term (IOL)	Heat-stable
Carboprost	15-methyl PGF2α	No	250 µg IM q15 min, max 2 mg	No	No	Asthma	2–8 °C
Dinoprostone	PGE2	No	No	Yes (gel/pessary)	No	Previous CS (relative)	Fridge

EXAM FAVOURITE

Classic traps: carbetocin for treatment (no, prevention only); ergometrine for retained placenta (no, traps it); oxytocin for a 9-week incomplete miscarriage (no, few receptors: misoprostol); carboprost in an asthmatic (no); ergometrine in pre-eclampsia (no: stroke/eclampsia); misoprostol IOL with a previous CS at term (no: Foley + oxytocin or CS).

Tranexamic acid

Tranexamic acid (TXA)

Class	Antifibrinolytic: synthetic lysine analogue
Mechanism	Blocks the lysine-binding sites on plasminogen → plasminogen cannot bind fibrin or become plasmin → the clot is not broken down . It does not make new clot; it protects the clot you have. Fibrinolysis is activated within the first hours after birth (placental separation releases tPA), which is why it works early

Indications (Obs/Gyn)	Treatment of PPH of any cause (tone, trauma, tissue, thrombin), with uterotonics; heavy menstrual bleeding; bleeding from cervical cancer; some units at CS for high-risk women (not routine)
Dose & route (Uganda)	PPH: 1 g (10 mL of 100 mg/mL) IV over 10 min (1 mL/min) within 3 h of birth ; second 1 g if bleeding continues after 30 min or restarts within 24 h . Heavy menstrual bleeding: 1 g orally 8-hourly (1–1.5 g 8–12-hourly) for up to 4–5 days from the start of each period. Cervical cancer bleeding: 1 g IV 8-hourly , then 1 g orally 8-hourly
Onset / PK	IV onset minutes; half-life ~2 h; renally excreted (reduce dose in renal impairment)
Side effects	Nausea, vomiting, diarrhoea; hypotension/dizziness if injected too fast ; rarely visual disturbance, seizures (very high doses)
Contraindications / cautions	Active thromboembolic disease (DVT, PE, stroke); caution with a history of VTE or thrombophilia; severe renal impairment (dose reduction); history of seizures; established DIC with predominant clotting
Monitoring	Time of birth vs time of dose (document both); continued bleeding; signs of VTE
Pregnancy / breastfeeding	Safe in pregnancy and breastfeeding
Storage / antidote	Room temperature. No antidote

Ward pearls

- **WOMAN trial (Lancet 2017, 20,060 women, 21 countries incl. Uganda): TXA reduced death from bleeding** (1.5% vs 1.9%), with the **biggest benefit when given within 3 h** (about a one-third reduction); **each 15-minute delay reduces the benefit by ~10%**; **no benefit after 3 h**; **no increase in thromboembolism**.
- Mnemonic "**1-10-3-30**": **1 g**, over **10 min**, within **3 h**, repeat after **30 min**.
- Give TXA **as soon as PPH is diagnosed, together with oxytocin** (E-MOTIVE bundle): don't wait for the uterotonics to fail.
- It is **not a uterotonic**: TXA plus a floppy uterus is still a floppy uterus. Massage and oxytocin first-line, all at once.

DRUG DOSING

For heavy menstrual bleeding, TXA is **non-hormonal** and taken **only on heavy days**, so it suits women trying to conceive. It reduces loss by about **40–50%**. Mefenamic acid 500 mg 8-hourly (an NSAID) can be combined with it. Exclude pregnancy, fibroids, polyps and cancer first.

Drugs for miscarriage, abortion care, IUFD and ectopic pregnancy

UGANDA CONTEXT

Post-abortion care (PAC) is always legal and always required in Uganda, whatever the cause of the abortion. Induced abortion is legally restricted, so use abortion-care regimens only for **legal indications** (e.g. to save the woman's life) under senior/unit policy. **Mifepristone is not consistently stocked** in public facilities; misoprostol-alone regimens are therefore common.

Mifepristone

Class	Synthetic anti-progestogen (19-norsteroid; RU-486)
Mechanism	Progesterone receptor antagonist → decidual necrosis and detachment, cervical softening , and a myometrium far more sensitive to prostaglandins (a "primer"). High doses also block glucocorticoid receptors
Indications (Obs/Gyn)	Missed miscarriage <14 weeks; IUFD (before misoprostol); medical abortion where legal
Dose & route (Uganda)	200 mg orally once , then misoprostol 24–48 h (1–2 days) later at the dose for the gestation
Onset / PK	Long half-life (~18–30 h); effect builds over 24–48 h
Side effects	Bleeding, cramps, nausea, headache; bleeding for up to 2 weeks after expulsion
Contraindications / cautions	Suspected ectopic (it does not treat ectopic), chronic adrenal failure / long-term steroids , severe uncontrolled asthma, inherited porphyria, bleeding disorder or anticoagulation, IUD in situ (remove first)
Monitoring	Bleeding, pain, completion (history ± scan or hCG after 2 weeks)
Pregnancy / breastfeeding	Teratogenic risk unclear, but counsel that a continuing pregnancy after failed treatment needs evacuation; breastfeeding acceptable
Storage / antidote	Room temperature. No antidote

Ward pearls

- **Mifepristone + misoprostol beats misoprostol alone:** higher complete-evacuation rates and a **shorter induction-to-delivery time** in IUFD.
- Where there is no mifepristone, **don't wait:** start the misoprostol-alone regimen.

Methotrexate (for ectopic pregnancy)

Class	Antimetabolite / folic acid antagonist (cytotoxic)
Mechanism	Inhibits dihydrofolate reductase → no tetrahydrofolate → no purine/thymidine synthesis → kills rapidly dividing trophoblast
Indications (Obs/Gyn)	Unruptured tubal ectopic meeting criteria; pregnancy of unknown location with rising hCG (referral centres); persistent trophoblast after salpingotomy; low-risk gestational trophoblastic neoplasia (different regimen)
Criteria (NICE, all needed)	Haemodynamically stable, no significant pain ; unruptured, adnexal mass <35 mm ; no fetal heartbeat ; hCG <1,500 IU/L (offer choice of methotrexate or surgery for 1,500–5,000 IU/L); no intrauterine pregnancy ; able to return for follow-up
Dose & route (Uganda)	50 mg/m² IM single dose (BSA = $\sqrt{(\text{height cm} \times \text{weight kg} \div 3,600)}$; e.g. 1.6 m ² → 80 mg IM). Baseline FBC, LFTs, creatinine, hCG, blood group
Monitoring	hCG on day 4 and day 7: a fall of ≥15% from day 4 to day 7 = success, then weekly until negative (4–6 weeks). hCG often rises between day 1 and day 4: expected. <15% fall → second dose 50 mg/m² (about 1 in 7 need it) or surgery
Side effects	"Separation pain" days 2–7; nausea, vomiting, stomatitis , conjunctivitis, photosensitivity , transient LFT rise; rarely marrow suppression , pneumonitis, alopecia; tubal rupture can still occur
Contraindications / cautions	Haemodynamic instability/rupture ; intrauterine or heterotopic pregnancy ; breastfeeding ; renal or hepatic disease (raised creatinine/LFTs); blood dyscrasias (anaemia, low WBC/platelets); immunodeficiency ; active peptic ulcer; active lung disease; cannot follow up

Patient advice	Avoid folic acid supplements (antidote effect: treatment fails), NSAIDs (reduce methotrexate clearance → toxicity; use paracetamol), alcohol (hepatotoxicity), sun exposure (photosensitivity) and sex until hCG is negative. Effective contraception for ~3 months (teratogenic)
Pregnancy / breastfeeding	Teratogenic ; contraindicated in breastfeeding
Storage / antidote	Cytotoxic handling and disposal. Antidote for toxicity: folinic acid (calcium leucovorin)

Ward pearls

- Memory aid (from the ectopic topic): **"No PAIN, No BEAT, Under 35 mm, Under 1,500, Can come back."**
Avoid-list: **SAFE-N** (Sun, Alcohol, Folate, Expectancy: no pregnancy for 3 months, NSAIDs).
- **Severe pain, dizziness or shoulder-tip pain after methotrexate = come back now:** rupture is still possible.
- **In most Ugandan hospitals the safer option is surgery** (salpingectomy): methotrexate and serial quantitative hCG are mainly at referral and private centres, and many women cannot return within hours.

NOTE

Anti-D: give to **Rh-negative, non-sensitised** women after ectopic pregnancy (with surgery), surgical evacuation, miscarriage ≥ 12 weeks, molar pregnancy, IUFD and delivery of an Rh-positive baby, **within 72 h**. Doses and Uganda vial sizes are in the **Quick Reference (section 7)**.

Cervical ripening and induction of labour: the drug summary

Step 1: score the cervix with the **Bishop score** (Quick Reference, section 9). ≥ 8 **favourable** → ARM + oxytocin. ≤ 6 **unfavourable** → ripen first.

Option	Type	How	Pros	Cons / cautions	Previous CS?
Membrane sweep	Mechanical	Finger sweeps membranes off the lower segment at ≥ 39 –41 weeks	Cheap, reduces need for formal IOL	Discomfort, spotting	Yes
Foley catheter	Mechanical	16–18 F through the cervix past the internal os, balloon 30–60 mL saline, taped to the thigh with gentle traction; remove at 12–24 h or when it falls out	Cheap, available, less tachysystole, no drug , safe with a live fetus	Slightly slower; infection risk if left long; not with low-lying placenta or ruptured membranes (relative)	Yes: preferred ripening method
Misoprostol	Pharmacological (PGE1)	25 µg vaginal 6-hourly or 25 µg oral 2-hourly	Cheap, heat-stable, effective, fewer CS for failed IOL	Tachysystole; can't be switched off quickly	No (third trimester)
Dinoprostone	Pharmacological (PGE2)	Gel 1–2 mg or 10 mg pessary	Pessary removable	Expensive, fridge, rarely in Uganda	UK: with caution; Uganda: avoid

Option	Type	How	Pros	Cons / cautions	Previous CS?
ARM + oxytocin	Mechanical + pharmacological	When Bishop ≥ 8 (or ≥ 6 after ripening)	Controllable, short half-life	Needs one-to-one monitoring	With caution (senior, theatre ready); never with a classical scar

Rules of thumb

1. **Start oxytocin at least 4 h after the last misoprostol dose** (combined effect $\rightarrow$ hyperstimulation).
2. **Previous CS: avoid misoprostol and (in Uganda) prostaglandins for IOL at term.** Use **Foley + oxytocin** at a hospital with theatre, with low-dose, slowly titrated oxytocin, and **stop at the first sign of scar pain, bleeding, maternal tachycardia or FHR abnormality.** **Classical scar or ≥ 2 CS $\rightarrow$ elective CS, no labour.**
3. **Grand multipara:** lower doses, slower titration, and no oxytocin if there is any hint of obstruction.
4. Contraindications to any IOL: **CPD/obstruction, transverse lie, placenta praevia, vasa praevia, cord presentation, previous classical CS, active genital herpes.**

CLINICAL PEARL

Why is misoprostol dangerous with a scar but oxytocin tolerated? **Oxytocin can be switched off** (half-life 3–5 min), whereas misoprostol, once absorbed, keeps working for hours and prostaglandins also soften the scar tissue. **"If you can't switch it off, don't put it into a scarred uterus."**

Antibiotics in pregnancy and gynaecology

Infection kills Ugandan mothers in three ways: **puerperal sepsis, septic (usually unsafe) abortion** and **pyelonephritis or chorioamnionitis that is treated too late**. Two questions decide almost every prescription on the ward: **"Is this drug safe for the fetus (and the breastfed baby)?"** and **"Does it cover the likely organisms?"** Pelvic infections are **polymicrobial**: Gram-positives (streptococci, including GBS; staphylococci), **Gram-negatives** (*E. coli*) and **anaerobes** (*Bacteroides*, *Clostridium*). That is why the classic regimen has **three drugs**, one for each group.

Safe versus avoid in pregnancy

Generally safe (use freely when indicated)	Why it is safe / the caveat
Penicillins (benzylpenicillin, ampicillin, amoxicillin, cloxacillin/flucloxacillin, benzathine penicillin)	Decades of use; no teratogenicity. Only issue is allergy
Cephalosporins (cefazolin, cefalexin, ceftriaxone, cefixime)	Safe in all trimesters. Avoid only after penicillin anaphylaxis
Macrolides : erythromycin, azithromycin	Safe. Erythromycin is the PPRM antibiotic
Metronidazole	Large meta-analyses show no teratogenic effect, including in the 1st trimester. Warn about alcohol (disulfiram-like reaction)
Nitrofurantoin	Good for cystitis/ASB. Avoid at term (from ~36–38 weeks), in labour and in G6PD deficiency (neonatal haemolysis); useless for pyelonephritis
Clindamycin	Safe. Used for GBS prophylaxis in penicillin allergy, BV and with quinine. Watch for <i>C. difficile</i> diarrhoea

Avoid or use with caution	Fetal/neonatal risk	Exam hook
Tetracyclines (doxycycline, tetracycline)	Yellow-brown teeth staining , impaired bone growth (after ~15 weeks); maternal liver toxicity	"Tetracycline = Teeth"
Fluoroquinolones (ciprofloxacin, levofloxacin)	Cartilage/arthropathy in animal studies	Avoid unless no alternative (e.g. MDR infection)
Aminoglycosides (gentamicin, streptomycin)	Fetal 8th-nerve damage (deafness) , clearly shown for streptomycin ; nephrotoxicity. Gentamicin: accepted as a short course in serious sepsis	Benefit beats risk in sepsis; keep courses short and monitored
Trimethoprim	Folate antagonist : neural tube and cardiac defects in the 1st trimester	Avoid in the 1st trimester
Sulfonamides / co-trimoxazole (as treatment)	Near term : displace bilirubin from albumin → kernicterus ; haemolysis in G6PD deficiency	HIV-positive women continue co-trimoxazole prophylaxis
Chloramphenicol	Grey baby syndrome near term (the neonatal liver cannot glucuronidate it → vomiting, grey skin, circulatory collapse)	Avoid in late pregnancy and breastfeeding
Co-amoxiclav in PPROM	Necrotising enterocolitis (NEC) in the preterm baby (ORACLE I trial)	Fine for other indications (e.g. UTI)

EXAM FAVOURITE

The favourite trio: **tetracyclines → teeth**, **chloramphenicol → grey baby**, **aminoglycosides (streptomycin) → deafness**. Add **sulfonamides near term → kernicterus** and **co-amoxiclav in PPROM → NEC**, and you have the whole viva.

Regimens by indication

Doses agree with the **Quick Reference** (section 6). Where Uganda differs, the box says so.

Indication	First-line regimen	Duration / key point
CS prophylaxis	Cefazolin 2 g IV (3 g if >120 kg) or ampicillin 2 g IV , 15–60 min before skin incision . Uganda often uses ceftriaxone 1–2 g IV ± metronidazole 500 mg IV	Single dose . Repeat if surgery >4 h or blood loss >1.5 L. Before incision, not after cord clamping
Manual removal of placenta / 3rd–4th degree tear	Ampicillin 2 g IV (or cefazolin) ± metronidazole 500 mg IV	Single dose
PPROM (no chorioamnionitis)	Erythromycin 250 mg orally 6-hourly	10 days or until labour. Never co-amoxiclav
Intrapartum GBS prophylaxis (preterm labour, ROM >18 h, intrapartum fever, previous GBS baby, GBS bacteriuria)	Benzylpenicillin 3 g (5 MU) IV , then 1.5–1.8 g (2.5–3 MU) IV 4-hourly ; or ampicillin 2 g IV then 1 g 4-hourly ; allergy: clindamycin 900 mg IV 8-hourly	Until delivery. Uganda uses a risk-based (not culture-based) approach
Chorioamnionitis	Ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV daily (+ metronidazole 500 mg IV 8-hourly if CS)	Deliver . Stop after vaginal birth if well; after CS continue until 48 h afebrile
Puerperal sepsis / maternal sepsis / septic abortion	Ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV once daily + metronidazole 500 mg IV 8-hourly (alternative: ceftriaxone 1–2 g IV daily + metronidazole)	IV until afebrile 48 h , then oral (amoxicillin + metronidazole) to complete ~7 days . Source control : evacuate RPOC, drain pus

Indication	First-line regimen	Duration / key point
Asymptomatic bacteriuria / cystitis	Nitrofurantoin 100 mg 12-hourly (not at term), amoxicillin 500 mg 8-hourly (only if sensitive), or cefalexin 500 mg 8–12-hourly	5–7 days; test-of-cure culture 1–2 weeks later
Pyelonephritis	Admit; ceftriaxone 1–2 g IV daily (or ampicillin + gentamicin)	Until afebrile 24–48 h, then oral to complete 10–14 days
Syphilis: early (<2 years)	Benzathine penicillin 2.4 MU deep IM (1.2 MU into each buttock)	Single dose
Syphilis: late latent / unknown duration (most ANC cases)	Benzathine penicillin 2.4 MU IM	Weekly × 3
Gonorrhoea + chlamydia in pregnancy	Ceftriaxone 500 mg IM single dose (or cefixime 400 mg orally where the syndromic kit uses it) + azithromycin 1 g orally single dose (or erythromycin 500 mg 6-hourly × 7 days)	No doxycycline in pregnancy. Treat the partner; test of cure for chlamydia
Bacterial vaginosis	Metronidazole 400–500 mg orally 12-hourly × 7 days (or clindamycin 300 mg 12-hourly × 7 days)	Treat symptomatic women in pregnancy
Trichomoniasis	Metronidazole 500 mg 12-hourly × 7 days (WHO preferred in women) or 2 g single dose	Treat partner. If breastfeeding after 2 g: express and discard for 12–24 h
Vulvovaginal candidiasis in pregnancy	Clotrimazole 100 mg pessary nightly × 7 days (or 500 mg pessary once, often repeated) + topical cream	Avoid oral fluconazole (high-dose/prolonged use linked with craniofacial and cardiac defects; even 150 mg has a miscarriage signal)
Mastitis	Cloxacillin/flucloxacillin 500 mg orally 6-hourly (erythromycin 250–500 mg 6-hourly if allergic)	10–14 days; keep breastfeeding; abscess → drain
PID (non-pregnant) outpatient	Ceftriaxone 500 mg–1 g IM once + doxycycline 100 mg 12-hourly × 14 days + metronidazole 400–500 mg 12-hourly × 14 days	Severe/tubo-ovarian abscess: admit, IV ceftriaxone + metronidazole, oral doxycycline. Always do a pregnancy test first

UGANDA CONTEXT

Uganda's syndromic STI packages (UCG 2023) are designed for places without a lab. In a **pregnant** woman with vaginal discharge syndrome, the **doxycycline** in the standard package is **swapped for azithromycin or erythromycin**. Always **treat the partner** and offer HIV/syphilis testing.

Drug card: Gentamicin

Class	Aminoglycoside
Mechanism	Binds the 30S ribosomal subunit → misreading of mRNA; concentration-dependent killing with a post-antibiotic effect (why once-daily dosing works)
Indications (Obs/Gyn)	Gram-negative cover in puerperal sepsis, septic abortion, chorioamnionitis , pyelonephritis (second line), PID (inpatient)
Dose & route (Uganda)	5 mg/kg IV (or IM) once daily (e.g. 300 mg for 60 kg). Older regimen still seen: 80 mg IV/IM 8-hourly (~1.5 mg/kg for a 55 kg woman). Obese: use adjusted body weight. Give IV over 3–5 min
Side effects	Nephrotoxicity (acute tubular necrosis, usually reversible), ototoxicity (vestibular and cochlear; often permanent), neuromuscular blockade

Contraindications / cautions	AKI or rising creatinine, myasthenia gravis. Caution with MgSO₄ (additive neuromuscular block), furosemide, other nephrotoxins
Pregnancy/breastfeeding	Crosses the placenta: theoretical fetal ototoxicity, so use short courses (≤5–7 days) when clearly needed. Compatible with breastfeeding (hardly absorbed from the baby's gut)
Monitoring	Urine output (catheter in sepsis), creatinine at baseline and every 1–2 days , trough level before the next dose where available (<1 mg/L for once-daily), ask about tinnitus/vertigo

DRUG DOSING

Once daily beats 8-hourly: the same or better killing (high peak ÷ MIC), **less kidney toxicity** (the kidney gets a drug-free interval), and fewer doses to miss on a busy ward. **80 mg 8-hourly under-doses** most women (240 mg/day vs 300 mg for 60 kg). If creatinine rises or urine output falls: **hold gentamicin** and switch to **ceftriaxone + metronidazole**.

Drug card: Metronidazole

Class	Nitroimidazole
Mechanism	Reduced inside anaerobes to toxic radicals that break DNA
Indications (Obs/Gyn)	Anaerobic cover in sepsis/septic abortion/post-CS infection; BV; trichomoniasis; PID; <i>C. difficile</i>
Dose & route (Uganda)	500 mg IV 8-hourly (sepsis); 400–500 mg orally 8–12-hourly ; trichomonas 2 g once or 500 mg 12-hourly × 7 days
Side effects	Metallic taste, nausea, dark urine; disulfiram-like reaction with alcohol (avoid during and 48 h after); peripheral neuropathy with long courses
Contraindications / cautions	Alcohol; severe liver disease (reduce dose)
Pregnancy/breastfeeding	Safe in all trimesters. Standard doses compatible with breastfeeding; after a 2 g single dose, pause breastfeeding 12–24 h
Monitoring	Clinical response; LFTs if prolonged in liver disease

Drug card: Benzathine penicillin G

Class	Long-acting (depot) penicillin
Mechanism	Blocks bacterial cell-wall synthesis; the depot releases penicillin slowly for 2–4 weeks , long enough for slow-dividing <i>Treponema pallidum</i>
Indications (Obs/Gyn)	Syphilis (mother and partner); rheumatic fever prophylaxis
Dose & route (Uganda)	2.4 MU deep IM (1.2 MU into each buttock). Early: once. Late latent/unknown: weekly × 3 (if a dose is >2 weeks late in pregnancy, restart the course)
Side effects	Painful injection; allergy/anaphylaxis (have adrenaline ready); Jarisch–Herxheimer reaction
Contraindications / cautions	Penicillin anaphylaxis (see below). Never IV (depot can cause embolic/cardiorespiratory arrest)
Pregnancy/breastfeeding	The only drug that reliably treats the fetus. Safe
Monitoring	Observe 30 min after injection; RPR titre at 3 and 6 months (a fourfold fall = response); examine and treat the baby at birth

RED FLAG

Jarisch–Herxheimer reaction: fever, chills, myalgia, headache and hypotension **2–24 h after the first penicillin dose**, from killed spirochaetes releasing endotoxin-like products. In pregnancy it can trigger **contractions, preterm labour and fetal distress**. It is **not an allergy**: give **paracetamol**, fluids, monitor the fetus, and **do not stop** the treatment.

NOTE

Penicillin allergy in pregnancy and syphilis: erythromycin (500 mg 6-hourly × 14 days) **does not cross the placenta reliably**, so the baby is untreated and must be treated at birth. Azithromycin has widespread resistance. Options are **penicillin desensitisation** (the gold standard, at a referral hospital) or **ceftriaxone 1 g IM/IV daily × 10–14 days** per protocol. Many "allergies" are not true allergies: take a careful history.

Drug card: Erythromycin

Class	Macrolide
Mechanism	Binds the 50S ribosomal subunit; bacteriostatic
Indications (Obs/Gyn)	PPROM (prolongs latency, reduces chorioamnionitis and neonatal infection, ORACLE I); chlamydia; penicillin allergy (mastitis, syphilis, penicillin V replacement in SCD)
Dose & route (Uganda)	PPROM: 250 mg orally 6-hourly × 10 days (or until labour). Chlamydia: 500 mg 6-hourly × 7 days
Side effects	Nausea, vomiting, abdominal cramps (motilin agonist), QT prolongation, cholestasis (estolate salt)
Contraindications / cautions	QT-prolonging drugs (quinine); liver disease
Pregnancy/breastfeeding	Safe. Small infantile pyloric stenosis signal with neonatal exposure
Monitoring	Signs of chorioamnionitis (temperature, pulse, FHR, uterine tenderness, liquor smell) every 4 h

Ward pearls: antibiotics

- **Timing saves lives:** in sepsis, give the **first dose within 1 hour**, after blood cultures if they can be drawn without delay. Do not wait for the lab.
- **Antibiotics never replace source control.** Retained products, an infected haematoma, a pelvic abscess or dead uterine tissue will keep her febrile whatever you prescribe.
- **Fever not settling after 48–72 h** → look for pus (ultrasound), retained products, wound, breasts, legs (DVT), **malaria** and septic pelvic thrombophlebitis; then consider resistant organisms (ESBL: ask for culture).
- Always ask **"What kind of reaction?"** to a penicillin allergy: a rash vs anaphylaxis changes whether a cephalosporin is allowed.
- **Doxycycline is fine for PID** in a non-pregnant woman, which is why a **pregnancy test comes before every PID prescription**.

Antimalarials in pregnancy

Malaria in pregnancy causes **maternal anaemia, hypoglycaemia, severe malaria (pregnant women are more likely to develop it), miscarriage, stillbirth, preterm birth and low birth weight**. The placenta hides parasites (VAR2CSA binding to chondroitin sulphate A), so a woman can have **placental malaria with a negative blood smear**. Prevention (**IPTp-SP + ITN**) is as important as treatment.

Drug card: Sulfadoxine–pyrimethamine (SP) for IPTp

Class	Antifolate combination (sulfonamide + dihydrofolate reductase inhibitor)
Mechanism	Sulfadoxine blocks dihydropteroate synthase , pyrimethamine blocks dihydrofolate reductase : sequential block of parasite folate synthesis
Indications (Obs/Gyn)	Intermittent preventive treatment in pregnancy (IPTp) for HIV-negative women (or HIV-positive women not on co-trimoxazole)
Dose & route (Uganda)	3 tablets (each sulfadoxine 500 mg / pyrimethamine 25 mg = 1,500/75 mg) orally, directly observed at the clinic. From 13 weeks , then monthly (at least 4 weeks apart) until delivery ; aim for ≥3 doses
Side effects	Usually none; rash; rarely Stevens–Johnson syndrome , marrow suppression
Contraindications / cautions	1st trimester ; sulfa allergy ; on co-trimoxazole (overlapping toxicity, no added benefit); folic acid ≥5 mg (antagonises the effect)
Pregnancy/breastfeeding	Safe in the 2nd and 3rd trimesters, including at term (WHO 2012 update). Compatible with breastfeeding
Monitoring	Record each dose on the ANC card; ask about rash

UGANDA CONTEXT

Give IPTp with the routine **iron + folic acid 400 mcg** (the low dose is fine). The problem is **folic acid 5 mg**: women with **SCD, epilepsy or a previous NTD** need 5 mg, so they get **daily proguanil 200 mg** instead, or SP with the folic acid **withheld for 1–2 weeks after each dose** (follow your unit protocol; see the sickle cell topic). An **HIV-positive** woman on **co-trimoxazole 960 mg daily** gets **no SP**, only an ITN.

Treating uncomplicated malaria by trimester

Trimester	First line	Alternative	Notes
1st	Artemether–lumefantrine (AL): WHO 2022 now recommends AL in the 1st trimester (large cohort data showed no higher risk of miscarriage, stillbirth or anomalies than quinine)	Oral quinine 600 mg (10 mg/kg salt) 8-hourly × 7 days + clindamycin 300 mg 8-hourly × 7 days	Many Ugandan wards and older UCG editions still teach quinine + clindamycin here. Check your unit's current UCG copy
2nd and 3rd	AL	Other ACT (e.g. artesunate–amodiaquine, dihydroartemisinin–piperaquine)	ACTs are safe; quinine is second choice

AL dose (adult ≥35 kg): 4 tablets (20/120 mg each) at 0 h, 8 h, then 12-hourly for 2 more days = 6 doses, 24 tablets, taken **with fatty food or milk** (lumefantrine absorption rises many-fold with fat). Repeat the dose if she vomits within 30 min.

RED FLAG

Avoid in pregnancy: primaquine (haemolysis of a possibly G6PD-deficient fetus; also avoid when breastfeeding an infant of unknown G6PD status), **doxycycline** (teeth/bone), **tafenoquine**. Doxycycline is therefore not an option for prophylaxis in pregnant travellers.

Drug card: Artesunate (severe malaria)

Class	Artemisinin derivative
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Mechanism	Endoperoxide bridge activated by haem iron → free radicals that kill parasites fast, including young ring forms (before they sequester)
Indications (Obs/Gyn)	Severe malaria in any trimester (impaired consciousness, convulsions, prostration, severe anaemia, hypoglycaemia, respiratory distress, jaundice, shock, AKI, hyperparasitaemia)
Dose & route (Uganda)	2.4 mg/kg IV (or IM) at 0, 12 and 24 h, then once daily until she can take oral drugs (minimum 3 doses / 24 h), then a full 3-day AL course . 60 mg vial: dissolve in 1 mL 5% sodium bicarbonate, then dilute (IV: 5 mL → 10 mg/mL; IM: 2 mL → 20 mg/mL)
Side effects	Well tolerated; post-artesunate delayed haemolysis 1–3 weeks later
Contraindications / cautions	Known hypersensitivity. No trimester restriction for severe malaria
Pregnancy/breastfeeding	Drug of choice in all trimesters: the risk of severe malaria to mother and fetus far outweighs any theoretical drug risk
Monitoring	GCS, RBS 4-hourly (malaria itself causes hypoglycaemia), Hb (and at 7, 14 and 28 days for delayed haemolysis), urine output, fetal heart

If artesunate is unavailable: **IM artemether 3.2 mg/kg loading, then 1.6 mg/kg daily**, or **IV quinine 20 mg/kg (salt) loading in 5% dextrose over 4 h, then 10 mg/kg 8-hourly** (each over 4 h). **Never give quinine as an IV bolus.**

Quinine: the hypoglycaemia drug

Quinine	
Uses	1st-trimester uncomplicated malaria (with clindamycin, where AL is not used); severe malaria when artesunate is unavailable
Key toxicity	Hypoglycaemia (quinine stimulates pancreatic insulin release ; pregnancy and parasites also consume glucose), cinchonism (tinnitus, deafness, nausea, blurred vision), QT prolongation, hypotension with fast infusion
What to do	Check RBS every 4 h (hourly if unconscious), infuse in 5% or 10% dextrose , treat glucose <3 mmol/L with 25–50 mL of 50% dextrose IV (or 10% dextrose 5 mL/kg)

EXAM FAVOURITE

"A pregnant woman on IV quinine becomes confused and sweaty" = **hypoglycaemia until proven otherwise**. Check the glucose **before** you think of cerebral malaria worsening or eclampsia.

Ward pearls: malaria

- **Test every febrile pregnant or postnatal woman** (mRDT or smear), but a **negative test does not exclude placental malaria**; treat the clinical picture.
- Severe malaria at an **HC III**: give the **first dose of IM artesunate** (2.4 mg/kg) and **refer**.
- Malaria causes **uterine contractions and fetal distress**: treat the fever (paracetamol) and monitor the fetus.
- Before discharge, check the **ITN** and the **IPTp** schedule on the ANC card.

HIV and eMTCT drugs

Drug card: TLD (tenofovir DF / lamivudine / dolutegravir)

Class	Two NRTIs (TDF, 3TC) + an integrase strand transfer inhibitor (DTG)
Mechanism	NRTIs are chain terminators of reverse transcriptase ; DTG blocks integration of viral DNA into the host genome
Indications (Obs/Gyn)	Lifelong ART for every HIV-positive pregnant or breastfeeding woman (test-and-treat, same day); PEP for 28 days; HBV co-infection (TDF + 3TC treat HBV)
Dose & route (Uganda)	One tablet (TDF 300 mg / 3TC 300 mg / DTG 50 mg) orally once daily. With rifampicin : add an extra DTG 50 mg (DTG twice daily) until 2 weeks after rifampicin stops
Side effects	TDF: renal tubular toxicity (Fanconi-type), falling eGFR, reduced bone density. DTG: weight gain , insomnia, headache; small creatinine rise from blocked tubular secretion (not true kidney damage). 3TC: well tolerated
Contraindications / cautions	CrCl <50 mL/min → no TDF (substitute AZT or ABC per guidelines). Polyvalent cations (iron, calcium, antacids, zinc) chelate DTG. Carbamazepine/phenytoin lower DTG. DTG raises metformin levels
Pregnancy/breastfeeding	Preferred regimen in pregnancy and breastfeeding (WHO 2019 onwards, Uganda). Fast viral suppression, high barrier to resistance
Monitoring	Viral load: first ANC visit if already on ART, 3 months after starting , then every 3 months until breastfeeding ends , and one available at 34–36 weeks ; baseline creatinine/CrCl ; weight, BP, glucose; adherence

NOTE

The dolutegravir neural tube defect story: in **2018**, Botswana's **Tsepamo** study found NTDs in about **0.9%** of babies of women on DTG **at conception** (4 of 426), several times the rate on other ART. With many more exposures the rate fell to about **0.1%**, **statistically the same as other ART**. WHO (2019) and Uganda therefore recommend **DTG for everyone**, including women who are pregnant or may become pregnant. Give **folic acid** as for any pregnancy. **Never stop or delay TLD** because of the NTD question.

DRUG DOSING

DTG + iron/calcium: take DTG **2 h before or 6 h after** iron, calcium or antacids, **or take them together with a meal**. A woman who takes her **TLD, FeFo and calcium together on an empty stomach** absorbs less DTG and is at risk of **non-suppression**. Ask how and when she takes her tablets.

Infant prophylaxis (Uganda consolidated guidelines)

Start **within 6 hours of birth**. Risk is set by the **mother's most recent viral load**.

Infant risk	Who	Regimen	Duration
Low risk	Mother on ART with a suppressed VL (below 1,000 copies/mL) near delivery	NVP once daily	6 weeks
High risk	Mother non-suppressed or VL unknown , ART <4 weeks before delivery, diagnosed in labour or while breastfeeding	AZT twice daily + NVP once daily for 6 weeks , then NVP alone for 6 weeks	12 weeks ; continue NVP if mother still not suppressed

Age / weight	NVP (10 mg/mL), once daily	AZT (10 mg/mL), twice daily
Birth–6 weeks, 2.0–2.49 kg	10 mg (1 mL)	10 mg (1 mL)

Age / weight	NVP (10 mg/mL), once daily	AZT (10 mg/mL), twice daily
Birth–6 weeks, ≥ 2.5 kg	15 mg (1.5 mL)	15 mg (1.5 mL)
6–12 weeks	20 mg (2 mL)	Stopped

Side effects: **AZT** → anaemia and neutropenia; **NVP** → rash and (rarely) hepatotoxicity. The HIV-exposed infant starts **co-trimoxazole from 6 weeks** (120 mg daily under 6 months) until HIV infection is excluded.

Drug card: Co-trimoxazole prophylaxis (CPT)

Class	Sulfonamide + trimethoprim (sulfamethoxazole 800 mg / trimethoprim 160 mg = 960 mg)
Mechanism	Sequential block of folate synthesis (like SP)
Indications (Obs/Gyn)	All HIV-positive pregnant and breastfeeding women in Uganda: prevents PCP, toxoplasmosis, malaria, bacterial pneumonia and diarrhoea
Dose & route (Uganda)	960 mg orally once daily
Side effects	Rash, Stevens–Johnson syndrome , marrow suppression, hepatitis
Contraindications / cautions	Sulfa allergy; never with IPTp-SP ; severe renal or liver disease
Pregnancy/breastfeeding	Continue throughout , including the 1st trimester and near term: the benefit outweighs the theoretical folate and kernicterus risks. Compatible with breastfeeding
Monitoring	Rash (stop at once if mucosal involvement), FBC if pale

Ward pearls: HIV

- On the **day of CS** she still takes her **TLD**. Never omit ART in labour.
- **New positive in labour** → **TLD now**, baby **high-risk** (AZT + NVP), DNA PCR per schedule.
- HIV-positive + HBsAg-positive: **TLD already covers HBV**. Stopping TDF can cause a hepatitis flare.
- **Efavirenz** is no longer considered a reason to avoid pregnancy; old "teratogen" teaching is outdated.

Anti-D, haematinics, supplements and vaccines

Drug card: Anti-D immunoglobulin

Class	Polyclonal human IgG against the RhD antigen
Mechanism	Coats RhD-positive fetal red cells in the maternal circulation so they are cleared before her immune system makes its own anti-D (prevents sensitisation and haemolytic disease of the fetus and newborn in the next pregnancy)
Indications (Obs/Gyn)	RhD-negative, non-sensitised women after birth of an Rh-positive baby, after sensitising events, and routine antenatal prophylaxis where available
Dose & route (Uganda)	1,500 IU (300 mcg) IM (the vial usually stocked) within 72 h (some benefit up to 10 days). UK doses: 250 IU before 20 weeks; 500 IU + Kleihauer from 20 weeks and after delivery. Antenatal prophylaxis: 1,500 IU at 28 weeks or 500 IU at 28 and 34 weeks
Side effects	Injection-site pain; rarely allergy; blood product (consent)
Contraindications / cautions	Already sensitised (anti-D antibodies present: no benefit); RhD-positive woman; never give to the baby

Pregnancy/ breastfeeding	Safe
Monitoring	Group the mother (booking) and the cord blood (baby's group, DAT); Kleihauer to size a large fetomaternal haemorrhage (1,500 IU covers about 15 mL of fetal red cells , 500 IU about 4 mL)

Event	Give anti-D?
Delivery of an Rh-positive (or ungrouped) baby	Yes , within 72 h
Miscarriage ≥12 weeks ; any surgical evacuation, ectopic, molar pregnancy	Yes
Threatened miscarriage <12 weeks with light bleeding	Not needed (UK); give if heavy/repeated bleeding
APH, abdominal trauma, ECV, amniocentesis, IUFD, after 20 weeks	Yes (+ Kleihauer where available)

UGANDA CONTEXT

Anti-D is **expensive and often out of stock** in public units. Find the **Rh-negative** women at booking, counsel them early and **plan the purchase before delivery**, so the 72-hour window is not missed.

Drug card: Iron + folic acid (FeFo)

Class	Haematinic (ferrous salt) + B vitamin
Mechanism	Replaces iron for haemoglobin synthesis (pregnancy needs ~1 g extra iron); folate for DNA synthesis in dividing cells and neural tube closure
Indications (Obs/Gyn)	Every pregnant woman (prevention); iron-deficiency anaemia (treatment); postpartum
Dose & route (Uganda)	Prevention: ferrous sulphate 200 mg (60 mg elemental iron) + folic acid 400 mcg once daily throughout pregnancy and 3 months postpartum . Treatment of anaemia: 120 mg elemental iron daily (e.g. ferrous sulphate 200 mg 12-hourly)
Side effects	Nausea, epigastric pain, constipation, black stools (warn her: not melaena), diarrhoea
Contraindications / cautions	SCD or thalassaemia without proven iron deficiency (iron overload); haemochromatosis. Absorption falls with tea, calcium, antacids ; raise with vitamin C. Separate from DTG
Pregnancy/breastfeeding	Safe
Monitoring	Hb at booking, 26–28 and 34–36 weeks ; expected reticulocyte peak at 7–10 days and Hb rise of about 1 g/dL every 2 weeks on treatment. No rise → poor adherence, malaria, worms, HIV, folate/B12 deficiency, blood loss, SCD

Parenteral iron (hospital level): for women who **cannot tolerate or will not take oral iron**, malabsorption, or **iron-deficiency anaemia late in pregnancy (>34 weeks)** when time is short. Give only in the **2nd/3rd trimester**, with resuscitation equipment at hand. Options: **iron sucrose 200 mg IV** per dose (repeat to the total deficit), **ferric carboxymaltose up to 1,000 mg (20 mg/kg) IV** in one infusion, or iron dextran (total-dose infusion; highest anaphylaxis risk). Total deficit (Ganzoni) $\approx \text{weight (kg)} \times (\text{target Hb} - \text{actual Hb, g/dL}) \times 2.4 + 500 \text{ mg}$ for stores. It raises Hb **no faster** than full-dose oral iron that is actually taken, but it is **certain**. **Transfuse** for severe anaemia with decompensation or Hb <7 g/dL near term.

Folic acid: 400 mcg or 5 mg?

Folic acid 5 mg daily (from 3 months before conception to 12 weeks, or throughout if haemolysis)	Folic acid 400 mcg daily
Previous NTD baby (or partner/family history)	Every other woman planning pregnancy and throughout pregnancy (in FeFo)
Epilepsy on antiepileptics (valproate, carbamazepine, phenytoin)	
Sickle cell disease and other haemolytic anaemias (throughout pregnancy)	
Diabetes (pre-existing)	
Obesity (BMI ≥ 30)	
Malabsorption (coeliac disease), drugs that are folate antagonists (e.g. methotrexate, long-term trimethoprim)	

Other supplements, deworming and vaccines (comparison)

Intervention	Regimen	Key point
Calcium	1.5–2 g elemental daily (divided) from 20 weeks	Reduces pre-eclampsia where dietary calcium is low. Take several hours apart from iron (and DTG)
Aspirin (not a supplement, but part of the ANC pack)	75–150 mg at night from 12 to 36 weeks	High risk of pre-eclampsia; start before 20 weeks
Deworming	Mebendazole 500 mg single dose or albendazole 400 mg single dose	After the 1st trimester only (teratogenic in animals). Hookworm is a major cause of iron-deficiency anaemia
Td vaccine	0.5 mL IM: Td1 first contact → Td2 ≥ 4 weeks later → Td3 ≥ 6 months → Td4 ≥ 1 year → Td5 ≥ 1 year	Td2 at least 2 weeks before delivery protects the newborn against neonatal tetanus . A woman with 5 documented doses needs no more

Sickle cell disease: the drug changes in pregnancy

Drug	What to do	Why
Hydroxyurea (hydroxycarbamide)	Stop at least 3 months before conception ; if she conceives on it, stop at once , counsel and arrange a detailed anomaly scan. Restart after delivery only if not breastfeeding (or per haematologist)	Inhibits DNA synthesis: teratogenic in animals
Folic acid	5 mg daily throughout	Chronic haemolysis consumes folate
Iron	Not routine : only if ferritin shows deficiency	Transfusion and haemolysis load iron
Penicillin V	250 mg orally twice daily (some protocols 500 mg); erythromycin 250 mg twice daily if allergic	Functional asplenia → pneumococcal sepsis
Malaria chemoprevention	Proguanil 200 mg daily (safe with folic acid 5 mg), or IPTp-SP with folic acid withheld 1–2 weeks after each dose (unit protocol) + ITN	Malaria is a major crisis trigger; folic acid 5 mg blunts SP
Aspirin	75 mg daily from 12 weeks	Placental disease, pre-eclampsia
ACE inhibitors, iron chelators	Stop before conception	Teratogenic / fetotoxic

Ward pearls: supplements

- The routine ANC pack is **FeFo + IPTp-SP + mebendazole + Td + ITN (+ calcium, aspirin if high risk)**. When a woman books late, **catch up everything at one visit**.
- **Black stools on iron** are expected; ask about **dizziness, haematemesis or abdominal pain** before blaming the iron.
- An **HIV-positive woman on TLD**: tell her to take **FeFo and calcium at a different time from TLD**, or all together **with food**.
- Always check the **Rhesus group before a woman leaves after any evacuation, ectopic or delivery**.

CLINICAL PEARL

Senior's favourite question: "Why can't she have SP?" Four answers: **first trimester, sulfa allergy, on co-trimoxazole, and on folic acid 5 mg** (relative: give proguanil or withhold the folic acid around each dose).

Contraception and reproductive hormones

Hormonal contraceptives work in three ways. They **stop ovulation** by negative feedback on the hypothalamus and pituitary. They **thicken cervical mucus** so sperm cannot get through. They **thin the endometrium**. Combined methods (oestrogen + progestogen) rely mainly on stopping ovulation. Progestogen-only methods rely more on the mucus, and how often they stop ovulation depends on the method: DMPA and implants almost always do, while the older POPs do so only about half the time.

Combined oral contraceptive (COC)

Class	Ethinylestradiol (usually 30 mcg) + a progestogen (levonorgestrel 150 mcg in Microgynon-type pills, the common Ugandan brand; norgestrel in Lo-Femenal)
Mechanism	Oestrogen + progestogen suppress FSH and LH, so no follicle and no LH surge . Also thickens mucus and thins the endometrium
Indications	Contraception; heavy or painful periods; PCOS (cycle control, acne, hirsutism); functional ovarian cysts
Dose & route	1 tablet orally daily for 21 days, then a 7-day break (or 7 iron "reminder" tablets in 28-day packs). Start day 1–5 of the cycle with no back-up needed. Starting later ("quick start") needs 7 days of condoms
Side effects	Nausea, breast tenderness, breakthrough bleeding (usually settles within 3 months), headache, mood change, VTE (about 2–3 times baseline; still lower than in pregnancy), a small rise in BP, a small rise in breast and cervical cancer risk. Protects against ovarian and endometrial cancer
Contraindications / cautions	See the MEC category 4 table below
Pregnancy/breastfeeding	Not needed in pregnancy (not teratogenic if taken by mistake). Breastfeeding: MEC 4 under 6 weeks and MEC 3 from 6 weeks to 6 months because oestrogen reduces milk supply . Not breastfeeding: MEC 4 under 21 days (VTE)
Monitoring/antidote	BP before starting and at each refill , BMI, smoking, migraine type. No antidote; stop the pill if there are VTE symptoms

WHO MEC category 4 for COCs (unacceptable health risk: do not use):

Area	Category 4 condition
Postpartum	Breastfeeding <6 weeks ; not breastfeeding <21 days

Area	Category 4 condition
Age and smoking	Age ≥ 35 and smokes ≥ 15 cigarettes/day (fewer than 15/day = MEC 3)
Blood pressure	Systolic ≥ 160 or diastolic ≥ 100 , or hypertension with vascular disease (140–159/90–99 = MEC 3)
Thrombosis	Current or past DVT/PE ; major surgery with prolonged immobilisation ; known thrombogenic mutations ; SLE with positive antiphospholipid antibodies
Vascular / heart	Ischaemic heart disease, stroke, complicated valvular heart disease (pulmonary hypertension, AF, endocarditis history), peripartum cardiomyopathy
Neurological	Migraine with aura (at any age)
Breast	Current breast cancer
Liver	Severe (decompensated) cirrhosis, hepatocellular adenoma or hepatoma, acute viral hepatitis (MEC 3/4 when starting)
Diabetes	With nephropathy, retinopathy, neuropathy or other vascular disease, or duration >20 years (MEC 3/4)
Multiple risk factors	Several arterial risk factors together (older age, smoking, diabetes, hypertension) (MEC 3/4)

EXAM FAVOURITE

The MEC scale: **1** = use freely; **2** = benefits generally outweigh risks; **3** = risks usually outweigh benefits (use only if nothing else is acceptable, with specialist follow-up); **4** = do not use. In a viva, say "**MEC 4, so absolutely contraindicated**", not just "not advised".

Missed pill rules (WHO, 30–35 mcg pills):

Situation	What to do
1–2 pills missed (up to 48 h late)	Take the missed pill now (even if that means 2 today), carry on. No back-up needed
≥ 3 pills missed (≥ 48 h late)	Take one now, carry on, and use condoms for 7 days
... and missed in week 3 (pills 15–21)	Skip the pill-free break : finish the active pills and start a new pack straight away
... and missed in week 1 with unprotected sex	Consider emergency contraception
Vomiting within 2 h, or severe diarrhoea	Treat as a missed pill

CLINICAL PEARL

The riskiest pills to miss are those **next to the pill-free week** (the start or end of the pack), because they stretch the hormone-free interval beyond 7 days, which is long enough for a follicle to grow.

Drug interactions: enzyme inducers. Drugs that induce liver **CYP3A4** speed up metabolism of oestrogen and progestogen, so contraceptive failure follows:

Drug	Effect on contraception	Advice
Rifampicin (TB), rifabutin	Very strong inducer. MEC 3 for COC, POP and implants	Use DMPA , a copper IUD or an LNG-IUS (all MEC 1). Continue for 28 days after stopping rifampicin

Drug	Effect on contraception	Advice
Efavirenz (older ART)	Lowers levonorgestrel and etonogestrel levels. Pregnancies reported with implants	DMPA or IUD preferred. Implant only with condoms and counselling
Nevirapine, ritonavir-boosted PIs	Change hormone levels (MEC 2)	Add condoms or choose DMPA or an IUD
Carbamazepine, phenytoin, phenobarbital, topiramate	Inducers (MEC 3)	DMPA or an IUD
Dolutegravir (TLD)	No clinically important interaction	All methods can be used
Broad-spectrum antibiotics (amoxicillin, doxycycline)	No proven effect (except rifamycins)	No extra precautions needed

UGANDA CONTEXT

Most women living with HIV in Uganda are now on **TLD (tenofovir/lamivudine/dolutegravir)**, which does **not** affect any contraceptive. The trap is the woman still on an **efavirenz-based regimen** or on **TB treatment**. Ugandan studies (Scarsi et al.) found that efavirenz lowers levonorgestrel levels from implants enough for pregnancies to occur. Note too that dolutegravir with rifampicin needs **50 mg twice daily**; check the ART clinic's plan.

Progestogen-only methods

Method	Drug & dose	Duration	Key facts
POP ("mini-pill")	Levonorgestrel 30 mcg or norethisterone 350 mcg daily, no break; desogestrel 75 mcg stops ovulation	Daily	Traditional POP: 3-hour window (desogestrel: 12 h). If late: take it now, use condoms for 48 h . Safe from day 0 postpartum
DMPA IM (Depo-Provera)	150 mg deep IM	Every 12 weeks (13 weeks). WHO allows reinjection up to 4 weeks late (2 weeks early) without back-up	The most-used method in Uganda. Amenorrhoea in about 50% by 1 year
DMPA-SC (Sayana Press)	104 mg SC (Uniject, can be self-injected)	Every 13 weeks	Uganda promotes self-injection at home, VHTs and drug shops
Implant: Jadelle	2 rods, levonorgestrel 75 mg each	5 years	Subdermal, inner upper arm; fertility returns within days of removal
Implant: Implanon NXT	1 rod, etonogestrel 68 mg	3 years (evidence for up to 5)	The most effective reversible method (failure <0.1%)
LNG-IUS (Mirena)	Levonorgestrel 52 mg, releasing ~20 mcg/day	5 years for HMB; licensed up to 8 years for contraception	First line for heavy menstrual bleeding ; endometrial protection in HRT

DMPA specifics: side effects include **irregular bleeding** at first, then amenorrhoea, **weight gain** (about 1–2 kg/year), and **reduced bone mineral density**, which is **reversible** after stopping (caution in teenagers under 18 and after 2 years, but it is not a reason to deny it). **Return of fertility is delayed:** on average **about 10 months after the last injection** (median 9–10 months), which is not infertility. Warn women who want a baby within the next year.

CLINICAL PEARL

Implants, the LNG-IUS and the copper IUD are **LARCs** (long-acting reversible contraception). Their typical-use failure rate is under 1%, versus about 7% for pills and 4% for DMPA, because they **do not depend on the user remembering**.

Copper IUD

Class	Non-hormonal intrauterine device (TCu380A)
Mechanism	Copper ions are toxic to sperm and ova (prevents fertilisation) and cause a sterile endometrial inflammation (prevents implantation)
Indications	Contraception for 10–12 years ; the most effective emergency contraception (>99%)
Dose & route	Inserted through the cervix. Postpartum: within 48 h (PPIUD, including at CS) or from 4 weeks
Side effects	Heavier, more painful periods , expulsion, perforation (about 1 in 1,000), infection in the first 20 days (from insertion)
Contraindications / cautions	MEC 4: pregnancy, current PID/purulent cervicitis/gonorrhoea/chlamydia, puerperal sepsis or post-septic abortion , unexplained vaginal bleeding (until assessed), cervical or endometrial cancer awaiting treatment , distorted cavity, pelvic TB, GTD with persistently raised hCG
Pregnancy/ breastfeeding	Fully compatible with breastfeeding. If pregnancy occurs with an IUD in place: exclude ectopic ; remove it if the strings are visible (lower miscarriage risk)
Monitoring/ antidote	Check the strings after each period; review at 3–6 weeks

Emergency contraception (EC)

Method	Dose / timing	Effectiveness	Notes
Copper IUD	Insert up to 5 days (120 h) after unprotected sex, or up to 5 days after the earliest expected ovulation	>99% , best	Also gives ongoing contraception
Levonorgestrel (Postinor-2, "P2")	1.5 mg orally once, within 72 h (WHO: up to 120 h, less effective)	~85% of expected pregnancies prevented; less effective with each day	Double dose (3 mg) if on enzyme inducers (rifampicin, efavirenz, carbamazepine) and in women with BMI >26 or weight >70 kg (FSRH). Safe in breastfeeding
Ulipristal acetate (ellaOne)	30 mg orally once, within 120 h	More effective than LNG on days 3–5	Not reliable with enzyme inducers. Do not start a hormonal method for 5 days after it (it blunts it). Express and discard breast milk for 1 week. Rarely available in Uganda

NOTE

Oral EC **delays ovulation**. It does **not** cause an abortion and does not harm an established pregnancy. Always pair EC with a pregnancy test at 3 weeks, STI/HIV risk assessment (PEP within 72 h if indicated) and a start of ongoing contraception.

Progestogens for pregnancy support and bleeding

Drug	Indication	Dose	Notes
Vaginal progesterone (micronised)	Short cervix ≤ 25 mm on scan at 16–24 weeks (singleton), or previous spontaneous preterm birth	200 mg vaginally at night until 34 weeks	Reduces preterm birth before 34 weeks (NICE NG25). Not effective in twins
Vaginal progesterone	Threatened miscarriage with a previous miscarriage (PRISM trial)	400 mg vaginally twice daily until 16 weeks	NICE 2021; modest gain in live births
17-OHPC (hydroxyprogesterone caproate, Makena)	Previously weekly IM for prior preterm birth	—	Withdrawn by the US FDA in 2023 (no benefit in PROLONG trial). Do not learn it as a current treatment
Norethisterone	Acute heavy bleeding; postponing a period	5 mg orally 8-hourly for 7–10 days (bleeding usually stops in 48 h); cyclical 5 mg 8-hourly days 5–26	Not a contraceptive at this dose. Avoid if VTE history (metabolised partly to ethinylestradiol)
Medroxyprogesterone acetate (oral)	Acute HMB; endometrial hyperplasia; withdrawal bleed before ovulation induction	10 mg orally 8-hourly for acute bleeding; 10 mg daily for 10 days for a withdrawal bleed	

Heavy menstrual bleeding (HMB): the drug ladder

First exclude pregnancy, anaemia, fibroids and polyps, and cancer (biopsy if ≥ 45 years or with risk factors).

Order	Drug	Dose	How it helps
1st line	LNG-IUS	52 mg device, 5 years	Reduces loss by ~90% ; contraception too
Non-hormonal (take only during the period)	Tranexamic acid	1 g orally 8-hourly for up to 4 days (can be 1.5 g)	Antifibrinolytic: reduces loss by ~50% . Avoid with active VTE
	Mefenamic acid	500 mg orally 8-hourly with food during menses	NSAID: reduces loss by ~25% and relieves pain
Hormonal	COC, cyclical norethisterone, DMPA	As above	Useful if contraception is wanted
Surgical	Endometrial ablation, myomectomy, hysterectomy		When drugs fail or with large fibroids

CLINICAL PEARL

Tranexamic acid + mefenamic acid can be combined, and neither is a contraceptive, so they suit a woman trying to conceive.

Fertility drugs

Drug	Mechanism	Dose	Key risks & monitoring
Letrozole (aromatase inhibitor), first line in PCOS	Less oestrogen, so less negative feedback, so more FSH	2.5 mg PO daily, days 3–7 (or 2–6); up by 2.5 mg per cycle to max 7.5 mg	Mostly mono-follicular ; endometrium spared. Higher live birth rate than clomiphene. Off-label for fertility

Drug	Mechanism	Dose	Key risks & monitoring
Clomiphene citrate (SERM)	Blocks hypothalamic oestrogen receptors, so the brain senses low oestrogen and releases more GnRH and FSH	50 mg PO daily, days 2–6 ; up by 50 mg per cycle to max 150 mg	Multiple pregnancy ~5–10%, OHSS (rare) , hot flushes, visual disturbance (stop the drug) , thin endometrium. Ultrasound follicle tracking at least in the first cycle; no more than 6 cycles
Metformin	Improves insulin resistance, lowers insulin-driven ovarian androgens	Build up to 500 mg 8-hourly (1.5 g/day)	Add in clomiphene resistance or obesity. GI upset, so start low
Gonadotrophins (hMG, recombinant FSH) + hCG trigger	Direct follicle stimulation	Daily SC injections under specialist care	Needs close ultrasound ± oestradiol monitoring ; highest OHSS and multiples risk. For WHO group I (hypothalamic) anovulation, clomiphene failure, IVF
Cabergoline (dopamine agonist)	D2 agonist, so prolactin falls	Hyperprolactinaemia: 0.25 mg PO twice weekly , titrate monthly	Nausea, dizziness, postural hypotension; valvulopathy at high doses. Also lowers OHSS risk in IVF
Bromocriptine	D2 agonist (older)	1.25 mg PO at night with food, up to 2.5 mg 8–12-hourly	More nausea than cabergoline. Preferred if pregnancy is planned (longest safety record)

RED FLAG

OHSS (ovarian hyperstimulation syndrome): abdominal distension, pain, vomiting, ascites, **haematocrit >45%**, oliguria and **VTE**, usually 3–10 days after the hCG trigger. Admit if severe: fluid balance, **enoxaparin 40 mg SC daily**, paracentesis for tense ascites, and **avoid NSAIDs**.

Lactation suppression (after stillbirth or neonatal death, or when the mother is advised not to breastfeed): **cabergoline 1 mg orally as a single dose within 24 h of birth** (if lactation is established: 0.25 mg 12-hourly for 2 days). **Avoid in hypertension and pre-eclampsia**. **Bromocriptine is no longer recommended postpartum** because of reports of **stroke, MI, seizures and psychosis**, especially in hypertensive women.

HRT (brief)

- **Why:** vasomotor symptoms, and **premature ovarian insufficiency or surgical menopause** (for example after a TAH-BSO for benign disease), where HRT should continue **until about age 51** to protect bone and heart.
- **What:** oestrogen **plus a progestogen if she has a uterus** (unopposed oestrogen causes endometrial hyperplasia and cancer); **oestrogen alone after hysterectomy**. **Transdermal** oestrogen carries **no extra VTE risk**, while oral does.
- **Contraindications:** oestrogen-dependent cancer (breast, endometrial), **undiagnosed vaginal bleeding**, active or recent VTE or arterial disease, active liver disease, pregnancy. **Risks:** a small, duration-dependent rise in breast cancer with combined HRT; VTE and stroke with oral oestrogen. **HRT is not a contraceptive.**

Ward pearls: contraception

- A woman on **rifampicin** needs **DMPA or an IUD**, not a pill or implant; continue the alternative for **28 days** after rifampicin ends.
- "Can I get pregnant soon after stopping the injection?" Tell her the truth: **about 10 months on average** after the last dose. For fast return, choose an **implant or IUD**.
- A breastfeeding mother at the 6-week visit: implant, IUD, POP or DMPA, **not the COC** (MEC 3 until 6 months).

- A woman with **BP 160/100** or **migraine with aura** can use **any progestogen-only method or an IUD**.
- For EC, the **copper IUD beats every pill** and works **up to 5 days**.

Analgesia and anaesthesia in obstetrics

Paracetamol

Class	Analgesic and antipyretic (not an NSAID)
Mechanism	Central COX inhibition; no anti-inflammatory or antiplatelet effect
Indications	First-line pain and fever at any gestation; postpartum and post-CS baseline analgesia
Dose & route	1 g orally or IV every 6 h (max 4 g/day); 15 mg/kg per dose if weight <50 kg
Side effects	Rare at therapeutic doses. Overdose causes hepatic necrosis
Contraindications / cautions	Severe liver disease, malnutrition, low body weight
Pregnancy/breastfeeding	Safe in all trimesters and in breastfeeding: the analgesic of choice. Use the lowest effective dose for the shortest time
Monitoring/antidote	Overdose: N-acetylcysteine

NSAIDs (ibuprofen, diclofenac, mefenamic acid, indomethacin)

Class	Non-selective COX inhibitors
Mechanism	Block prostaglandin synthesis: analgesic, anti-inflammatory, antiplatelet
Indications	Postpartum and post-CS pain (with paracetamol), dysmenorrhoea, HMB (mefenamic acid). Indomethacin is used as a specialist tocolytic before 32 weeks
Dose & route	Ibuprofen 400 mg orally 8-hourly with food; diclofenac 50 mg 8-hourly orally, 75 mg IM 12-hourly, or 100 mg PR at the end of a CS
Side effects	Dyspepsia and GI bleeding, renal impairment, bronchospasm, fluid retention
Contraindications / cautions	Asthma (NSAID-sensitive), peptic ulcer, severe pre-eclampsia with renal impairment or oliguria, PPH with AKI or coagulopathy, dehydration/sepsis with oliguria , thrombocytopenia
Pregnancy/breastfeeding	Avoid from 20 weeks: fetal renal impairment and oligohydramnios (US FDA 2020). Contraindicated from 30 weeks: premature closure of the ductus arteriosus (fetal pulmonary hypertension). Also linked to miscarriage around conception. Safe in breastfeeding (ibuprofen is preferred)
Monitoring /antidote	Urine output and creatinine in sick women; no antidote

Opioids in labour and after CS

Drug	Dose	Onset / duration	Key points
Pethidine (still standard in Uganda)	50–100 mg IM (about 1 mg/kg), repeat 4-hourly , max 2 doses in labour usually	15–20 min / 2–4 h	Give with an anti-emetic (e.g. promethazine 25 mg IM or metoclopramide 10 mg IV). Metabolite norpethidine accumulates: neonatal respiratory depression and poor suckling, worst when born 2–4 h after the dose . Avoid in renal failure and seizure risk (norpethidine is pro-convulsant)

Drug	Dose	Onset / duration	Key points
Morphine	5–10 mg IM (0.1 mg/kg) 4-hourly, or titrated IV 1–2 mg	20 min / 3–4 h	Better analgesia; post-CS first 24 h. Nausea, itch, sedation
Tramadol	50–100 mg orally/IM/IV 6–8-hourly (max 400 mg/day)		Moderate pain. Lowers the seizure threshold; less respiratory depression
Naloxone (mother)	0.4 mg IV , repeat every 2–3 min	Short acting: re-dosing may be needed	For opioid respiratory depression (RR <10) or sedation
Naloxone (baby)	0.1 mg/kg IM (ventilate first)		Only after airway and ventilation are secured. Avoid if the mother is opioid-dependent (neonatal withdrawal seizures)

RED FLAG

A baby born limp and apnoeic after maternal pethidine needs **ventilation with bag and mask first** (Helping Babies Breathe: within the Golden Minute). Naloxone is an adjunct after effective ventilation, not the first step.

UGANDA CONTEXT

Epidurals are rarely available outside a few private and national referral units, so pethidine, **non-drug support** (a birth companion, mobility, massage, upright positions) and **TENS** (rare) are the realistic options. Offer analgesia: pain relief in labour is part of respectful maternity care.

Lignocaine (lidocaine)

Class	Amide local anaesthetic (also a class Ib anti-arrhythmic)
Mechanism	Blocks voltage-gated sodium channels in nerve membranes, so no action potential
Indications	Episiotomy and perineal repair (infiltration), pudendal block (for instrumental delivery or repair), cervical/paracervical block (MVA), diluting IM MgSO ₄ and ceftriaxone
Dose & route	1% = 10 mg/mL (2% = 20 mg/mL). Max plain: 3 mg/kg (about 18–20 mL of 1% for 60 kg). Max with adrenaline: 7 mg/kg . Episiotomy/repair: about 10 mL of 1% . Pudendal block: 10 mL of 1% each side just below and behind the ischial spine , through the sacrospinous ligament
Side effects	Local pain; local anaesthetic systemic toxicity (LAST) if given intravascularly or in overdose
Contraindications / cautions	True amide allergy (rare); heart block. Never use adrenaline-containing solution on end-arteries (digits) or when IV injection is likely
Pregnancy/ breastfeeding	Safe in pregnancy and breastfeeding
Monitoring/ antidote	Aspirate before every injection ; add up the total dose (including the lignocaine in MgSO ₄ and any re-infiltration). Antidote for LAST: 20% lipid emulsion

LAST: recognise it early.

Stage	Features
Early CNS (warning)	Circumoral numbness or tingling, metallic taste , tinnitus, light-headedness, visual blurring, slurred speech, anxiety
Late CNS	Muscle twitching, seizures , reduced consciousness, coma
Cardiovascular	Hypertension and tachycardia at first, then bradycardia, arrhythmias (VT/VF), hypotension, asystole

Management of LAST (AAGBI, stepwise):

1. **Stop injecting. Call for help** (anaesthetist).
2. **Airway**: 100% oxygen; ventilate if needed (hyperventilation helps: acidosis worsens toxicity). Left lateral tilt if still pregnant.
3. **Seizures**: **diazepam 10 mg IV** slowly (or midazolam). In Uganda MgSO₄ may also be to hand, but a benzodiazepine is first line.
4. **IV access**, monitor ECG and BP.
5. **Cardiac arrest**: start CPR (ACLS). Arrhythmias may be refractory; **prolonged CPR is justified** (lipid can take >1 h to work). Use smaller adrenaline doses (≤ 1 mcg/kg); **avoid lignocaine as an anti-arrhythmic** (use amiodarone).
6. **Give 20% lipid emulsion** (Intralipid) for cardiovascular instability or arrest: **1.5 mL/kg IV bolus over 1 min** (about 100 mL for 70 kg), then an **infusion of 15 mL/kg/h**. Repeat the bolus **up to twice at 5-min intervals** and double the infusion to **30 mL/kg/h** if still unstable. **Maximum cumulative 12 mL/kg**.

CLINICAL PEARL

Bupivacaine is far more cardiotoxic than lignocaine (VF that is hard to reverse). Its max dose is **2 mg/kg**. Never use bupivacaine for IV regional block.

Spinal anaesthesia and post-spinal hypotension

Drug	Dose	Notes
Hyperbaric ("heavy") bupivacaine 0.5%	2–2.5 mL (10–12.5 mg) intrathecal at L3/4 or L4/5 $\pm$ fentanyl 10–25 mcg	Dextrose makes it heavier than CSF, so the block follows gravity (tilt changes the level)
Phenylephrine (first choice)	50–100 mcg IV bolus , or infusion	Pure α -agonist: reverses vasodilatation; less fetal acidosis than ephedrine. Causes reflex bradycardia
Ephedrine	6 mg IV bolus , repeat	$\alpha + \beta$: use when HR is low . Crosses the placenta: fetal acidosis with large doses

Post-spinal hypotension (the **commonest** complication) comes from **sympathetic block** (vasodilatation) plus **aortocaval compression**. Prevent and treat it with **left lateral tilt 15°**, a crystalloid **co-load of 500–1,000 mL**, and a vasopressor early (aim for SBP near baseline). Untreated, it causes maternal nausea, fainting and **fetal bradycardia/acidosis**.

Ketamine

Class	Dissociative anaesthetic (NMDA antagonist)
Mechanism	Blocks NMDA receptors: analgesia and "dissociation"; sympathetic stimulation
Indications	Induction in haemorrhagic shock (supports BP); short procedures (MVA, manual removal of placenta, repair) where no anaesthetist is present; analgesia
Dose & route	Induction 1–2 mg/kg IV ; analgesic doses 0.25–0.5 mg/kg IV

Side effects	Raises BP and HR , raised ICP, emergence hallucinations (reduce with diazepam), hypersalivation, laryngospasm
Contraindications / cautions	Severe pre-eclampsia/eclampsia and hypertension (BP surge, raised ICP), cardiac ischaemia, psychosis. Protective airway reflexes are not guaranteed : aspiration can still happen
Pregnancy/breastfeeding	Crosses the placenta; high doses increase uterine tone
Monitoring/antidote	SpO ₂ , BP, airway; no antidote

General anaesthesia (GA) in obstetrics

Pregnancy makes GA dangerous: **aspiration** (progesterone relaxes the lower oesophageal sphincter, delayed gastric emptying in labour, raised intra-abdominal pressure), **difficult intubation** (oedema, large breasts, weight), **rapid desaturation** (lower FRC, higher O₂ use), **awareness**, and **uterine atony** (volatile agents relax the uterus).

Aspiration prophylaxis before CS (especially under GA):

Drug	Dose	Purpose
Omeprazole	40 mg IV (or 40 mg orally the night before and morning of an elective CS)	Raises gastric pH
Ranitidine (H ₂ blocker)	50 mg IV or 150 mg orally	Same; withdrawn in many countries since 2020 (NDMA contamination), so use omeprazole or famotidine
Metoclopramide	10 mg IV	Speeds gastric emptying and raises lower oesophageal sphincter tone
Sodium citrate 0.3 M	30 mL orally just before induction	Immediately neutralises acid (non-particulate)

Then **rapid sequence induction** (pre-oxygenation, thiopental or propofol, **suxamethonium**, cricoid pressure, cuffed tube). Remember **MgSO₄ potentiates non-depolarising muscle relaxants**.

Ward pearls: analgesia and anaesthesia

- Before any perineal repair, calculate the **maximum lignocaine volume** for her weight and say it aloud.
- A drowsy woman after pethidine with RR 8 → **naloxone 0.4 mg IV** and oxygen.
- The first sign of LAST is often the patient saying her **lips tingle** or she has a **metallic taste**.

Other medical drugs in pregnancy

Insulin

Class	Human insulin: soluble (Actrapid), isophane/NPH (Insulatard), premixed 30/70 (Mixtard)
Mechanism	Replaces or supplements endogenous insulin. Does not cross the placenta
Indications	Pre-existing type 1 and type 2 diabetes; GDM not controlled on diet ± metformin; start at once if fasting glucose ≥7.0 mmol/L at diagnosis; DKA; labour (IV infusion)
Dose & route	SC. Start about 0.3–0.5 units/kg/day (0.2–0.3 for mild GDM). High fasting glucose → bedtime isophane 0.1–0.2 units/kg at 22:00 ; high post-meal → pre-meal soluble 2–4 units. Titrate by 2 units every 2–3 days . Labour: variable-rate IV insulin with dextrose, target 4–7 mmol/L
Side effects	Hypoglycaemia (worst in the first trimester in type 1), weight gain, lipohypertrophy

Contraindications / cautions	Requirements rise sharply after 20–24 weeks ; steroids raise glucose for 3–5 days; a fall of >15–20% in late pregnancy may mean placental insufficiency
Pregnancy/breastfeeding	The gold standard in pregnancy and safe in breastfeeding. Stop it at delivery in GDM ; in type 1 cut to pre-pregnancy doses at once
Monitoring/antidote	Capillary glucose (fasting, 1 h post-meal). Hypo <4.0 mmol/L: 15–20 g fast sugar , recheck in 15 min; unable to swallow: IV 20% dextrose 75–100 mL or glucagon 1 mg IM

Oral agents in GDM:

Drug	Dose	Pros	Cons
Metformin	500 mg with the evening meal, titrate to 2 g/day	Oral, cheap, no hypoglycaemia, less maternal weight gain	GI upset; crosses the placenta (long-term child data reassuring but incomplete); stop if eGFR <30, sepsis, dehydration. Dolutegravir raises metformin levels (use a lower total dose and titrate carefully, e.g. many clinicians keep to about 1 g/day)
Glibenclamide	2.5–5 mg daily, max 15–20 mg	Oral	Crosses the placenta : more macrosomia and neonatal hypoglycaemia . Second line only

Enoxaparin and heparins (thromboprophylaxis)

Class	LMWH (enoxaparin); unfractionated heparin (UFH)
Mechanism	Activates antithrombin : LMWH mainly inhibits factor Xa ; UFH inhibits Xa and thrombin (IIa)
Indications	Antenatal and postnatal VTE prophylaxis by risk score; after CS; OHSS; treatment of DVT/PE; antiphospholipid syndrome (with aspirin); mechanical valves (specialist)
Dose & route	Prophylaxis (by booking weight): <50 kg 20 mg; 50–90 kg 40 mg; 91–130 kg 60 mg; 131–170 kg 80 mg — SC once daily . Treatment: 1 mg/kg SC 12-hourly . UFH prophylaxis: 5,000 IU SC 12-hourly ; treatment by IV infusion with APTT monitoring
Side effects	Bruising, bleeding, heparin-induced thrombocytopenia (rare with LMWH), osteoporosis (long-term UFH)
Contraindications / cautions	Active bleeding, platelets <75 × 10⁹/L , severe renal impairment (use UFH), uncontrolled hypertension
Pregnancy/breastfeeding	Does not cross the placenta : the anticoagulant of choice in pregnancy. Safe in breastfeeding
Monitoring/antidote	Platelets at baseline; anti-Xa in extremes of weight or renal disease. Protamine fully reverses UFH, only partly reverses LMWH

LMWH and regional anaesthesia (spinal/epidural):

Timing	Prophylactic dose	Treatment dose
Last dose before a spinal/epidural	≥12 h	≥24 h
First dose after a spinal or catheter removal	≥4 h	≥4 h (often longer after surgery; senior decision)
Remove the epidural catheter	≥12 h after the last dose	≥24 h after the last dose

DRUG DOSING

Tell women on LMWH: "If you think you are in labour or you bleed, do not inject; come in." That keeps the spinal option open.

Warfarin: crosses the placenta. **Warfarin embryopathy** with exposure at **6–12 weeks** (nasal hypoplasia, stippled epiphyses / chondrodysplasia punctata), later **CNS abnormalities** and **fetal intracranial haemorrhage** (especially at delivery). Switch to LMWH **before 6 weeks**. The exception is some women with **mechanical heart valves**, where a specialist may continue warfarin (fewer valve thromboses; dose ≤ 5 mg/day). **Warfarin is safe in breastfeeding**. **DOACs** (rivaroxaban, apixaban) are **avoided in pregnancy and breastfeeding**.

Anti-emetics for nausea, vomiting and hyperemesis gravidarum

Drug	Dose	Line	Key issues
Cyclizine (antihistamine)	50 mg PO/IM/IV 8-hourly	1st	Drowsiness; long safety record
Promethazine (antihistamine)	25 mg PO/IM 8-hourly	1st	Widely available in Uganda; sedation
Prochlorperazine / chlorpromazine (phenothiazines)	12.5 mg IM/IV 8-hourly; 10–25 mg PO 8-hourly	1st	Extrapyramidal effects, sedation
Metoclopramide (D2 antagonist)	10 mg PO/IM/IV 8-hourly, max 5 days	2nd	Extrapyramidal reactions / oculogyric crisis (young women especially): stop and give procyclidine 5–10 mg IV/IM (or an anticholinergic available locally, e.g. biperiden)
Ondansetron (5-HT3 antagonist)	4–8 mg PO/IV 8–12-hourly	2nd	Small cleft lip/palate signal with first-trimester use (about 3 extra per 10,000): counsel, but do not withhold if needed. Constipation, QT prolongation
Hydrocortisone → prednisolone	100 mg IV 12-hourly, then 40–50 mg PO daily , tapered	3rd	Refractory cases only, specialist decision

Hyperemesis essentials (RCOG GTG 69):

1. IV fluids: **0.9% sodium chloride with potassium chloride** (guided by daily U&E). **Avoid dextrose-only fluids at first.**
2. **Thiamine before any dextrose** (100 mg IV, as Pabrinex or plain thiamine, then **oral thiamine 25–50 mg 8-hourly** or 100 mg daily). Dextrose in a thiamine-depleted woman can precipitate **Wernicke's encephalopathy**: confusion, **ophthalmoplegia/nystagmus**, ataxia.
3. **Thromboprophylaxis with LMWH** while admitted (dehydration + immobility).
4. Check for **molar pregnancy and multiple pregnancy** (scan), UTI, thyroid (transient gestational thyrotoxicosis needs **no** antithyroid drugs).
5. Give an **H2 blocker or PPI** if there is reflux or gastritis.

Antiepileptic drugs (AEDs)

Drug	Pregnancy risk	Notes
Sodium valproate	Highest: major malformations about 10% (neural tube defects 1–2%, cardiac, cleft, hypospadias); neurodevelopmental problems in 30–40% (lower IQ, autism)	Avoid in girls and women who could become pregnant unless nothing else works, under a pregnancy-prevention programme. Never stop suddenly in pregnancy: switch under specialist care
Phenytoin	Fetal hydantoin syndrome: craniofacial features, nail and distal phalanx hypoplasia , growth and developmental delay	Enzyme inducer (contraception failure)
Carbamazepine	NTD about 0.5–1% , cleft, cardiac	Enzyme inducer
Phenobarbital	Cardiac defects, cleft; neonatal sedation/withdrawal	Still used in Uganda; enzyme inducer
Lamotrigine	Among the safest	Levels fall by up to 50% in pregnancy (more glucuronidation): check levels or clinically adjust; reduce back after delivery (toxicity)
Levetiracetam	Among the safest	Clearance rises in pregnancy; renal excretion

DRUG DOSING

Every woman with epilepsy on AEDs who might conceive needs **folic acid 5 mg daily**, ideally **3 months before conception** and through the first trimester. Also: **monotherapy at the lowest effective dose**, a detailed **anomaly scan**, and **vitamin K 1 mg IM for the baby** at birth (as for every newborn). A fit in pregnancy after 20 weeks is **eclampsia until proven otherwise**: give MgSO₄.

AEDs and **breastfeeding**: encourage it (most are compatible); watch the baby for **sedation and poor feeding** (phenobarbital, benzodiazepines, lamotrigine at high levels).

Thyroid drugs

Drug	Use	Dose	Pregnancy points
Levothyroxine	Hypothyroidism	Usual dose, e.g. 50–150 mcg daily	Increase by 25–30% as soon as pregnancy is confirmed (e.g. take 2 extra doses per week); the fetus depends on maternal T4 in the first trimester (brain development). Check TSH every 4–6 weeks ; target TSH <2.5 mU/L . Take on an empty stomach, apart from iron and calcium . Return to the pre-pregnancy dose after delivery
Propylthiouracil (PTU)	Hyperthyroidism in the first trimester	50–150 mg 8-hourly (lowest effective dose)	Less teratogenic than carbimazole but hepatotoxic , so switch in the second trimester
Carbimazole / methimazole	Hyperthyroidism from the second trimester	5–40 mg daily	First-trimester embryopathy : aplasia cutis (scalp defects), choanal and oesophageal atresia , omphalocele

Both PTU and carbimazole cross the placenta (fetal goitre, hypothyroidism if over-treated): aim for **free T4 at the upper end of normal**. **Graves' disease with high TRAb** → watch for **fetal and neonatal thyrotoxicosis**. Both drugs are compatible with breastfeeding at modest doses.

Galactagogues

Domperidone (D2 antagonist, raises prolactin) **10 mg orally 8-hourly** for about **7 days** (off-label) is used when supply stays low **after correcting attachment and feeding frequency**. It can prolong the **QT interval**: avoid in cardiac disease, with QT-prolonging drugs, or at high doses. Metoclopramide works similarly but causes depression and extrapyramidal effects. **Fix the latch before prescribing**.

Ward pearls: medical drugs

- GDM: diet first, then **metformin**, then **insulin**; stop them all at delivery and **check glucose** 6–13 weeks postpartum.
- A woman on **enoxaparin** who wants a spinal: when was the **last dose**? Prophylactic 12 h, treatment 24 h.
- **Thiamine before dextrose** in every vomiting, dehydrated pregnant woman.
- **Never stop an AED abruptly** because a woman is pregnant: uncontrolled seizures (especially SUDEP) are more dangerous than most drugs.

Drug safety: teratogens, breastfeeding and pregnancy pharmacokinetics

How timing matters

Period (from conception)	Weeks from LMP	Effect of a harmful drug
0–2 weeks (pre-implantation)	2–4	"All or nothing" : miscarriage or complete recovery
3–8 weeks (organogenesis)	5–10	Structural malformations : the critical window
After 8 weeks (fetal period)	>10	Growth, function (renal, brain), and neonatal effects near term

The **baseline** risk of a major congenital anomaly in any pregnancy is **2–3%**; quote risks as an increase over this.

Teratogen table

Drug	Effect	Critical window
Warfarin	Embryopathy : nasal hypoplasia, stippled epiphyses ; later CNS defects, fetal haemorrhage	6–12 weeks ; bleeding throughout, especially at delivery
Sodium valproate	Neural tube defects (spina bifida), cardiac, cleft, hypospadias, neurodevelopmental delay/autism	First trimester (NTD at 3–4 weeks post-conception); neurodevelopment throughout
Phenytoin	Fetal hydantoin syndrome : craniofacial, hypoplastic nails/distal phalanges , growth and mental delay	First trimester
Carbamazepine	NTD , craniofacial, cardiac	First trimester
Isotretinoin (and other retinoids)	Retinoid embryopathy : ear anomalies (microtia), craniofacial, cardiac outflow , CNS, thymus; high miscarriage	First trimester; stop 1 month before conception (acitretin: 3 years)
ACE inhibitors / ARBs	Fetal renal failure , oligohydramnios , skull ossification defects (hypocalvaria), limb contractures, lung hypoplasia	Second and third trimesters (switch before or as soon as pregnancy is known)
Tetracyclines (doxycycline)	Yellow-brown tooth discolouration , enamel hypoplasia, inhibited bone growth	After about 15 weeks (teeth calcify)

Drug	Effect	Critical window
Methotrexate	Aminopterin syndrome: skull ossification defects, craniofacial, limb; miscarriage	First trimester; stop 3 months before conception (per older guidance; ask the specialist)
Misoprostol (failed abortion)	Möbius syndrome (facial and 6th/7th nerve palsy), limb (transverse) defects	First trimester (vascular disruption)
Lithium	Ebstein's anomaly (tricuspid valve displaced; absolute risk small); neonatal toxicity, polyhydramnios , diabetes insipidus	First trimester; toxicity at term
Thalidomide	Phocomelia (limb reduction), ear, cardiac, gut	Days 20–36 after conception
Diethylstilbestrol (DES)	Daughters: clear-cell adenocarcinoma of the vagina , vaginal adenosis, T-shaped uterus , cervical insufficiency	Before 18 weeks (historical, 1940s–1971)
Alcohol	Fetal alcohol spectrum disorder: smooth philtrum, thin upper lip, short palpebral fissures, microcephaly, FGR, learning problems	Throughout: no safe amount
Statins	Previously classed as teratogenic (weak evidence); FDA removed the contraindication in 2021	Usually stopped before or in pregnancy (cholesterol is needed for fetal development; little benefit in 9 months)
Carbimazole	Aplasia cutis, choanal and oesophageal atresia	First trimester
Aminoglycosides (streptomycin, kanamycin)	Fetal ototoxicity (8th nerve deafness)	Throughout (single-dose or short gentamicin courses for sepsis are acceptable)
Folate antagonists (trimethoprim, high-dose SP)	NTD, cardiac	First trimester (cotrimoxazole for HIV prophylaxis is continued : the benefit outweighs the risk)

EXAM FAVOURITE

Classic pairings: **Ebstein – lithium**; **Möbius – misoprostol**; **phocomelia – thalidomide**; **clear-cell vaginal carcinoma – DES**; **grey baby – chloramphenicol** (neonate); **kernicterus – sulphonamides** (near term, displace bilirubin); **stippled epiphyses – warfarin**; **aplasia cutis – carbimazole**.

NOTE

Dolutegravir: an early (2018) Botswana signal of NTDs faded with more data (Tsepamo). WHO recommends **TLD for all, including women who may become pregnant**. Efavirenz is no longer considered teratogenic either. Keep women on effective ART; stopping it is the real danger to the baby.

Breastfeeding: safe or not?

Most drugs pass into milk in tiny amounts. Worry more when the drug is **toxic in small doses**, the baby is **preterm or jaundiced**, or the drug **suppresses lactation**. **LactMed** (free online) is the reference.

Generally compatible	Avoid or use with caution (why)
Paracetamol, ibuprofen , diclofenac	Codeine and tramadol : ultra-rapid metabolisers → infant opioid toxicity (codeine deaths reported; contraindicated in breastfeeding in the UK)

Generally compatible	Avoid or use with caution (why)
Penicillins, cephalosporins, macrolides , metronidazole (single 2 g dose: pause feeding for 12–24 h), gentamicin , nitrofurantoin (not if baby G6PD-deficient or <1 month jaundiced)	Chloramphenicol (grey baby syndrome, marrow suppression); tetracyclines long courses (short courses acceptable); sulphonamides in jaundiced, preterm or G6PD-deficient babies
Nifedipine, methyldopa , labetalol, amlodipine, enalapril	Amiodarone (iodine: neonatal hypothyroidism)
Heparin, LMWH , warfarin	DOACs (limited data)
Insulin, metformin , glibenclamide	Lithium (infant toxicity; specialist only), high-dose aspirin (Reye's syndrome risk)
Levothyroxine , PTU, carbimazole (modest doses)	Radioactive iodine ; cytotoxic chemotherapy , methotrexate in cancer doses
Sertraline , most AEDs (lamotrigine, levetiracetam, carbamazepine)	Benzodiazepines repeated, phenobarbital (infant sedation, poor feeding)
ARVs (TLD) , cotrimoxazole, antimalarials (ACTs, quinine)	Oestrogen-containing COCs (early), ergometrine , cabergoline , bromocriptine : reduce milk supply
Progestogen-only contraception, IUDs, LNG emergency contraception	Ulipristal (express and discard for 1 week)

Pharmacokinetic changes in pregnancy

Change	Mechanism	Effect on drugs	Example
↑ Volume of distribution	Plasma volume +40–50% , total body water +8 L, more fat	Lower peak levels of water-soluble drugs; may need bigger loading doses	Gentamicin, MgSO ₄ (hence the big loading dose)
↑ Renal clearance	GFR rises ~50% from early pregnancy	Faster elimination of renally cleared drugs	Levetiracetam , lithium, penicillins, LMWH (weight-based dosing), digoxin
↓ Albumin (to ~28–37 g/L)	Haemodilution	More free (active) drug of highly protein-bound drugs; total levels mislead	Phenytoin (interpret free levels), valproate
↑ Hepatic metabolism of some drugs	Oestrogen/progesterone induce CYP3A4 , CYP2D6 and UGT (glucuronidation) ; CYP1A2 falls	Lower levels	Lamotrigine (UGT: levels fall up to 50%), nifedipine, methadone. Caffeine and theophylline levels rise (CYP1A2 ↓)
Delayed gastric emptying and ↓ GI motility (progesterone), especially in labour	Smooth muscle relaxation; opioids	Slower, less predictable oral absorption ; aspiration risk under GA	Give IV/IM routes in labour emergencies; aspiration prophylaxis
Placenta and fetus	Lipid-soluble, small (<500 Da), un-ionised, unbound drugs cross	Fetal exposure; fetal ion trapping in an acidotic fetus	Pethidine, local anaesthetics, benzodiazepines cross; heparin and insulin do not (large molecules)

After delivery these changes reverse within days to weeks: doses that were **increased** in pregnancy (lamotrigine, levothyroxine, insulin) must be **reduced** again, or toxicity or hypoglycaemia follows.

How to counsel a woman about a drug in pregnancy

1. **Ask first:** every woman of reproductive age, "Could you be pregnant? Are you planning a pregnancy? Are you breastfeeding?" Check LMP and do a urine hCG if unsure.
2. **Weigh both sides:** the risk of the drug **versus the risk of the untreated disease** (seizures, HIV transmission, depression, malaria, VTE, eclampsia). Untreated illness often harms the baby more than the drug.
3. **Use numbers:** "Every pregnancy has a **2–3%** chance of a major anomaly; this drug raises it to about X%." Absolute risks, not "doubles the risk".
4. **Choose well:** an older drug with a good safety record, **monotherapy**, the **lowest effective dose**, for the **shortest time**; avoid the critical window where possible.
5. **Plan ahead** (pre-conception): switch teratogens (valproate, warfarin, ACE inhibitors, isotretinoin, methotrexate) **before** conception; **folic acid** (400 mcg, or **5 mg** if on AEDs, previous NTD, diabetes, sickle cell); effective contraception until then.
6. **Never tell her to stop suddenly** without a plan; if she was **already exposed**, reassure where appropriate, arrange a **detailed anomaly scan** and document the counselling.
7. **Breastfeeding:** usually continue both breastfeeding and the drug; watch the baby for sedation, poor feeding or jaundice.

UGANDA CONTEXT

Many women in Uganda take **herbal medicines** in pregnancy (including local preparations taken to "open the way" or speed labour) and buy drugs over the counter at drug shops. Ask about them specifically: uterotonic herbs cause **tachysystole, uterine rupture and fetal distress**, and some herbal products are hepatotoxic.

Ward pearls: drug safety

- The dangerous time is **weeks 3–8 after conception**: often **before she knows she is pregnant**. So ask every woman before you prescribe.
- **Heparin and insulin are big molecules and do not cross the placenta**, which is exactly why they are the drugs of choice in pregnancy.
- In breastfeeding, the drugs to remember are the ones that **dry up milk** (oestrogen, ergometrine, dopamine agonists) and the ones that **sedate or poison the baby** (codeine, lithium, chloramphenicol, chemotherapy).

Management of drug emergencies on the labour ward

These are the drug-induced emergencies you must be able to recognise and start treating before the senior arrives.

1. Magnesium toxicity

1. **Stop** the infusion or withhold the next IM dose.
2. **ABC**: open the airway, **oxygen**, left lateral position; call the anaesthetist if RR <12 or SpO₂ is falling.
3. **Calcium gluconate 10%, 10 mL (1 g) IV slowly over 10 min** for respiratory depression, cardiac arrhythmia or severe drowsiness.
4. **Ventilate** (bag-valve-mask, then intubation) if apnoeic; calcium works only briefly.
5. Check **urine output, creatinine and magnesium level**; restart at a lower dose only once reflexes return and urine output is ≥ 30 mL/h.

2. Oxytocin hyperstimulation (tachysystole) with fetal distress

1. **Stop the oxytocin** (or remove any remaining vaginal misoprostol/dinoprostone).
2. Turn her onto her **left side**; give **oxygen** only if she is hypoxic; give an IV fluid bolus if hypotensive.
3. If the fetal heart does not recover: acute tocolysis with **terbutaline 0.25 mg SC** (or salbutamol per local protocol), and prepare for urgent delivery.
4. Restart oxytocin (at half the previous rate) only after the fetal heart is normal and contractions have settled, and after a senior review.

3. Water intoxication (oxytocin's antidiuretic effect)

Headache, vomiting, confusion, fits; sodium <130 mmol/L. **Stop oxytocin**, restrict fluids, use saline-based fluids for any further infusion, check sodium; fits → treat and seek senior/ICU help. Prevent it by running oxytocin in **normal saline or Ringer's**, not large volumes of 5% dextrose.

4. Local anaesthetic systemic toxicity (LAST)

1. **Stop injecting**; call for help.
2. **ABC**, oxygen; treat fits with **diazepam 10 mg IV** (or MgSO₄ if eclampsia is possible).
3. **20% lipid emulsion: 1.5 mL/kg IV bolus** over 1 min, then **15 mL/kg/h**; repeat the bolus up to twice at 5-min intervals if unstable (where available).
4. Cardiac arrest: CPR with left uterine displacement; consider perimortem caesarean after 4 min if ≥20 weeks.

5. Anaphylaxis (antibiotics, oxytocin, iron sucrose, blood)

Adrenaline 0.5 mg IM (0.5 mL of 1:1000) into the anterolateral thigh, repeat every 5 min if needed; lie her flat with **left lateral tilt**; high-flow oxygen; **1 L crystalloid bolus**; then chlorphenamine 10 mg IV and hydrocortisone 200 mg IV as adjuncts. Record the reaction and never re-challenge.

6. Neonatal respiratory depression after maternal opioids

Start **neonatal resuscitation first** (warm, dry, position the airway, ventilate with bag and mask). Naloxone **0.1 mg/kg IM** is only considered if the baby is breathing poorly **after** effective ventilation has restored heart rate and colour, and never if the mother is opioid-dependent.

7. Post-spinal hypotension

Left lateral tilt, fluid co-load, **phenylephrine 50–100 µg IV boluses** (or infusion) or **ephedrine 6 mg IV boluses**; treat a fall of >20% from baseline or SBP <90 mmHg to protect placental perfusion.

8. Hypoglycaemia (insulin, IV quinine, after steroids stopped)

Glucose <4 mmol/L (some units use <3.0): conscious → oral glucose; unconscious or unable to swallow → **50 mL of 50% dextrose IV** (or 100–150 mL of 10% dextrose), then a dextrose infusion and hourly glucose.

Emergency drug boxes

UGANDA CONTEXT

Many Ugandan labour wards keep ready-packed boxes. Check them at the start of every shift. If you can list the contents of these two boxes, you understand the two biggest killers of Ugandan mothers.

Eclampsia box	PPH box
MgSO ₄ 50% ampoules (at least 20 g, i.e. 4 × 10 mL vials of 5 g)	Oxytocin 10 IU × 5 (kept cold)
Water for injection to make 20%	Misoprostol 200 µg × 4
2% lignocaine (for the IM injections)	Tranexamic acid 1 g × 2
Calcium gluconate 10%, 10 mL	Ergometrine 0.5 mg (cold; not for hypertensives)
Hydralazine 20 mg ampoule and/or nifedipine 10 mg tablets	Ringer's lactate / normal saline 1 L × 2, giving sets
Diazepam 10 mg (if no MgSO ₄)	Two large-bore cannulae (14–16 G), blood bottles, dry glass tube for clotting test
Syringes, needles, cannula, urinary catheter and urine bag, reflex hammer	Urinary catheter, condom-catheter tamponade kit, calibrated drape

Ward-round & exam pearls

Model presentation: "Mrs A, 30, P3, day 1 after caesarean for severe pre-eclampsia. She is on **MgSO₄ maintenance**, last dose at 06:00, due at 10:00: RR 18, reflexes present, urine 45 mL/h, so the dose can be given. BP is controlled on **nifedipine MR 20 mg 12-hourly**; methyldopa has been stopped because of the risk of postnatal depression. She had **ergometrine avoided** at delivery and received **oxytocin and TXA**. She is on **enoxaparin 40 mg SC daily** from 12 h after surgery. Plan: stop MgSO₄ at 24 h after delivery, continue BP checks, review in the postnatal clinic."

Viva questions with answers

1. *"What is the antidote to magnesium sulphate and how do you give it?"* Calcium gluconate 10%, 10 mL (1 g) IV slowly over 10 min, with airway support.
2. *"Why is diazepam no longer first line for eclampsia?"* The Collaborative Eclampsia Trial (1995) showed MgSO₄ reduced recurrent fits by about half versus diazepam and by about two-thirds versus phenytoin, with a trend towards fewer maternal deaths; Magpie (2002) showed prophylactic MgSO₄ halves the risk of eclampsia.
3. *"Which uterotonic would you avoid in a woman with asthma? And in pre-eclampsia?"* Carboprost (bronchospasm); ergometrine (vasoconstriction, hypertensive crisis, stroke).
4. *"Why not sublingual nifedipine?"* Rapid, uncontrolled BP fall → placental hypoperfusion and fetal distress, maternal myocardial or cerebral ischaemia.
5. *"Which antibiotic is avoided in PPRM and why?"* Co-amoxiclav: ORACLE trial showed increased necrotising enterocolitis in the newborn.
6. *"Name three drugs that interact with hormonal contraception."* Rifampicin, efavirenz (implants lose efficacy) and enzyme-inducing antiepileptics (carbamazepine, phenytoin). Dolutegravir does **not** interact.
7. *"What is the safest antimalarial in severe malaria in the first trimester?"* IV artesunate (WHO recommends it in all trimesters).
8. *"How long do antenatal steroids take to work?"* Maximum benefit from 24 h after the first dose to 7 days.

OSCE tips

- In a prescribing station, write **drug name (generic), dose, route, frequency, duration, date/time, your signature**, and state the monitoring.
- In a MgSO₄ station, **say the dilution out loud**: "8 mL of 50% (4 g) plus 12 mL water = 20 mL of 20%."
- Always ask about **allergies and asthma** first; examiners watch for it.

Mnemonics & memory aids

- **MgSO₄ checks: "R-R-U"**: Respirations ≥ 16 , Reflexes present, Urine ≥ 30 mL/h.
- **Magnesium levels climb "4-6-12"**: reflexes go at about **4–5**, breathing at about **6–7.5**, the heart at **>12 mmol/L**. (The knee goes before the lungs, the lungs before the heart.)
- **Ergometrine contraindications: "HHH + T"**: Hypertension/pre-eclampsia, Heart disease, Hands (Raynaud's/peripheral vascular disease), undelivered Twin (and retained placenta).
- **Carboprost: "Carbo-PROST closes the chest"**: never in asthma.
- **Teratogens: "Warfarin Voices ACE TOM's Leaving"**: Warfarin, Valproate, ACE inhibitors/ARBs, Tetracyclines, isotretinoin, Methotrexate/misoprostol, Lithium.
- **Safe antibiotics in pregnancy: "Pregnant Cows Eat Many Nice Carrots"**: Penicillins, Cephalosporins, Erythromycin/azithromycin, Metronidazole, Nitrofurantoin (not at term), Clindamycin.
- **Steroid numbers: "6-12-4, 12-24-2"**: dexamethasone **6 mg 12-hourly $\times 4$** ; betamethasone **12 mg 24-hourly $\times 2$** .

High-yield summary

Numbers & doses to know

Item	Value
MgSO ₄ loading (Pritchard)	4 g of 20% IV over 5–10 min + 5 g of 50% IM each buttock (14 g)
MgSO ₄ maintenance	5 g of 50% IM 4-hourly (Pritchard) or 1 g/h IV (Zuspan), for 24 h after delivery/last fit
MgSO ₄ recurrent fit	2 g of 20% IV over 5 min
Making 20% MgSO ₄	8 mL of 50% + 12 mL water = 20 mL (4 g)
Therapeutic Mg ²⁺	2–3.5 mmol/L ; reflexes lost 4–5 ; respiratory depression 6–7.5 ; cardiac arrest >12
Calcium gluconate	10 mL of 10% (1 g) IV over 10 min
Nifedipine IR (acute)	10 mg orally , repeat after 20–30 min
Hydralazine	5 mg IV slowly , every 20–30 min, max 20 mg
Labetalol IV	20 → 40 → 80 → 80 mg every 10 min, max 300 mg
Methyldopa	250–750 mg 8-hourly , max 3 g/day (maintenance only)
Aspirin (PE prevention)	75–150 mg at night from 12 weeks (ideally before 16) to 36 weeks
Oxytocin AMTSL	10 IU IM within 1 min of birth
Oxytocin PPH	10 IU IV slowly + 20 IU in 1 L at 60 drops/min
Ergometrine	0.2 mg IM , max 1 mg/24 h
Carboprost	250 µg IM every 15 min , max 8 doses (2 mg)
Misoprostol	PPH treatment 800 µg SL ; prevention 600 µg oral ; IOL 25 µg
Tranexamic acid	1 g IV over 10 min within 3 h ; repeat once
Dexamethasone	6 mg IM 12-hourly $\times 4$ (24–34 weeks)
Nifedipine tocolysis	20 mg oral , then 10–20 mg 6–8-hourly for 48 h
Sepsis	Ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg daily + metronidazole 500 mg IV 8-hourly

Item	Value
CS prophylaxis	Cefazolin 2 g IV 15–60 min before incision
Syphilis	Benzathine penicillin 2.4 MU IM (× 3 weekly if late/unknown)
Anti-D (Uganda vial)	1,500 IU (300 µg) IM within 72 h
IPTp-SP	3 tablets monthly from 13 weeks; folic acid 0.4 mg , not 5 mg
Severe malaria	IV artesunate 2.4 mg/kg at 0, 12, 24 h , then daily
Lignocaine max	3 mg/kg plain; 7 mg/kg with adrenaline
Enoxaparin prophylaxis	40 mg SC daily (weight-adjusted)
Anaphylaxis	Adrenaline 0.5 mg IM (0.5 mL of 1:1000)

Classic exam traps

- **Trap:** "Give ergometrine for PPH in a woman with pre-eclampsia." **Right:** oxytocin, misoprostol and TXA; ergometrine is contraindicated.
- **Trap:** "Methyldopa for acute severe hypertension." **Right:** it takes hours to work; use nifedipine IR, hydralazine or labetalol.
- **Trap:** "Give the next MgSO₄ dose because it is due." **Right:** check R-R-U first; withhold if reflexes are absent, RR <16 or urine <30 mL/h.
- **Trap:** "Diazepam is an acceptable first-line anticonvulsant in eclampsia." **Right:** MgSO₄ is first line; diazepam only if MgSO₄ is unavailable.
- **Trap:** "Co-amoxiclav for PPROM." **Right:** erythromycin (co-amoxiclav increases NEC).
- **Trap:** "Folic acid 5 mg with SP." **Right:** high-dose folic acid antagonises SP; give 0.4 mg.
- **Trap:** "Misoprostol 800 µg to induce labour at term." **Right:** 25 µg; high doses cause tachysystole and uterine rupture.
- **Trap:** "Carboprost is the next step in an asthmatic with atonic PPH." **Right:** avoid carboprost (bronchospasm); use oxytocin, ergometrine (if normotensive), misoprostol, TXA, then mechanical/surgical measures.

Questions seniors ask

1. **"What do you check before giving magnesium?"** Respiratory rate ≥16, patellar reflexes present, urine output ≥30 mL/h; and have calcium gluconate at the bedside.
2. **"She has lost her knee jerks. What now?"** Withhold MgSO₄, check RR, SpO₂ and urine output; if breathing is depressed, give calcium gluconate 1 g IV over 10 min and support ventilation.
3. **"Which uterotonic is first line and why?"** Oxytocin: effective, few contraindications, and short half-life so it can be titrated; ergometrine and prostaglandins are second line.
4. **"Why run oxytocin in saline rather than dextrose?"** Oxytocin has an antidiuretic (ADH-like) effect; large volumes of dextrose cause water intoxication and hyponatraemia.
5. **"When do you give antenatal steroids and when not?"** 24–34 weeks, birth likely within 7 days, no maternal infection, and adequate newborn care; not routinely ≥34 weeks.
6. **"What antihypertensive would you use after delivery in a breastfeeding mother?"** Nifedipine MR, amlodipine, labetalol or enalapril; stop methyldopa (postnatal depression).
7. **"Which contraception for a woman on rifampicin?"** Copper IUD, LNG-IUS or DMPA; avoid relying on implants, COC or POP (enzyme induction).
8. **"What is the maximum dose of lignocaine you can infiltrate?"** 3 mg/kg plain (about 18 mL of 1% for a 60 kg woman), 7 mg/kg with adrenaline.

Flashcards

56 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Three mechanisms by which MgSO_4 prevents eclamptic fits	1. NMDA-receptor antagonism (raises seizure threshold) 2. Calcium antagonism → cerebral vasodilatation (relieves vasospasm) 3. Reduced blood-brain-barrier permeability (less cerebral oedema) It is not an antihypertensive.
2	How do you make 4 g of 20% MgSO_4 from the 50% ampoule?	8 mL of 50% (4 g) + 12 mL water for injection = 20 mL of 20%. Give IV over 5–10 min . (50% = 0.5 g/mL; 20% = 0.2 g/mL.)
3	Pritchard regimen: loading and maintenance	Loading: 4 g of 20% IV over 5–10 min + 10 g of 50% IM (5 g deep IM each buttock, each with 1 mL 2% lignocaine) = 14 g . Maintenance: 5 g of 50% IM every 4 h , alternate buttocks, for 24 h after delivery or last fit (whichever is later).
4	Zuspan regimen and dose for a recurrent fit on MgSO_4	Zuspan: 4 g IV over 5–20 min, then 1 g/h IV infusion . Recurrent fit: 2 g of 20% IV over 5 min (4 mL of 50% + 6 mL water), continue the regimen; if fits persist, look for another cause.
5	The three checks before every MgSO_4 dose (R-R-U)	• Respiratory rate $\geq 16/\text{min}$ • Reflexes (patellar) present (biceps if spinal) • Urine $\geq 30 \text{ mL/h}$ ($\geq 100 \text{ mL}$ in 4 h) RR < 16 or absent reflexes → withhold. Oliguria → withhold IM dose / halve infusion to 0.5 g/h.
6	Serum Mg^{2+} levels: therapeutic range and the toxicity ladder	• 2–3.5 mmol/L (4.8–8.4 mg/dL): therapeutic • 4–5 mmol/L: loss of patellar reflexes • 6–7.5 mmol/L: respiratory depression/paralysis • 7.5: conduction defects • >12 mmol/L: cardiac arrest "Reflexes, then breathing, then the heart."
7	Antidote to MgSO_4 toxicity: drug, dose, rate	Calcium gluconate 10%, 10 mL (1 g) IV slowly over 10 min , plus oxygen and bag-mask ventilation if breathing is depressed. Keep it at the bedside before the first MgSO_4 dose.
8	Why does oliguria matter so much on MgSO_4 ?	Mg is excreted almost entirely by the kidneys (half-life ~4 h normally). Oliguria ($< 30 \text{ mL/h}$) → accumulation → toxicity. Withhold/reduce the dose, check creatinine $\pm$ serum Mg, and do not give big fluid boluses (pulmonary oedema).
9	Absolute contraindication to MgSO_4 , and two important drug interactions	Myasthenia gravis (myasthenic crisis); also caution in heart block and renal failure. Interactions: neuromuscular blockers (prolonged block: tell the anaesthetist) and aminoglycosides (gentamicin: additive neuromuscular block). Nifedipine interaction is theoretical/rare: give both and monitor.
10	Hydralazine for acute severe hypertension: dilution and dose	Dilute 20 mg to 20 mL (1 mg/mL). Give 5 mg (5 mL) IV slowly over 2–5 min ; repeat 5–10 mg every 20–30 min if BP still $\geq 160/110$; max 20 mg per episode . Side effects: tachycardia, headache, hypotension → fetal distress.

#	Question	Answer
11	Nifedipine for acute severe hypertension: dose and the key safety rule	Nifedipine IR 10 mg orally, swallowed whole ; repeat 10–20 mg after 20–30 min if still $\geq 160/110$ (~30–40 mg in the first hour). Never sublingual or bitten : precipitous hypotension, fetal distress, MI/stroke.
12	Methyldopa: dose, 3 side effects, and why switch postpartum	250–750 mg 8-hourly (max 3 g/day) ; slow onset (never for acute BP). Side effects: sedation, depression, positive Coombs / haemolytic anaemia , hepatitis, postural hypotension. Switch within 2 days postpartum because of postnatal depression (use nifedipine MR, amlodipine or enalapril).
13	Antihypertensives to AVOID in pregnancy and why	• ACE inhibitors / ARBs: fetopathy (renal failure, oligohydramnios, skull hypoplasia, death) • Atenolol: fetal growth restriction • Diuretics: reduce plasma volume (only for pulmonary oedema) Enalapril is fine postpartum , including breastfeeding.
14	Antenatal corticosteroids: regimens, window and repeat course (WHO 2022)	Dexamethasone 6 mg IM 12-hourly $\times 4$ or betamethasone 12 mg IM 24-hourly $\times 2$. Window 24–34 weeks (Uganda usually 28–34), birth likely within 7 days, accurate GA, no infection, adequate newborn care. One repeat course if ≥ 7 days since the first and still at risk <34 weeks. Monitor glucose in diabetics.
15	Oxytocin for AMTSL: dose, route and timing?	10 IU IM (or IV slowly) within 1 minute of birth of the baby (after excluding another baby).
16	Oxytocin regimen for treatment of PPH?	10 IU IV slowly (or IM) , then infusion 20–40 IU in 1 L NS/RL , e.g. 20 IU in 1 L at 60 drops/min , maintenance 40 drops/min . Max about 3 L of oxytocin-containing fluid (water intoxication).
17	WHO MCPC oxytocin augmentation starting regimen and maximum?	2.5 IU in 500 mL NS/RL at 10 drops/min , increase by 10 drops/min every 30 min to max 60 drops/min , aiming for 3–5 contractions in 10 min, each >40 s . If inadequate: 5 IU in 500 mL from 30 drops/min. The 10 IU/500 mL step is for primigravidae only .
18	2.5 IU oxytocin in 500 mL at 10 drops/min (20 drops/mL set): how many mIU/min?	$5 \text{ mIU/mL} \times 0.5 \text{ mL/min} = \mathbf{2.5 \text{ mIU/min}}$. At 60 drops/min = 15 mIU/min .
19	Half-life of oxytocin and why it matters?	3–5 minutes . Stopping the drip reverses hyperstimulation within minutes; steady state takes ~30–40 min, so increase no more often than every 30 min ; a bolus wears off fast, so PPH needs an infusion.
20	First steps for oxytocin-induced tachysystole with fetal bradycardia?	Stop the oxytocin, left lateral , IV crystalloid (not dextrose), oxygen per protocol, call a senior, continuous FHR. If persisting: terbutaline 250 μg SC (or salbutamol infusion). No recovery $\rightarrow$ deliver .
21	Why can oxytocin cause water intoxication and how do you prevent it?	It resembles ADH (differs by 2 amino acids) and has an antidiuretic effect at high doses. Prevent by using NS/RL not 5% dextrose , avoiding prolonged high doses and limiting oxytocin fluid to ~3 L. Signs: headache, vomiting, confusion, fits , $\text{Na}^+ < 130 \text{ mmol/L}$.
22	Ergometrine: dose, maximum and 5 contraindications?	0.2 mg IM (or slow IV), repeat after 15 min then 4-hourly; max 1 mg (5 doses)/24 h . CI: hypertension, pre-eclampsia, cardiac disease, retained placenta, undelivered second twin (also boosted PIs such as ritonavir). Store 2–8 $^{\circ}\text{C}$.
23	Carboprost: dose, maximum and absolute contraindication?	250 μg deep IM every 15 min, max 8 doses (2 mg) . Asthma (bronchospasm). Side effects: diarrhoea, vomiting, fever, desaturation.
24	Misoprostol doses after birth: prevention vs treatment of PPH?	Prevention: 600 μg orally . Treatment: 800 μg sublingually . Expect shivering and fever.

#	Question	Answer
25	Misoprostol for induction of labour with a live fetus at term (no scar)?	25 µg vaginally every 6 h or 25 µg orally every 2 h (1 × 200 µg tablet in 200 mL water, give 25 mL). Not with a previous CS. Wait ≥4 h after the last dose before oxytocin.
26	Heat-stable carbocytin: dose and what it is NOT for?	100 µg IM or slow IV, single dose for prevention of PPH. Not for treatment of PPH, not for induction/augmentation, never before the baby is born.
27	Tranexamic acid in PPH: the "1-10-3-30" rule?	1 g IV over 10 min , within 3 h of birth, repeat 1 g after 30 min if still bleeding (or if it restarts within 24 h). WOMAN trial: no benefit after 3 h; each 15-min delay cuts benefit ~10%.
28	Methotrexate for ectopic: dose, follow-up and success rule?	50 mg/m² IM single dose. hCG day 4 and day 7 ; a ≥15% fall from day 4 to day 7 = success, then weekly to negative. <15% → second dose or surgery. Avoid folate, NSAIDs, alcohol, sun; contraception 3 months .
29	Name the three antibiotics (with doses) for puerperal sepsis or septic abortion, and what each covers.	Ampicillin 2 g IV 6-hourly (Gram-positives, streptococci) + gentamicin 5 mg/kg IV once daily (Gram-negatives, <i>E. coli</i>) + metronidazole 500 mg IV 8-hourly (anaerobes). IV until afebrile 48 h, then oral to complete ~7 days. Plus source control (evacuate RPOC, drain pus).
30	Gentamicin once-daily dose for a 60 kg woman, and the older alternative regimen?	5 mg/kg once daily = 300 mg IV/IM . Older regimen: 80 mg 8-hourly (only 240 mg/day, so it under-doses most women). Monitor urine output and creatinine ; hold if AKI.
31	Antibiotic for PPROM, dose and duration. Which antibiotic must you avoid and why?	Erythromycin 250 mg orally 6-hourly for 10 days (or until labour). Avoid co-amoxiclav : associated with necrotising enterocolitis in the preterm baby (ORACLE I).
32	When and what do you give for caesarean section antibiotic prophylaxis?	Single dose 15–60 min BEFORE skin incision (not after cord clamping): cefazolin 2 g IV (3 g if >120 kg) or ampicillin 2 g IV . Uganda often uses ceftriaxone 1–2 g IV ± metronidazole . Repeat if surgery >4 h or blood loss >1.5 L.
33	Match the antibiotic to its fetal/neonatal harm: tetracycline, chloramphenicol, streptomycin, sulfonamides near term, trimethoprim in the 1st trimester.	Tetracycline → teeth staining/bone. Chloramphenicol → grey baby syndrome. Streptomycin (aminoglycosides) → 8th-nerve deafness. Sulfonamides near term → kernicterus. Trimethoprim → folate antagonism, neural tube defects.
34	Syphilis in pregnancy: treatment for early vs late latent/unknown duration?	Benzathine penicillin 2.4 MU deep IM (1.2 MU each buttock). Early (<2 years): single dose. Late latent or unknown duration: weekly × 3. Treat the partner; treat/examine the baby at birth.
35	What is the Jarisch–Herxheimer reaction and why does it matter in pregnancy?	Fever, chills, myalgia, hypotension 2–24 h after the first penicillin dose for syphilis (endotoxin-like release from dying spirochaetes). In pregnancy it can cause contractions, preterm labour and fetal distress. Not an allergy : paracetamol, fluids, fetal monitoring; continue treatment .
36	IPTp-SP: dose, when to start, interval, and four reasons not to give it.	3 tablets SP (1,500/75 mg) by DOT, from 13 weeks, monthly (≥4 weeks apart) until delivery; aim ≥3 doses. Not in the 1st trimester , not with sulfa allergy , not with co-trimoxazole , and avoid with folate acid 5 mg (use 400 mcg).
37	Severe malaria in pregnancy: drug, dose and schedule?	IV (or IM) artesunate 2.4 mg/kg at 0, 12 and 24 h, then daily until she can take oral drugs (minimum 3 doses), then a full 3-day AL course. All trimesters. Check RBS 4-hourly; Hb at 7, 14, 28 days (delayed haemolysis).

#	Question	Answer
38	Why does quinine cause hypoglycaemia, and what do you do about it?	Quinine stimulates pancreatic insulin release ; pregnancy and parasites also consume glucose. Infuse in 5–10% dextrose, check RBS 4-hourly (hourly if unconscious), give 50% dextrose 25–50 mL IV if glucose <3 mmol/L.
39	What happened to the dolutegravir neural tube defect signal, and what is the current guidance?	Tsepamo (Botswana, 2018): ~ 0.9% NTDs with DTG at conception. With more data it fell to ~ 0.1% , the same as other ART . WHO (2019) and Uganda: TLD for all , including women who are or may become pregnant; give routine folic acid.
40	Infant ARV prophylaxis in Uganda: low-risk vs high-risk?	Start within 6 h of birth . Low risk (mother suppressed): NVP once daily × 6 weeks . High risk (non-suppressed/unknown VL, ART <4 weeks, diagnosed in labour or breastfeeding): AZT twice daily + NVP once daily × 6 weeks, then NVP alone × 6 weeks (12 weeks). ≥2.5 kg: 15 mg (1.5 mL) of each.
41	Anti-D after delivery in Uganda: dose, route, timing and who should NOT get it?	1,500 IU (300 mcg) IM within 72 h of delivery of an Rh-positive baby (some benefit up to 10 days). Not for Rh-positive women, women already sensitised (anti-D antibodies present), and never to the baby .
42	Indications for folic acid 5 mg (instead of 400 mcg)?	Previous NTD (or family history), epilepsy on antiepileptics , sickle cell disease/haemolytic anaemia , pre-existing diabetes , BMI ≥30 , malabsorption. In SCD continue throughout pregnancy ; beware the SP interaction (use proguanil or withhold around SP doses).
43	Four WHO MEC category 4 contraindications to the COC?	Any of: breastfeeding <6 weeks postpartum (not breastfeeding <21 days); migraine with aura ; BP ≥160/100 ; age ≥35 and smoking ≥15/day ; current/past VTE ; ischaemic heart disease or stroke; complicated valvular disease; current breast cancer; SLE with antiphospholipid antibodies; major surgery with prolonged immobilisation.
44	COC missed pill rule for ≥3 missed pills (≥48 h late)?	Take one now, carry on, condoms for 7 days . If in week 3 skip the pill-free break; if in week 1 with unprotected sex, consider emergency contraception . (1–2 missed: take it now, no back-up needed.)
45	Best contraception for a woman on rifampicin?	DMPA or an IUD (copper or LNG-IUS), all MEC 1. COC, POP and implants are MEC 3 (enzyme induction). Continue for 28 days after rifampicin stops.
46	Which ARV has no interaction with hormonal contraception, and which one does?	Dolutegravir (TLD): no interaction . Efavirenz lowers levonorgestrel/etonogestrel levels (implant failures reported in Uganda).
47	DMPA: doses, interval and two key side effects?	Depo-Provera 150 mg IM or Sayana Press 104 mg SC every 12–13 weeks (up to 4 weeks late allowed). Reversible bone density loss and delayed return of fertility (about 10 months after the last injection); also irregular bleeding and weight gain.
48	Emergency contraception options and time limits?	Copper IUD up to 5 days (>99%, best); levonorgestrel 1.5 mg within 72 h (3 mg with enzyme inducers or BMI >26/weight >70 kg); ulipristal 30 mg within 120 h .
49	Clomiphene: dose, monitoring and main risks?	50 mg daily days 2–6 , up to 150 mg ; ultrasound tracking at least in the first cycle ; max 6 cycles . Risks: multiple pregnancy (5–10%) , OHSS , visual disturbance, hot flushes, thin endometrium.
50	How do you suppress lactation after a stillbirth?	Cabergoline 1 mg orally, single dose, within 24 h of birth (avoid in hypertension/pre-eclampsia), plus a firm bra and analgesia. Not bromocriptine (stroke, MI, seizures).

#	Question	Answer
51	Maximum doses of lignocaine?	Plain: 3 mg/kg; with adrenaline: 7 mg/kg. 1% = 10 mg/mL, so about 18–20 mL of 1% for 60 kg. Aspirate first.
52	Dose of 20% lipid emulsion for local anaesthetic toxicity?	1.5 mL/kg IV bolus over 1 min , then 15 mL/kg/h infusion; repeat bolus up to twice at 5 min and double the rate to 30 mL/kg/h if needed; max 12 mL/kg .
53	Why avoid NSAIDs in late pregnancy?	From 20 weeks : fetal renal impairment and oligohydramnios . From 30 weeks : premature closure of the ductus arteriosus . Safe postpartum and in breastfeeding (if no asthma, renal impairment or PPH with AKI).
54	Enoxaparin and a spinal: timing rules?	Last prophylactic dose ≥12 h before; last treatment dose ≥24 h before; first dose ≥4 h after the spinal or catheter removal.
55	Warfarin embryopathy: window and features?	Exposure at 6–12 weeks : nasal hypoplasia and stippled epiphyses (chondrodysplasia punctata); later CNS defects and fetal haemorrhage. Switch to LMWH before 6 weeks. Safe in breastfeeding.
56	Four pharmacokinetic changes of pregnancy?	↑ volume of distribution (plasma volume +40–50%); ↑ GFR (~50%) and renal clearance; ↓ albumin (more free drug, e.g. phenytoin); ↑ hepatic metabolism of some drugs (lamotrigine via UGT); delayed gastric emptying (labour: erratic oral absorption, aspiration).

SELF-TEST

Single best answer questions

36 questions. The answers and explanations follow at the end.

1. A 23-year-old primigravida at 36 weeks with eclampsia is on Pritchard MgSO_4 at Mengo Hospital. Eight hours after loading, before the 4-hourly IM dose, she is drowsy with slurred speech. RR is 10/min, SpO_2 90% on air, patellar reflexes are absent, and she has passed 40 mL of urine in 4 hours. BP 150/100. What is the MOST appropriate immediate management?

- A Give the IM dose because she is still at risk of fits, then recheck in 1 hour
- B Withhold MgSO_4 , give oxygen and support ventilation, and give calcium gluconate 1 g IV over 10 min
- C Give furosemide 40 mg IV to improve urine output, then give half the IM dose
- D Give a 1 L fluid bolus of normal saline for oliguria
- E Give diazepam 10 mg IV to replace MgSO_4 as the anticonvulsant

2. A midwife at an HC IV asks you to help prepare the IV part of the Pritchard loading dose. The ward stocks only 10 mL ampoules of 50% MgSO_4 and water for injection. Which preparation is correct?

- A 20 mL of 50% MgSO_4 given IV undiluted over 5 min
- B 4 mL of 50% MgSO_4 + 16 mL water, given IV over 5–10 min
- C 8 mL of 50% MgSO_4 + 12 mL water, given IV over 5–10 min
- D 8 mL of 50% MgSO_4 + 12 mL water, given IV as a rapid push over 1 min
- E 10 mL of 50% MgSO_4 + 10 mL water, given IV over 20 min

3. A 30-year-old G3P2 at 34 weeks with severe pre-eclampsia has BP 172/114 on two readings 15 minutes apart. She received her MgSO_4 loading dose 30 minutes ago. She has no IV antihypertensive available, and she is alert and able to swallow. What is the MOST appropriate next step?

- A Methyldopa 500 mg orally now, and recheck BP in 1 hour
- B No antihypertensive is needed because MgSO_4 will lower the BP
- C Nifedipine immediate-release 10 mg sublingually
- D Furosemide 40 mg IV
- E Nifedipine immediate-release 10 mg orally, swallowed, recheck BP in 20–30 min

4. A 26-year-old at 31 weeks presents with regular painful contractions and cervix 3 cm dilated. She is afebrile, membranes are intact, there is no bleeding, and the FHR is 140/min. Her GA is confirmed by a 12-week scan. Your HC IV has no NICU and the regional referral hospital (with neonatal care) is 2 hours away. What is the BEST plan?

- A Salbutamol IV infusion for 48 hours and admit locally
- B Dexamethasone 6 mg IM and nifedipine 20 mg orally, then refer in utero with notes of dose times
- C Await spontaneous delivery locally; steroids are only useful if the full course is completed
- D Indomethacin 100 mg rectally and continue for 7 days
- E MgSO_4 as the tocolytic of choice, then observe

5. A 28-year-old with eclampsia is on a Zuspan MgSO_4 infusion at 1 g/h after delivery. Her RR is 18/min and patellar reflexes are present, but her urine output over the last 2 hours is 20 mL/h. Creatinine is 140 $\mu\text{mol/L}$. BP 148/96, chest clear. What is the MOST appropriate action regarding MgSO_4 ? Advanced

- A Stop MgSO_4 completely and switch to diazepam
- B Continue 1 g/h and give a 2 L fluid challenge
- C Give calcium gluconate 1 g IV now
- D Reduce the infusion to 0.5 g/h, continue hourly R-R-U checks and check serum Mg if available
- E Increase to 2 g/h because renal involvement means more severe disease

- 6.** A 32-year-old woman with a history of chronic hypertension and moderate persistent asthma presents at 33 weeks with BP 168/112 and proteinuria 3+. Nifedipine and hydralazine are in stock; labetalol has just been delivered to the pharmacy. Which drug should be AVOIDED for acute BP control in this woman?
- A Labetalol IV
 - B Hydralazine IV
 - C Nifedipine IR oral
 - D MgSO_4
 - E Oxytocin
- 7.** On day 2 after a caesarean section for pre-eclampsia, a 25-year-old breastfeeding mother is on methyldopa 500 mg 8-hourly. Her BP is 152/98. She says she feels low and tearful. What is the MOST appropriate change to her antihypertensive?
- A Increase methyldopa to 750 mg 8-hourly
 - B Add hydrochlorothiazide to methyldopa
 - C Stop methyldopa and start nifedipine MR 20 mg 12-hourly (or enalapril)
 - D Stop all antihypertensives because BP always normalises after delivery
 - E Start atenolol 50 mg daily
- 8.** A 29-year-old with severe pre-eclampsia on Pritchard MgSO_4 is taken for emergency caesarean section under general anaesthesia. The anaesthetist plans suxamethonium and a non-depolarising muscle relaxant, and gentamicin is prescribed postoperatively. Which statement BEST describes the interaction concern? Advanced
- A MgSO_4 antagonises muscle relaxants, so higher doses are needed
 - B Gentamicin reverses magnesium toxicity
 - C There is no interaction; MgSO_4 acts only on the brain
 - D MgSO_4 should be switched to phenytoin before any general anaesthetic
 - E MgSO_4 potentiates neuromuscular blockade, and gentamicin adds to it, so reduce relaxant doses and monitor neuromuscular function and breathing
- 9.** A woman with eclampsia was given the full Pritchard loading dose at an HC III at 10:00 and referred to Mengo Hospital, arriving at 13:00. She has had no further fits, RR 18/min, reflexes present, and 150 mL of urine in the catheter bag since 10:00. The referral note says the next dose was not sent. What is the MOST appropriate MgSO_4 plan?
- A Repeat the full 14 g loading dose because she has changed facility
 - B Give 5 g of 50% IM at 14:00 (4 h after loading), then 4-hourly for 24 h after delivery or the last fit
 - C Give no more MgSO_4 because she has not fitted again
 - D Start diazepam maintenance because the loading dose was given elsewhere
 - E Give 2 g of 20% IV now and every hour until delivery
- 10.** A 26-year-old G2P1 at term is on oxytocin augmentation (5 IU in 500 mL, now 50 drops/min) at Mengo Hospital for slow progress at 6 cm. The midwife counts 6 contractions in 10 minutes, each lasting 70 seconds, and the fetal heart rate is 90 bpm for the last 3 minutes. What is the MOST appropriate immediate action?
- A Reduce the infusion to 30 drops/min and reassess in 30 minutes
 - B Perform an immediate artificial rupture of membranes
 - C Stop the oxytocin infusion and turn her onto her left side
 - D Give terbutaline 250 μg SC and continue the oxytocin at the same rate
 - E Proceed directly to caesarean section without any intrauterine resuscitation

11. A 30-year-old P3 with severe pre-eclampsia (on MgSO₄) and known asthma has an atonic PPH 20 minutes after vaginal birth. She received oxytocin 10 IU IM for AMTSL, and an oxytocin infusion and tranexamic acid 1 g IV have just been started. The uterus remains boggy despite massage. BP is 158/104. Which additional uterotonic is MOST appropriate?

- A Misoprostol 800 µg sublingually
- B Ergometrine 0.2 mg IV
- C Carboprost 250 µg deep IM
- D Syntometrine 1 mL IM
- E Heat-stable carbetocin 100 µg IV

12. A 32-year-old G3P2 at 41 weeks 3 days has had one previous lower-segment caesarean section for fetal distress. She wants a vaginal birth. Membranes are intact, the fetus is cephalic with a normal heart rate, and her Bishop score is 3. She is at a regional referral hospital with theatre available. What is the MOST appropriate method of cervical ripening?

- A Misoprostol 25 µg vaginally every 6 hours
- B Misoprostol 25 µg orally every 2 hours
- C Oxytocin 2.5 IU in 500 mL at 10 drops/min with intact membranes
- D Foley catheter balloon ripening, followed by ARM and carefully titrated oxytocin
- E Misoprostol 200 µg vaginally once

13. A primigravida is on an oxytocin infusion of 5 IU in 500 mL normal saline running at 40 drops/min through a standard giving set (20 drops/mL). The intern is asked by the consultant on the ward round what dose she is receiving. What is the correct answer?

- A 5 mIU/min
- B 10 mIU/min
- C 15 mIU/min
- D 40 mIU/min
- E 20 mIU/min

14. A 28-year-old nulliparous woman at a private hospital in Kampala has an unruptured right tubal ectopic pregnancy (mass 22 mm, no heartbeat), minimal pain, stable vital signs and a day-1 hCG of 1,300 IU/L. She received methotrexate 50 mg/m² IM. Her hCG is 1,650 IU/L on day 4 and 1,560 IU/L on day 7. She remains well with mild pain. What is the MOST appropriate next step? Advanced

- A Reassure her that treatment has worked and check hCG weekly
- B Give a second dose of methotrexate 50 mg/m² IM and repeat the hCG monitoring
- C Start folic acid 5 mg daily to reduce side effects
- D Perform immediate laparotomy for presumed rupture
- E Give misoprostol 800 µg vaginally to expel the ectopic

15. A 24-year-old G2P1 at 34 weeks is confirmed by two ultrasound scans to have an intrauterine fetal death. She has no uterine scar, BP 118/76, platelets and bedside clotting time are normal, and her Bishop score is 2. Mifepristone is not available. Which regimen is MOST appropriate to induce labour?

- A Misoprostol 400 µg sublingually every 4–6 hours
- B Misoprostol 200 µg vaginally every 4 hours
- C Misoprostol 800 µg vaginally once, repeated after 3 hours
- D Misoprostol 25 µg vaginally every 6 hours
- E Oxytocin 10 IU IV bolus followed by 20 IU in 1 L at 60 drops/min

- 16.** A 35-year-old P5 delivers at an HC IV. Forty minutes after birth she has lost 700 mL measured on the calibrated drape; the uterus is soft. The midwife starts uterine massage and an oxytocin infusion. The HC IV has tranexamic acid in stock. What is the MOST appropriate way to use it?
- A Give tranexamic acid only if bleeding continues after all uterotonics have failed
 - B Give 1 g IV as a rapid push over 1 minute
 - C Give 1 g IV over 10 minutes now, with a second 1 g if bleeding continues after 30 minutes
 - D Wait for a bedside clotting test to show no clot at 7 minutes before giving it
 - E Give 1 g orally 8-hourly for 5 days
- 17.** A 19-year-old primigravida has been on oxytocin augmentation for 14 hours at a district hospital; the oxytocin was diluted in 5% dextrose and she has received about 5 L of IV fluid. She becomes confused, vomits and has a generalised seizure. BP is 124/78 and urine protein is negative. Which investigation result would MOST support the likely cause? Advanced
- A Serum sodium 118 mmol/L
 - B Platelet count $60 \times 10^9/L$
 - C Random blood glucose 2.1 mmol/L
 - D Serum magnesium 4.5 mmol/L
 - E Haemoglobin 7.5 g/dL
- 18.** A 29-year-old G2P1 at 9 weeks by dates is found on transvaginal scan to have an embryo with CRL 18 mm and no cardiac activity (confirmed by a second sonographer). She is not bleeding, is haemodynamically stable, and chooses medical management. The hospital stocks both mifepristone and misoprostol. Which regimen is MOST appropriate?
- A Misoprostol 25 µg vaginally every 6 hours
 - B Misoprostol 600 µg orally once
 - C Oxytocin 20 IU in 1 L normal saline at 60 drops/min
 - D Ergometrine 0.2 mg IM, then MVA if not complete
 - E Mifepristone 200 mg orally, then misoprostol 800 µg vaginally or sublingually 1–2 days later
- 19.** A 24-year-old G2P1 at 31 weeks presents to Mengo with a gush of clear fluid 6 hours ago. Sterile speculum shows pooling of liquor. Temperature 36.8°C, pulse 84, uterus non-tender, FHR 140 bpm, no contractions. Dexamethasone has been started. Which antibiotic plan is MOST appropriate?
- A Co-amoxiclav 625 mg orally 8-hourly for 10 days
 - B No antibiotics unless she develops fever
 - C Erythromycin 250 mg orally 6-hourly for 10 days or until labour
 - D Ampicillin 2 g IV 6-hourly plus gentamicin 5 mg/kg IV daily
 - E Doxycycline 100 mg orally 12-hourly for 7 days
- 20.** A 28-year-old P2, day 3 after an emergency caesarean section for obstructed labour at an HC IV, has a temperature of 39.0°C, pulse 116, BP 104/66, a tender uterus and foul-smelling lochia. She weighs 60 kg, urine output is good and creatinine is 62 µmol/L. Blood cultures have been taken. Which antibiotic regimen is MOST appropriate?
- A Ampicillin 2 g IV 6-hourly + gentamicin 300 mg IV once daily + metronidazole 500 mg IV 8-hourly
 - B Oral amoxicillin 500 mg 8-hourly + metronidazole 400 mg 8-hourly for 5 days
 - C Ceftriaxone 1 g IV once only, then review the culture result
 - D Ampicillin 2 g IV 6-hourly + gentamicin 80 mg IV once daily
 - E Ciprofloxacin 400 mg IV 12-hourly + doxycycline 100 mg 12-hourly

21. A 26-year-old G3P2 at 20 weeks attends ANC. She is HIV-positive on TLD with a suppressed viral load and takes co-trimoxazole 960 mg once daily. She has not had any IPTp. What is the MOST appropriate malaria prevention?

- A Give IPTp-SP 3 tablets today and monthly until delivery
- B Give IPTp-SP but stop co-trimoxazole for one week around each dose
- C Weekly chloroquine prophylaxis
- D Continue co-trimoxazole and an insecticide-treated net; no SP
- E Daily doxycycline 100 mg

22. A 22-year-old primigravida at 10 weeks is brought in confused and vomiting, temperature 39.4°C, RBS 4.8 mmol/L, Hb 8.2 g/dL. The mRDT is positive for *P. falciparum*. She weighs 55 kg. What is the MOST appropriate treatment?

- A Oral artemether-lumefantrine 4 tablets at 0 and 8 h, then 12-hourly for 2 days
- B IV artesunate 132 mg (2.4 mg/kg) at 0, 12 and 24 h, then daily until she can take oral treatment
- C Oral quinine 600 mg 8-hourly + clindamycin 300 mg 8-hourly for 7 days
- D IV quinine 1,100 mg (20 mg/kg) as a bolus over 5 minutes
- E Wait until 13 weeks, then give IPTp-SP

23. A 30-year-old G2P1 at 28 weeks has been on TLD for 4 months. Her viral load is 1,800 copies/mL. She says she never misses a dose: every morning on an empty stomach she swallows her TLD together with her ferrous sulphate-folic acid tablet and the calcium tablet she was given for pre-eclampsia prevention. What is the MOST appropriate next step? Advanced

- A Switch to a second-line protease inhibitor regimen today
- B Stop the calcium and iron for the rest of the pregnancy
- C Stop dolutegravir because of the neural tube defect risk
- D Add a second daily dose of dolutegravir 50 mg
- E Counsel her to take TLD 2 h before (or 6 h after) iron and calcium, or all with food, and repeat the VL after enhanced adherence counselling

24. A 19-year-old primigravida at 18 weeks has a reactive rapid syphilis test at booking. She has no symptoms and has never been treated. She is not allergic to penicillin. What is the MOST appropriate management?

- A Erythromycin 500 mg orally 6-hourly for 14 days
- B Wait for an RPR titre before deciding whether to treat
- C Benzathine penicillin 2.4 MU IM weekly for 3 weeks, and treat her partner
- D Benzathine penicillin 2.4 MU IM single dose; the partner does not need treatment
- E Doxycycline 100 mg orally 12-hourly for 28 days

25. Six hours after her first injection of benzathine penicillin for secondary syphilis, a 25-year-old at 30 weeks has a temperature of 38.4°C, myalgia, headache and irregular mild uterine tightenings. There is no rash, wheeze or angio-oedema. BP 110/70, FHR 158 bpm. What is the MOST appropriate management? Advanced

- A Paracetamol, fluids, CTG/fetal monitoring and observation; continue the penicillin course as planned
- B Label her penicillin-allergic and switch to erythromycin
- C Give adrenaline 0.5 mg IM and hydrocortisone IV
- D Start nifedipine tocolysis and dexamethasone immediately
- E Start ampicillin and gentamicin for presumed chorioamnionitis

26. A 27-year-old P1, blood group O RhD-negative, has just had a normal delivery at Mengo. Her indirect Coombs test at 28 weeks was negative. The cord blood group is A RhD-positive. She received no anti-D during pregnancy. What is the MOST appropriate action?

- A No action, because ABO incompatibility protects her
- B Give anti-D 1,500 IU IM to the baby
- C Wait until the next pregnancy and give anti-D at 28 weeks
- D Give anti-D immunoglobulin 1,500 IU (300 mcg) IM to the mother within 72 hours
- E Give anti-D only if she plans more children and has a positive Kleihauer test

27. A 23-year-old G1 at 24 weeks complains of intense vulval itching and a thick white curd-like discharge without odour. Vaginal pH is 4.2 and the wet mount shows hyphae. What is the MOST appropriate treatment?

- A Fluconazole 150 mg orally, repeated weekly until delivery
- B Clotrimazole 100 mg vaginal pessary nightly for 7 days
- C Metronidazole 2 g orally as a single dose
- D Doxycycline 100 mg 12-hourly for 7 days
- E No treatment, because candidiasis is normal in pregnancy

28. A 28-year-old P2 living with HIV, on TLD, has just started rifampicin-based treatment for pulmonary TB at a Kampala HC IV. She has been using the combined pill and wants to avoid pregnancy. Which contraceptive advice is MOST appropriate?

- A Continue her 30 mcg combined pill; rifampicin does not affect it
- B Insert a Jadelle implant today
- C Give DMPA 150 mg IM every 12 weeks (or offer a copper IUD)
- D Switch to a levonorgestrel progestogen-only pill
- E Double her combined pill to two tablets daily

29. During repair of a second-degree tear, an intern has infiltrated 25 mL of 2% lignocaine into a 55 kg woman. She says her lips are tingling and she has a metallic taste; one minute later she has a generalised tonic-clonic seizure. Pulse 110, BP 150/90. What is the MOST appropriate immediate management? Advanced

- A Stop injecting, call for help, give 100% oxygen and support the airway, and give IV diazepam
- B Give a magnesium sulphate loading dose for eclampsia
- C Finish the repair quickly so that she can be positioned
- D Give naloxone 0.4 mg IV
- E Give lignocaine 1 mg/kg IV to prevent ventricular arrhythmias

30. A 36-year-old P4 who smokes 20 cigarettes a day and has migraine with visual aura asks for contraception. BP 132/84. She does not want more children for now but is unsure about sterilisation. Which method is MOST suitable?

- A Combined oral contraceptive pill
- B Combined contraceptive patch
- C Combined vaginal ring
- D Combined pill taken continuously without breaks
- E Copper IUD or LNG-IUS

- 31.** A term baby is born at Mengo Hospital 2.5 hours after the mother received pethidine 100 mg IM. The baby is floppy and not breathing after drying and stimulation. The heart rate is 90/min. What is the NEXT step?
- A Give naloxone 0.1 mg/kg IM immediately
 - B Start bag-and-mask ventilation with room air
 - C Give naloxone to the mother
 - D Start chest compressions
 - E Give IV dextrose 10% 2 mL/kg
- 32.** A 24-year-old G2P1 at 10 weeks has vomited for 3 weeks. She is dehydrated, drowsy and confused, with horizontal nystagmus and an unsteady gait. Urine ketones 3+. Before you start IV fluids containing dextrose, which drug MUST be given?
- A Metoclopramide 10 mg IV
 - B Ondansetron 8 mg IV
 - C Hydrocortisone 100 mg IV
 - D Thiamine IV
 - E Potassium chloride 40 mmol IV bolus
- 33.** A 30-year-old at 38 weeks, on enoxaparin 40 mg SC daily for a previous DVT, took her dose 6 hours ago. She now needs a caesarean for breech presentation in early labour; mother and baby are stable (Category 3). What is the BEST plan for anaesthesia? Advanced
- A Proceed now with a spinal, since prophylactic LMWH is safe at any time
 - B Give protamine to fully reverse the enoxaparin, then spinal
 - C Wait until at least 12 hours after the last dose, then spinal, if mother and baby remain stable
 - D Proceed with general anaesthesia now as the routine choice
 - E Check the APTT and proceed with a spinal if it is normal
- 34.** A newborn has a flat, hypoplastic nose, and X-rays show stippled epiphyses. The mother has a mechanical mitral valve and continued her anticoagulant throughout the first trimester. Which drug is MOST likely responsible?
- A Warfarin
 - B Enoxaparin
 - C Unfractionated heparin
 - D Aspirin
 - E Phenytoin
- 35.** A 26-year-old with juvenile myoclonic epilepsy, well controlled on sodium valproate for 8 years, tells you she and her husband want to start trying for a baby. She is on no contraception. What is the MOST appropriate advice? Advanced
- A Stop valproate today and start trying to conceive
 - B Continue valproate and take folic acid 400 mcg daily
 - C Continue valproate but avoid pregnancy until she is 30
 - D Double the valproate dose because levels fall in pregnancy
 - E Use contraception meanwhile, start folic acid 5 mg daily, and refer to a neurologist to switch to lamotrigine or levetiracetam before conceiving

36. A 29-year-old G3P2 at 32 weeks presents with musculoskeletal low back pain after carrying water. She asks for the diclofenac her neighbour uses. There are no red flags. What is the BEST analgesic plan?

- A Diclofenac 50 mg orally 8-hourly for 5 days
- B Paracetamol 1 g orally 6-hourly (max 4 g/day), with physiotherapy advice
- C Ibuprofen 400 mg orally 8-hourly, as it is safer than diclofenac
- D Codeine 60 mg 6-hourly for 2 weeks
- E Pethidine 100 mg IM

Answers & explanations

1. B · Withhold MgSO₄, give oxygen and support ventilation, and give calcium gluconate 1 g IV over 10 min

B is correct: absent reflexes, RR 10/min and falling SpO₂ in an oliguric woman = **magnesium toxicity** (respiratory depression). Stop MgSO₄, call for help, oxygen and bag-mask support, and **calcium gluconate 10% 10 mL (1 g) IV over 10 min**. **A** is wrong: giving more magnesium could cause respiratory or cardiac arrest. **C** is wrong: furosemide is a senior decision after stabilisation, not the first step, and no further MgSO₄ should be given now. **D** is wrong: oliguria in pre-eclampsia is usually renal vasospasm; a litre bolus risks pulmonary oedema and does not treat the respiratory depression. **E** is wrong: diazepam would worsen respiratory depression and is inferior to MgSO₄.

2. C · 8 mL of 50% MgSO₄ + 12 mL water, given IV over 5–10 min

C is correct: 4 g of 50% (0.5 g/mL) = 8 mL; made up to 20 mL gives **20% (0.2 g/mL)**, given over **5–10 min**. **A** is wrong: 20 mL of 50% is **10 g**, and 50% should never be given as an IV bolus. **B** is wrong: 4 mL of 50% contains only 2 g (that is the recurrent-fit dose, and it would be 10% not 20%). **D** is wrong: the right mixture but too fast; rapid IV magnesium causes hypotension, arrhythmia and arrest. **E** is wrong: 10 mL of 50% is 5 g (25%), not the 4 g loading dose.

3. E · Nifedipine immediate-release 10 mg orally, swallowed, recheck BP in 20–30 min

E is correct: BP ≥160/110 is an emergency to treat within 30–60 min; **oral nifedipine IR 10 mg swallowed** works in 10–20 min and needs no IV access; repeat 10–20 mg if still ≥160/110. **A** is wrong: methyldopa takes 4–6 h and has no role in acute control. **B** is wrong: MgSO₄ is an anticonvulsant, not an antihypertensive. **C** is wrong: sublingual/bitten nifedipine causes precipitous, uncontrolled hypotension and fetal distress. **D** is wrong: diuretics reduce the contracted plasma volume in pre-eclampsia and are used only for pulmonary oedema.

4. B · Dexamethasone 6 mg IM and nifedipine 20 mg orally, then refer in utero with notes of dose times

B is correct: start **dexamethasone** (first of four 6 mg doses), use **nifedipine** to buy time, and **transfer the mother in utero** to a unit with newborn care; the uterus is the best incubator. **A** is wrong: betamimetics are no longer recommended (tachycardia, pulmonary oedema, hyperglycaemia). **C** is wrong: steroids help even if birth occurs before the course is complete, and a 31-week baby should be born where it can be cared for. **D** is wrong: indomethacin must not exceed 48 h (ductal closure, oligohydramnios) and is specialist use. **E** is wrong: MgSO₄ is not an effective tocolytic; it is given for **neuroprotection** when birth <32 weeks is expected within 24 h.

5. D · Reduce the infusion to 0.5 g/h, continue hourly R-R-U checks and check serum Mg if available

D is correct: with **oliguria** and rising creatinine but normal RR and reflexes, magnesium is accumulating; **halve the infusion (0.5 g/h)**, keep hourly checks, measure serum Mg if possible, and restrict fluids. **A** is wrong: she still needs seizure prophylaxis and has no signs of toxicity; diazepam is inferior. **B** is wrong: oliguria in pre-eclampsia is not from dehydration, and 2 L risks pulmonary oedema; full-dose magnesium will accumulate. **C** is wrong: calcium gluconate is for toxicity (respiratory depression/arrest), which she does not have. **E** is wrong: a higher dose in renal impairment will cause toxicity.

6. A · Labetalol IV

A is correct: labetalol is a non-selective β -blocker and can precipitate **bronchospasm in asthma**; it is also avoided in heart failure. **B** is wrong: hydralazine is appropriate (watch tachycardia and hypotension). **C** is wrong: oral nifedipine (swallowed) is a first-line option. **D** is wrong: MgSO_4 is indicated for seizure prophylaxis in severe pre-eclampsia and is not an antihypertensive; asthma is not a contraindication. **E** is wrong: oxytocin is not an antihypertensive and is the correct uterotonic in hypertension (ergometrine is the one to avoid).

7. C · Stop methyldopa and start nifedipine MR 20 mg 12-hourly (or enalapril)

C is correct: methyldopa should be **switched within 2 days postpartum** because it causes **depression**; nifedipine MR, amlodipine or **enalapril** (safe in breastfeeding) are preferred. **A** is wrong: increasing methyldopa worsens the mood risk. **B** is wrong: diuretics are not first line and may reduce milk supply. **D** is wrong: postpartum BP often peaks on days 3–6; stopping abruptly risks stroke and eclampsia. **E** is wrong: atenolol is not a preferred postpartum agent and has no advantage here.

8. E · MgSO_4 potentiates neuromuscular blockade, and gentamicin adds to it, so reduce relaxant doses and monitor neuromuscular function and breathing

E is correct: magnesium reduces acetylcholine release at the neuromuscular junction, **potentiating both depolarising and non-depolarising blockers**; **aminoglycosides** have an additive effect. Use lower relaxant doses, monitor with a nerve stimulator, and watch breathing postoperatively. **A** is wrong: the effect is potentiation, not antagonism. **B** is wrong: calcium gluconate is the antidote; gentamicin worsens weakness. **C** is wrong: magnesium acts at the neuromuscular junction too, which is why reflexes disappear in toxicity. **D** is wrong: phenytoin is inferior (Collaborative Eclampsia Trial) and switching is not needed.

9. B · Give 5 g of 50% IM at 14:00 (4 h after loading), then 4-hourly for 24 h after delivery or the last fit

B is correct: the maintenance dose is due **4 h after loading** (14:00); her R-R-U checks are satisfactory (urine ~37 mL/h), so continue **5 g IM 4-hourly** for 24 h after delivery or the last fit. **A** is wrong: a second loading dose risks toxicity; always read the referral note. **C** is wrong: prophylaxis must continue for 24 h after delivery/last fit, since fits often recur. **D** is wrong: diazepam is inferior and there is no reason to change. **E** is wrong: 2 g IV is only for a **recurrent fit**, not a maintenance schedule.

10. C · Stop the oxytocin infusion and turn her onto her left side

C is correct: this is tachysystole (>5 contractions in 10 min, each >60 s) with fetal bradycardia. Oxytocin has a half-life of 3–5 minutes, so stopping it (not just slowing it) plus left lateral positioning, IV crystalloid and a senior review is the first step; tocolysis follows if it persists. **A** is wrong: slowing the rate leaves a hyperstimulating dose running while the fetus is compromised. **B** is wrong: ARM does nothing for hyperstimulation and risks cord prolapse. **D** is wrong: giving a tocolytic while continuing the drug causing the problem is contradictory. **E** is wrong: intrauterine resuscitation takes seconds and often restores the FHR; CS is for failure to recover, not the first step.

11. A · Misoprostol 800 μg sublingually

A is correct: misoprostol (PGE_1) is safe in hypertension and does not cause bronchospasm; 800 μg SL is the PPH treatment dose. **B** is wrong: ergometrine is contraindicated in pre-eclampsia/hypertension (stroke, eclampsia). **C** is wrong: carboprost ($\text{PGF}_{2\alpha}$) causes bronchospasm and is contraindicated in asthma. **D** is wrong: Syntometrine contains ergometrine 0.5 mg, so the same contraindication applies. **E** is wrong: carbetocin is licensed and recommended for prevention only, not treatment, and adds nothing to an oxytocin receptor already being stimulated.

12. D · Foley catheter balloon ripening, followed by ARM and carefully titrated oxytocin

D is correct: with a uterine scar, mechanical ripening with a Foley catheter (no drug, low tachysystole risk) followed by ARM and low-dose oxytocin under senior supervision is the recommended route; she must be monitored for scar pain, bleeding and FHR changes. A and B are wrong: misoprostol for third-trimester induction in a scarred uterus increases the risk of uterine rupture and is avoided (WHO, Uganda practice). C is wrong: oxytocin on an unripe cervix (Bishop 3) with intact membranes usually fails and prolongs exposure; ripen first. E is wrong: 200 µg is a mid-trimester dose, dangerous in the third trimester in any uterus and especially with a scar.

13. E · 20 mIU/min

E is correct: 5 IU in 500 mL = 5,000 mIU ÷ 500 mL = 10 mIU/mL. 40 drops/min ÷ 20 drops/mL = 2 mL/min. 10 × 2 = 20 mIU/min. A is wrong: this confuses IU in the bag with mIU/min. B is wrong: this is the concentration per mL (10 mIU/mL), not per minute, or the rate at 20 drops/min. C is wrong: 15 mIU/min would be 30 drops/min of this solution (or 60 drops/min of 2.5 IU/500 mL). D is wrong: 40 is the drop rate in drops/min, not the dose; it would be 40 mIU/min only if the bag held 20 mIU/mL (10 IU in 500 mL).

14. B · Give a second dose of methotrexate 50 mg/m² IM and repeat the hCG monitoring

B is correct: the fall from day 4 (1,650) to day 7 (1,560) is only about 5%, less than the 15% required; she is stable with mild pain, so a second dose of methotrexate (or surgery, depending on her choice and follow-up capacity) is indicated. A is wrong: the day 1–4 rise is expected, but success is judged by a ≥15% fall between day 4 and day 7, which has not happened. C is wrong: folic acid antagonises methotrexate and causes treatment failure; it is on the avoid list. D is wrong: she is stable with mild pain and no signs of rupture, so emergency laparotomy is not indicated now (though it would be if she deteriorated). E is wrong: misoprostol has no role in ectopic pregnancy; the pregnancy is outside the uterus.

15. D · Misoprostol 25 µg vaginally every 6 hours

D is correct: after 28 weeks the uterus is very sensitive to prostaglandins, so the FIGO/WHO regimen is 25 µg vaginally 6-hourly (or 25 µg orally 2-hourly), never more than 50 µg per dose. A is wrong: 400 µg is the 14–28-week dose; at 34 weeks it risks hyperstimulation and rupture. B is wrong: 200 µg is a mid-trimester dose and should never be given in the third trimester with the fetus in utero. C is wrong: 800 µg is the first-trimester missed-miscarriage dose (or PPH treatment after birth). E is wrong: that is the PPH regimen; an oxytocin bolus and high-dose infusion on an unripe cervix is ineffective and dangerous, and induction oxytocin must be low-dose and titrated.

16. C · Give 1 g IV over 10 minutes now, with a second 1 g if bleeding continues after 30 minutes

C is correct: in the WOMAN trial and E-MOTIVE bundle, TXA 1 g IV over 10 minutes is given as soon as PPH is diagnosed, within 3 hours of birth, for any cause, with a second 1 g after 30 minutes if bleeding continues. A is wrong: benefit falls about 10% with every 15-minute delay; TXA is given together with uterotonics, not after they fail. B is wrong: rapid injection causes hypotension; give it over 10 minutes (1 mL/min). D is wrong: TXA works by preventing breakdown of existing clot and should not wait for evidence of coagulopathy. E is wrong: that is the heavy menstrual bleeding style of dosing, far too slow for acute haemorrhage.

17. A · Serum sodium 118 mmol/L

A is correct: prolonged oxytocin (which has an ADH-like antidiuretic effect) given in large volumes of 5% dextrose (free water) causes water intoxication with dilutional hyponatraemia, presenting with headache, vomiting, confusion and seizures. Management: stop the oxytocin, restrict fluids, check U&E and treat the seizure. B is wrong: thrombocytopenia suggests HELLP, but she is normotensive with no proteinuria. C is wrong: large volumes of dextrose would cause hyperglycaemia rather than hypoglycaemia. D is wrong: she has not received MgSO₄, and magnesium toxicity causes loss of reflexes and respiratory depression, not seizures. E is wrong: anaemia alone does not cause seizures and does not explain the story.

18. E · Mifepristone 200 mg orally, then misoprostol 800 µg vaginally or sublingually 1–2 days later

E is correct: for missed miscarriage under 14 weeks, WHO 2022 prefers mifepristone 200 mg orally followed 1–2 days later by misoprostol 800 µg vaginally/sublingually/buccally, which gives higher complete evacuation than misoprostol alone. A is wrong: 25 µg is the third-trimester/term dose and is far too small at 9 weeks. B is wrong: 600 µg orally once is the regimen for incomplete miscarriage (open os, partly expelled), not a missed miscarriage with a closed cervix. C is wrong: the uterus has few oxytocin receptors before about 20 weeks, so oxytocin is ineffective. D is wrong: ergometrine is used for bleeding while awaiting evacuation, not to expel a missed miscarriage, and would not empty the uterus.

19. C · Erythromycin 250 mg orally 6-hourly for 10 days or until labour

C is correct: erythromycin 250 mg 6-hourly for 10 days prolongs latency and reduces chorioamnionitis and neonatal infection after PPROM (ORACLE I). A is wrong: co-amoxiclav is linked to necrotising enterocolitis in preterm babies. B is wrong: antibiotics are indicated in PPROM even without infection signs. D is wrong: ampicillin + gentamicin is for chorioamnionitis, which she does not have. E is wrong: doxycycline is a tetracycline and is avoided in pregnancy (teeth and bone).

20. A · Ampicillin 2 g IV 6-hourly + gentamicin 300 mg IV once daily + metronidazole 500 mg IV 8-hourly

A is correct: post-CS endometritis/puerperal sepsis is polymicrobial and needs IV cover for Gram-positives (ampicillin), Gram-negatives (gentamicin 5 mg/kg = 300 mg once daily) and anaerobes (metronidazole). B is wrong: she is septic (fever, tachycardia), so she needs IV therapy first; oral step-down comes after 48 h afebrile. C is wrong: a single dose with no anaerobic cover is inadequate; do not wait for cultures. D is wrong: it misses anaerobes and 80 mg once daily is a gross under-dose. E is wrong: fluoroquinolones are not first line and doxycycline adds nothing for puerperal sepsis; she may also be breastfeeding.

21. D · Continue co-trimoxazole and an insecticide-treated net; no SP

D is correct: co-trimoxazole itself protects against malaria, and combining it with SP (another sulfonamide/antifolate) adds no benefit and raises the risk of Stevens–Johnson syndrome and marrow toxicity. She needs an ITN. A is wrong: SP must not be given with co-trimoxazole. B is wrong: co-trimoxazole prophylaxis should not be interrupted; it protects against PCP, toxoplasmosis and bacterial infection. C is wrong: chloroquine is not used for prophylaxis in Uganda because of resistance. E is wrong: doxycycline is contraindicated in pregnancy.

22. B · IV artesunate 132 mg (2.4 mg/kg) at 0, 12 and 24 h, then daily until she can take oral treatment

B is correct: confusion means severe (cerebral) malaria, and IV artesunate 2.4 mg/kg at 0, 12, 24 h then daily is the treatment of choice in ALL trimesters, followed by a full oral ACT course. A is wrong: oral AL is for uncomplicated malaria, and she is vomiting and confused. C is wrong: oral quinine + clindamycin is for uncomplicated 1st-trimester malaria. D is wrong: quinine is a second-line drug here and must never be given as a bolus; the loading dose is infused over 4 h in dextrose (risk of hypotension, arrhythmia and hypoglycaemia). E is wrong: IPTp is prevention, not treatment, and waiting would be fatal.

23. E · Counsel her to take TLD 2 h before (or 6 h after) iron and calcium, or all with food, and repeat the VL after enhanced adherence counselling

E is correct: iron and calcium (polyvalent cations) chelate dolutegravir in the gut and reduce its absorption when taken together fasting. Separate them (DTG 2 h before or 6 h after) or take them together with a meal, give adherence counselling, and repeat the VL. A is wrong: one raised VL after 4 months with a clear, fixable cause is not treatment failure; switching needs repeated non-suppression after IAC. B is wrong: she still needs iron and calcium; only the timing must change. C is wrong: the NTD signal fell to the background rate, and at 28 weeks the neural tube closed long ago. D is wrong: twice-daily DTG is for women on rifampicin, not for cation interactions.

24. C · Benzathine penicillin 2.4 MU IM weekly for 3 weeks, and treat her partner

C is correct: an asymptomatic woman with no known infection date has latent syphilis of unknown duration, treated with benzathine penicillin 2.4 MU IM weekly × 3; the partner must be treated or she will be reinfected. A is wrong: erythromycin does not reliably cross the placenta, so the fetus is not treated; it is only for true penicillin allergy. B is wrong: in pregnancy treat on the same day; do not delay for a lab result. D is wrong: a single dose is for early syphilis (<2 years), and partner treatment is essential. E is wrong: doxycycline is contraindicated in pregnancy.

25. A · Paracetamol, fluids, CTG/fetal monitoring and observation; continue the penicillin course as planned

A is correct: this is a Jarisch–Herxheimer reaction (fever, myalgia, contractions within 24 h of the first dose), which is self-limiting. Treat symptoms, monitor mother and fetus, and continue treatment. B is wrong: it is not an allergy; switching to erythromycin would leave the fetus untreated. C is wrong: there are no features of anaphylaxis. D is wrong: mild tightenings during a JHR usually settle; tocolysis and steroids are only considered if true preterm labour develops. E is wrong: the timing after penicillin and absence of uterine tenderness or offensive liquor make chorioamnionitis unlikely.

26. D · Give anti-D immunoglobulin 1,500 IU (300 mcg) IM to the mother within 72 hours

D is correct: a non-sensitised RhD-negative mother of an RhD-positive baby needs anti-D within 72 h of delivery; in Uganda the usual vial is 1,500 IU (300 mcg) IM. A is wrong: ABO incompatibility gives only partial protection; it is not a reason to withhold anti-D. B is wrong: anti-D is never given to the baby. C is wrong: sensitisation happens at this delivery, so waiting is too late. E is wrong: every non-sensitised woman should receive it (future plans can change), and the Kleihauer test guides extra doses rather than the decision to give the standard dose.

27. B · Clotrimazole 100 mg vaginal pessary nightly for 7 days

B is correct: vulvovaginal candidiasis in pregnancy is treated with a topical azole, and a longer 7-day course of clotrimazole works better in pregnancy. A is wrong: oral fluconazole is avoided in pregnancy (high-dose or prolonged use is linked to craniofacial and cardiac defects, and even single doses have a miscarriage signal). C is wrong: metronidazole treats BV and trichomoniasis, not Candida. D is wrong: doxycycline is contraindicated in pregnancy and does not treat fungi. E is wrong: although candidiasis is common in pregnancy, symptomatic infection should be treated.

28. C · Give DMPA 150 mg IM every 12 weeks (or offer a copper IUD)

C is right: rifampicin is a potent CYP3A4 inducer. DMPA (high dose, parenteral) and IUDs are WHO MEC 1 with rifampicin; she should keep using it for 28 days after TB treatment ends. A is wrong: rifampicin makes the COC unreliable (MEC 3); dolutegravir itself does not interact, but rifampicin does. B is wrong: implant levels also fall with rifampicin (MEC 3). D is wrong: the POP is the most vulnerable to enzyme induction. E is wrong: doubling the pill is not a recommended strategy and still leaves her at risk; choose an unaffected method.

29. A · Stop injecting, call for help, give 100% oxygen and support the airway, and give IV diazepam

A is right: this is local anaesthetic systemic toxicity (500 mg = about 9 mg/kg, three times the 3 mg/kg plain maximum). Stop the drug, call for help, secure the airway with oxygen (hypoxia and acidosis worsen toxicity) and stop the seizure with a benzodiazepine; give 20% lipid emulsion if there is cardiovascular instability or arrest. B is wrong: she has just delivered and had no hypertension or proteinuria; the timing with the injection and the prodrome point to LAST (eclampsia is a differential to exclude, not the first treatment). C is wrong: continuing wastes time and may add more drug. D is wrong: naloxone reverses opioids, not local anaesthetics. E is wrong: giving more lignocaine worsens toxicity; amiodarone is used for arrhythmias in LAST.

30. E · Copper IUD or LNG-IUS

E is right: an IUD (copper or LNG-IUS) is MEC 1–2 for smokers and for migraine with aura, is long-acting and reversible. A, B, C and D are all combined (oestrogen-containing) methods: migraine with aura is MEC 4 at any age, and age ≥ 35 with ≥ 15 cigarettes/day is also MEC 4, because of the risk of ischaemic stroke and MI. The route (patch, ring) or continuous use does not remove the oestrogen risk.

31. B · Start bag-and-mask ventilation with room air

B is right: per Helping Babies Breathe, a baby who is not breathing after drying and clearing the airway needs ventilation within the Golden Minute; most babies respond, including after pethidine. A is wrong as the next step: naloxone is only an adjunct once ventilation is effective, and it is dangerous if the mother is opioid-dependent. C is wrong: treating the mother now does not help the born baby. D is wrong: compressions start only if HR is below 60/min despite 30 s of effective ventilation. E is wrong: hypoglycaemia is not the cause of apnoea here, and airway/breathing come first.

32. D · Thiamine IV

D is right: confusion, nystagmus/ophthalmoplegia and ataxia in hyperemesis suggest Wernicke's encephalopathy. Thiamine must be given before any dextrose, which can use up the last thiamine and worsen it. A and B are useful anti-emetics but do not treat or prevent Wernicke's. C: steroids are a third-line option for refractory hyperemesis, not for this emergency. E is wrong and dangerous: potassium is added to the infusion according to U&E, never given as an IV bolus.

33. C · Wait until at least 12 hours after the last dose, then spinal, if mother and baby remain stable

C is right: a spinal is usually delayed until at least 12 h after a prophylactic LMWH dose (24 h after a treatment dose) because of the risk of spinal haematoma. In a stable Category 3 case, waiting is safer than GA. If the situation became urgent, a senior would weigh GA against the risk. A is wrong: the 12-hour rule applies. B is wrong: protamine only partly reverses LMWH and is not a routine way to allow a spinal. D is wrong: GA carries aspiration and airway risks and is not the routine choice in a stable woman. E is wrong: the APTT does not reflect LMWH (anti-Xa) activity.

34. A · Warfarin

A is right: warfarin embryopathy (nasal hypoplasia, stippled epiphyses) follows exposure at 6–12 weeks. B and C are wrong: heparins are large molecules that do not cross the placenta, which is why they are preferred in pregnancy. D is wrong: low-dose aspirin is not teratogenic. E is wrong: phenytoin causes fetal hydantoin syndrome (craniofacial features and hypoplastic nails/distal phalanges), and she was taking an anticoagulant.

35. E · Use contraception meanwhile, start folic acid 5 mg daily, and refer to a neurologist to switch to lamotrigine or levetiracetam before conceiving

E is right: valproate carries about a 10% risk of major malformation (including NTDs) and a 30–40% risk of neurodevelopmental problems, so it should be switched before conception under specialist care, with high-dose folic acid (5 mg) and effective contraception in the meantime. A is wrong: stopping abruptly risks status epilepticus and SUDEP. B is wrong: folic acid 400 mcg does not remove the valproate risk, and women on AEDs need 5 mg. C is wrong: delaying does not solve the problem; switching does. D is wrong: raising the valproate dose increases teratogenic risk (the dose-rising drug in pregnancy is lamotrigine).

36. B · Paracetamol 1 g orally 6-hourly (max 4 g/day), with physiotherapy advice

B is right: paracetamol is the analgesic of choice at any gestation. A and C are wrong: all NSAIDs are avoided after 20 weeks (fetal renal impairment and oligohydramnios) and contraindicated after 30 weeks (premature closure of the ductus arteriosus); ibuprofen is no safer here. D is wrong: regular opioids for weeks risk dependence, constipation and neonatal withdrawal; they are not first line for simple back pain. E is wrong: a parenteral opioid is excessive for mechanical back pain and is not a take-home plan.

Clinical cases

Work through each step before reading the model answer.

Case 1: Eclampsia on MgSO₄: the reflexes disappear

A 19-year-old primigravida at 37 weeks is brought to Mengo Hospital after a generalised fit at home. BP 176/118, pulse 104, RR 20, SpO₂ 96%. Urine protein 3+. She is post-ictal but breathing. The HC III gave nothing. She weighs 62 kg.

1. What are your immediate actions and which MgSO₄ regimen will you use?

Call for help; left lateral position, airway, suction, **oxygen**; IV access (bloods: FBC, creatinine, LFT, grouping, bedside clotting test, glucose, mRDT). **Pritchard loading dose: 4 g of 20% IV over 5–10 min** (8 mL of 50% + 12 mL water) **plus 10 g of 50% IM** (5 g in each buttock with 1 mL 2% lignocaine). Draw up **calcium gluconate 10% 10 mL** at the bedside. Insert a **Foley catheter** with hourly urine measurement.

2. Her BP remains 174/116 after the loading dose. What do you give?

MgSO₄ is not an antihypertensive. Once she can swallow safely: **nifedipine IR 10 mg orally** (swallowed), or while drowsy **hydralazine 5 mg IV slowly** (20 mg diluted to 20 mL, 5 mL over 2–5 min), repeating every 20–30 min to a max of 20 mg. Target **140–150/90–100**; avoid a precipitous fall (fetal distress). Monitor FHR. Plan delivery once stable.

3. She delivers by caesarean section. At 02:00 (before the 4-hourly IM dose) she is drowsy, RR 11/min, SpO₂ 91%, patellar reflexes absent, urine 15 mL/h for 3 hours. Interpret and act.

Magnesium toxicity from **oliguria** (renal excretion). **Withhold** the dose; call the senior and anaesthetist; ABC, **oxygen**, bag-mask ventilation if RR keeps falling; **calcium gluconate 10%, 10 mL (1 g) IV slowly over 10 min**. Continuous SpO₂ and ECG if available. (She had a spinal? then test biceps reflex; general anaesthetic relaxants would also be potentiated.)

4. What investigations and cause-finding do you do?

Serum Mg²⁺ if available (loss of reflexes ≈ 4–5 mmol/L; respiratory depression ≈ 6–7.5 mmol/L), **creatinine, urea, potassium**, FBC and LFT (HELLP), fluid-balance chart review (hidden bleeding, over-restriction or overload), and a **chart review for dosing errors** (double loading at referral, mL/g confusion, 50% given IV).

5. Thirty minutes later RR is 18, SpO₂ 97%, and reflexes have returned. Urine is still 20 mL/h. When and how do you restart MgSO₄?

She still needs prophylaxis for **24 h after delivery**. Restart only when **reflexes present, RR ≥16 and urine ≥30 mL/h**; with persisting oliguria, give a **reduced dose** (e.g. 0.5 g/h by infusion, or delay/halve the IM dose per unit protocol, senior decision) with **hourly R-R-U checks**. Restrict fluids to ~80 mL/h; avoid big boluses; consider renal review if creatinine keeps rising.

6. What postpartum drug plan and counselling does she need before discharge?

Continue antihypertensives (**nifedipine MR 20 mg 12-hourly** or amlodipine/enalapril; not methyldopa long term postpartum), BP 4-hourly for ≥3 days, repeat bloods at 24–48 h, oxytocin not ergometrine, BP review within 1 week. Counsel: recurrence risk, **aspirin 75–150 mg from 12 weeks** and **calcium** in the next pregnancy, lifelong cardiovascular risk, and contraception without oestrogen while hypertensive.

Take-away: Magnesium is excreted by the kidneys: oliguria turns a therapeutic dose into a toxic one. Check R-R-U before every dose, keep calcium gluconate 1 g at the bedside, and remember the ladder: reflexes go

first, then breathing, then the heart.

Case 2: Augmentation that went too fast

Mrs K, a 34-year-old para 5+0 at 39 weeks, is admitted to Mengo Hospital labour ward in active labour. At 7 cm her progress stalls for 3 hours with 2 weak contractions in 10 minutes; membranes ruptured spontaneously with clear liquor. The night intern starts oxytocin 5 IU in 500 mL of 5% dextrose at 20 drops/min and increases it by 20 drops/min every 15 minutes. One hour later she has 6 contractions in 10 minutes, each lasting about 70 seconds, she is distressed with pain between contractions, and the fetal heart rate is 95 bpm with late decelerations.

1. Before looking at the fetal heart, what was wrong with the decision to augment and the regimen used?

- **Grand multipara (para 5):** slow progress is **obstruction until proven otherwise**; she needed a senior review for CPD (moulding, caput, descent, Bandl's ring, haematuria) before any oxytocin.
- **Starting dose too high and escalation too fast:** WHO MCPC starts **2.5 IU in 500 mL at 10 drops/min** and increases by **10 drops/min every 30 min** (oxytocin takes ~30 min to reach steady state).
- **5% dextrose** as diluent risks **water intoxication**; use NS or RL.
- Oxytocin in a grand multip should be used only with great caution and close one-to-one monitoring.

2. What is happening now, and what do you do in the next 2 minutes?

Uterine tachysystole (hyperstimulation) with fetal distress.

1. **Stop the oxytocin immediately** (half-life 3–5 min).
2. **Left lateral position.**
3. **IV crystalloid** (Ringer's lactate), oxygen per protocol.
4. **Call the senior/obstetrician**; continuous FHR auscultation.
5. Vaginal exam: exclude cord prolapse, check dilatation and for signs of obstruction.
6. Watch for **rupture**: pain between contractions, scar-like tenderness, contractions suddenly stopping, bleeding, maternal tachycardia.

3. Ten minutes later the contractions are still 5–6 in 10 minutes and the FHR is 100 bpm. What drug can help, and at what dose?

Acute tocolysis: terbutaline 250 µg (0.25 mg) SC (or IV slowly over 5 min), or salbutamol 10 mg in 1 L IV fluid at 10 drops/min (WHO MCPC). Watch for maternal tachycardia and palpitations. Meanwhile **prepare for delivery**: if the FHR does not recover, she needs a **category 1 caesarean section** (she is not fully dilated).

4. The FHR recovers to 140 bpm with relaxation between contractions. Review finds moulding 1+ and no caput; the head is 2/5 palpable. The senior decides to restart oxytocin. How should it be done? Calculate the starting dose in mIU/min.

- Restart only after senior review, at the **lowest dose: 2.5 IU in 500 mL NS/RL at 10 drops/min**, increasing by **10 drops/min every 30 min** to a max of 60 drops/min; for a multipara, **do not go to a stronger solution**: failure means CS.
- Dose: $2,500 \text{ mIU} \div 500 \text{ mL} = 5 \text{ mIU/mL}$; $10 \text{ drops/min} \div 20 = 0.5 \text{ mL/min}$; $5 \times 0.5 = 2.5 \text{ mIU/min}$.
- **One-to-one care:** contractions and FHR every 15–30 min, recorded on the partograph with IU/L and drops/min.

5. She delivers a 3.9 kg baby vaginally 3 hours later. What is her PPH risk and what prevention do you give?

High risk of atonic PPH: grand multipara, large baby, **prolonged oxytocin exposure** (receptor desensitisation), long labour. Give **AMTSL: oxytocin 10 IU IM within 1 minute**, CCT, check tone every 15 min for 2 h, and **continue an oxytocin infusion** (e.g. 20 IU in 1 L) for a few hours. Have **misoprostol 800 µg SL** and **TXA 1 g IV** ready, and a calibrated drape in place.

6. Thirty minutes later she has lost 600 mL and her uterus is boggy. Her BP is 150/100 with 2+ protein (previously unrecognised pre-eclampsia). Which drugs do you give and which do you avoid?

Start the **E-MOTIVE bundle together**: massage, **oxytocin 10 IU IV slowly + infusion**, **TXA 1 g IV over 10 min** (within 3 h of birth), IV fluids, examine the genital tract and placenta, escalate. Add **misoprostol 800 µg SL**. **Avoid ergometrine and Syntometrine** (hypertension/pre-eclampsia: stroke, eclampsia). Carboprost only if available and she is **not asthmatic**. Also treat the pre-eclampsia (**MgSO₄**, antihypertensive if $\geq 160/110$).

Take-away: Oxytocin is life-saving after birth but dangerous before it: in a grand multipara, exclude obstruction first, start low (2.5 mIU/min), go slow (every 30 min), count every contraction, and at the first sign of tachysystole stop the drip, turn her left lateral and consider terbutaline. Afterwards, expect atonic PPH, and always check the BP before ergometrine.

Case 3: Fever after caesarean section: choosing and dosing antibiotics

Mrs A., a 26-year-old P2 weighing 62 kg, is on day 4 after an emergency caesarean section at an HC IV for obstructed labour after 20 hours of labour with membranes ruptured for 16 hours. She received ceftriaxone 1 g IV at the start of the operation. Today she has a temperature of 38.9°C, pulse 118/min, BP 96/60 mmHg, RR 24/min, a tender bulky uterus, offensive lochia and a slightly red wound. She is breastfeeding. She has no known drug allergies. She is referred to Mengo Hospital.

1. What are your immediate priorities in the first hour?

Recognise **maternal sepsis** (fever, tachycardia, borderline BP, RR ≥ 22). **ABC**, oxygen if SpO₂ <94%, two large-bore cannulae, **blood cultures** (and wound swab, high vaginal swab), FBC, creatinine, lactate if available, **mRDT/smear**, RBS, group and cross-match. **Fluid bolus 500 mL Ringer's lactate** and reassess. **Catheterise for hourly urine output**. Start **IV antibiotics within 1 hour**. Call a senior.

2. Which antibiotics would you start, and why this combination?

Ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV once daily + metronidazole 500 mg IV 8-hourly. Post-CS endometritis is **polymicrobial**: ampicillin covers streptococci and many Gram-positives, gentamicin covers Gram-negatives (*E. coli*), metronidazole covers anaerobes (*Bacteroides*). An acceptable alternative is ceftriaxone 1–2 g IV daily + metronidazole 500 mg IV 8-hourly.

3. Calculate her gentamicin dose. The ward has 80 mg/2 mL ampoules. How do you give it, and why not 80 mg 8-hourly?

5 mg/kg × 62 kg = 310 mg, rounded to **320 mg (4 ampoules = 8 mL) IV over 3–5 min once daily** (300 mg is also acceptable). Once-daily dosing gives a high peak (concentration-dependent killing), a post-antibiotic effect and **less nephrotoxicity**. **80 mg 8-hourly = 240 mg/day**, an under-dose for her weight. Gentamicin is compatible with breastfeeding.

4. On day 2 of treatment her creatinine has risen from 68 to 118 µmol/L and urine output is 20 mL/h. What do you do with the antibiotics?

This is **acute kidney injury** in sepsis. **Stop gentamicin** (nephrotoxic and ototoxic, and it accumulates when the kidneys fail) and switch Gram-negative cover to **ceftriaxone 1–2 g IV daily**, continuing **metronidazole** (ampicillin can continue or be dropped once ceftriaxone is running). Optimise fluids with careful balance, avoid NSAIDs, check creatinine and electrolytes daily, and review drug doses.

5. After 72 hours she is still febrile. What are you looking for?

Antibiotics do not replace source control. Look for: **retained products or pelvic abscess / infected haematoma** (ultrasound), **wound infection or collection** (open and drain), **uterine necrosis** or dehiscence with peritonitis (laparotomy, possibly hysterectomy), **mastitis**, **DVT/septic pelvic thrombophlebitis**, **malaria**, UTI/pyelonephritis, chest infection, and **resistant organisms** (review culture results; ESBL → escalate per microbiology, e.g. meropenem at referral level).

6. She improves and is afebrile for 48 hours. How do you complete treatment, and what do you tell her about breastfeeding?

Switch to oral therapy, e.g. **amoxicillin 500 mg 8-hourly (or amoxicillin–clavulanate 625 mg 8-hourly) + metronidazole 400 mg 8-hourly** to complete about **7 days** in total (longer for an abscess). **Continue breastfeeding**: penicillins, cephalosporins, gentamicin and standard-dose metronidazole are all compatible. Advise hygiene, danger signs (fever, offensive lochia, wound pus), and review in clinic.

Take-away: Puerperal sepsis needs the triple regimen (ampicillin + gentamicin 5 mg/kg once daily + metronidazole) within 1 hour, calculated by weight, with daily creatinine and urine output: stop gentamicin at the first sign of AKI and switch to ceftriaxone. If fever persists at 48–72 h, look for the source rather than just changing the antibiotic.

Case 4: Seizure during a perineal repair: local anaesthetic toxicity

At 02:00 on the Mengo labour ward, a 21-year-old primipara weighing **50 kg** has just had a spontaneous vaginal delivery with a second-degree tear. A junior clerk, unable to find 1% lignocaine, uses a 20 mL vial of **2% lignocaine**, and a colleague had already given 5 mL of 1% for the episiotomy. Midway through the repair the woman says her mouth feels numb, she has ringing in her ears and a 'metal' taste. Blood loss was about 300 mL; BP was 118/72 an hour ago with no proteinuria.

1. What is the likely diagnosis, and how much lignocaine has she received relative to the safe maximum?

Local anaesthetic systemic toxicity (LAST). Dose given: $20 \text{ mL} \times 20 \text{ mg/mL} = 400 \text{ mg} + 5 \text{ mL} \times 10 \text{ mg/mL} = 50 \text{ mg} = 450 \text{ mg} (9 \text{ mg/kg})$. The maximum for plain lignocaine is **3 mg/kg = 150 mg** (15 mL of 1%). She has had **three times** the maximum. Early CNS symptoms (circumoral numbness, tinnitus, metallic taste) come before seizures and cardiac toxicity.

2. What do you do in the first minute?

1. **Stop injecting** immediately.
2. **Shout for help** (midwife, doctor on call, anaesthetist); bring the resuscitation trolley.
3. **Airway and oxygen**: 100% O₂ by face mask, suction ready, recovery position if drowsy.
4. **IV access** (two cannulas), attach pulse oximeter, BP and ECG if available.
5. Tell the team the total dose given and ask someone to fetch **20% lipid emulsion** (theatre).

3. She now has a generalised tonic-clonic seizure. How do you manage it, and what differential must you consider?

Protect her from injury, support the airway, oxygen, and give **diazepam 10 mg IV slowly** (or midazolam; rectal diazepam 20 mg if no IV). Check a capillary glucose. **Differential: postpartum eclampsia** (check BP and urine protein; if hypertensive or proteinuric give **MgSO₄**), hypoglycaemia, epilepsy, amniotic fluid embolism. Here the timing with a large local anaesthetic dose and the prodrome make LAST most likely.

4. Her pulse falls to 40/min and BP is 70/40, then she becomes pulseless with a broad-complex rhythm. What now? Calculate the lipid doses.

Start CPR (ACLS); ventilate with oxygen. Give 20% lipid emulsion: bolus 1.5 mL/kg = 75 mL over 1 min, then infusion 15 mL/kg/h = 750 mL/h. If still unstable after 5 min, repeat the bolus (maximum twice more) and double the infusion to 30 mL/kg/h (1,500 mL/h). Maximum cumulative dose 12 mL/kg = 600 mL. Use small adrenaline doses (≤ 1 mcg/kg), and amiodarone, not lignocaine, for arrhythmias. Continue CPR for a long time (at least 60 min): recovery can be slow.

5. She recovers after the lipid. What is her aftercare?

Transfer to a **high-dependency** bed with continuous monitoring (ECG, SpO₂, BP) for at least **12 h** (cardiac toxicity can recur). Check glucose, electrolytes and, after lipid, **amylase** (pancreatitis). Complete the repair later under a proper block or anaesthesia with a correctly calculated dose. Watch for PPH (she has been stressed and immobile). Keep the baby with her when safe and support breastfeeding. **Debrief** her and her family honestly.

6. How could this have been prevented, and what system changes would you suggest?

- **Calculate the maximum dose before starting** (3 mg/kg plain; 50 kg → 150 mg = **15 mL of 1%** or **7.5 mL of 2%**) and **add every dose** given by anyone.
- If only 2% is available, **dilute it** to 1% with sterile saline, or use half the volume.
- **Aspirate before each injection** and inject slowly in small aliquots.
- Supervision of junior staff; stock **1% lignocaine** in delivery packs.
- Keep **lipid emulsion** where local anaesthetics are used (labour ward, theatre) and train staff.
- **Report** the incident through the hospital's quality/safety system (a learning, no-blame review).

Take-away: Lignocaine 1% = 10 mg/mL and 2% = 20 mg/mL; the plain maximum is **3 mg/kg**. Calculate before you inject, aspirate, and recognise LAST early (numb lips, metallic taste, tinnitus). Manage it with stop, call, oxygen, benzodiazepine for seizures, CPR and **20% lipid 1.5 mL/kg then 15 mL/kg/h**.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.